SCIENTIFIC JOURNAL PUBLICATION TREND TRACKING SYSTEM

Group 6

I. INTRODUCTION

1 Overview

1.1 Project Information

- Project Name: Scientific Journal Publication Trend Tracking System
- Group Name: Group 6
- Software Type: Web Application
- GitHub Repository:

  `https://github.com/NguyenNgocBaoTram10/Scientific-Journal-Publication-Trend-Tracking-System`

1.2 Project team

Full Name	Role	Student ID	Email
Lê Thanh Tú	Leader	049206004778	tult4778@ut.edu.vn
Nguyễn Ngọc Bảo Trâm	Member	082306008015	tramnnb8015@ut.edu.vn
Lê Thị Huyền Trân	Member	077306003635	tranlth3635@ut.edu.vn
Trần Trung Hiếu	Member	089206008473	hieutt8473@ut.edu.vn
Trần Văn Học	Member	052206005618	hoctv5618@ut.edu.vn
Nguyễn Đoàn Nhật Quang	Member	079206010585	quangndn0585@ut.edu.vn

Bảng 1: Project Team Information

2 Project Scope and Limitations

2.1 Major Features

2.1.1 Major Features for User

- Register and log in to the system.
- Search scientific publications using keywords, authors, journals, or research fields.
- View detailed publication information.
- Save publications to bookmarks.

- Follow research topics of interest.
- Receive notifications about new publications and research trends.
- View publication trend analysis and statistical dashboards.
- Generate and export analytical reports (PDF/CSV).

2.1.2 Major Features for Administrator

- Log in and log out.
- Manage user accounts.
- Configure academic API sources (Semantic Scholar, OpenAlex, and Crossref).
- Synchronize publication data from external APIs.
- Monitor the data synchronization process.
- Manage system notifications.
- Generate system reports.
- Configure system settings.

2.2 Project Limitations

2.2.1 Limited Use of Full-Text Data

- The system only uses publication metadata such as title, abstract, keywords, authors, and publication year.
- It does not process full-text content, which limits the depth of analysis and understanding of research papers.

2.2.2 Restricted Data Scope

- The system focuses only on selected research fields (e.g., Computer Science and Artificial Intelligence).
- Therefore, it does not fully represent trends across all scientific disciplines.

2.2.3 Dependence on External APIs

- The system relies on external APIs such as Semantic Scholar, OpenAlex, and Crossref.
- Any API downtime, structural changes, or request limitations may affect system stability and data collection performance.

2.2.4 Simplified Trend Analysis and Update Frequency

- Trend analysis is mainly based on publication frequency and growth rate.
- It does not apply advanced Machine Learning or AI models for prediction.
- Data is synchronized periodically (daily or weekly), so real-time trends may not be immediately reflected.

II. PROJECT MANAGEMENT PLAN

1 Overview

1.1 Scope and estimation

#	WBS Item	Complexity	Duration (Days)
1. Project Initiation			
1.1	Define project scope	Medium	1
1.2	Collect and analyze requirements	Complex	2
2. Project Management and Environment Setup			
2.1	Set up development environment (GitHub, Database, Backend, Frontend)	Simple	1
2.2	Design database (ERD, Database Schema)	Complex	2
2.3	Configure security (JWT, RBAC, Password Hashing)	Complex	2
3. System Design			
3.1	Design Use Case Diagram and Activity Diagram	Medium	1
3.2	Design Sequence Diagram	Medium	1
3.3	Design DFD and ERD	Medium	1
4. System Development			
4.1	User Registration and Login	Medium	2
4.2	Search Scientific Publications	Complex	2
4.3	View Publication Details	Medium	1
4.4	Bookmark Management	Medium	1
4.5	Research Topic Following	Medium	1
4.6	Notification Management	Medium	1
4.7	Publication Trend Analysis	Complex	3
4.8	Develop Statistical Dashboard	Complex	2
4.9	Display Trending Topics	Medium	1
4.10	Generate Statistical Reports (PDF/CSV)	Complex	2

#	WBS Item	Complexity	Duration (Days)
5. Administration Functions			
5.1	User Management	Complex	2
5.2	Configure Academic API Sources	Complex	2
5.3	Synchronize Data from Semantic Scholar, OpenAlex, and Crossref	Complex	3
6. Testing			
6.1	Prepare Test Plan	Medium	1
6.2	Design Test Cases	Medium	2
6.3	Execute Testing	Complex	2
6.4	Fix Defects After Testing	Complex	2
7. Project Finalization			
7.1	Complete Project Documentation	Medium	2
7.2	Deploy the System	Medium	1
		Total Duration	**45 Working Days**

1.2 Project Objective

- **Timeline:** The project must be completed before **July 3, 2026**.
- **Allocated Effort:** 45 man-days.

1.3 Project Risks

#	Risk Description	Impact	Possibilit	Response Plan
1	External academic APIs (Semantic Scholar, OpenAlex, Crossref) become unavailable or change their endpoints.	The system cannot retrieve publication data, causing incomplete search results and inaccurate trend analysis.	High	Implement API fallback mechanisms, monitor API status regularly, and cache previously retrieved data whenever possible.

#	Risk Description	Impact	Possibility	Response Plan
2	Large volumes of publication data lead to slow system performance.	Users experience long response times when searching publications or viewing dashboards.	Medium	Optimize database indexing, implement pagination, caching, and asynchronous background synchronization.
3	Security vulnerabilities expose user accounts or stored data.	Unauthorized access, data leakage, and loss of user trust.	Medium	Use JWT authentication, RBAC authorization, password hashing, HTTPS, and regular security testing.
4	Trend analysis results become inaccurate because of outdated or incomplete datasets.	Generated reports and statistics may mislead users' research decisions.	Medium	Schedule automatic data synchronization and validate collected data before analysis.
5	System integration between frontend, backend, and database encounters compatibility issues.	Development delays and unexpected system failures during deployment.	Medium	Perform integration testing continuously and maintain version compatibility among all components.
6	Insufficient testing before deployment leaves critical defects unresolved.	System crashes, incorrect functionality, and poor user experience.	High	Prepare comprehensive Test Plans, Test Cases, conduct functional, integration, and user acceptance testing before release.
7	Project schedule delays due to underestimated implementation effort.	Failure to complete the project within the planned timeline.	Medium	Monitor project progress weekly, prioritize critical features, and adjust the implementation schedule when necessary.

2 Project Deliverables

Sprint	Week	Duration	Deliverables	Note
Sprint 1	Week 1	14/05 – 20/05/2026	• Define project scope • Analysis requirement	
Sprint 2	Week 2	21/05 – 27/05/2026	• Finish Product Overview • Finish User Requirement • Finish Functional Requirement	
Sprint 3	Week 3	28/05 – 03/06/2026	• Finish Database Design • Finish Software design description	
Sprint 4	Week 4	04/06 – 10/06/2026	• Finish Software requirement specification	
Sprint 5	Week 5	11/06 – 17/06/2026	• Finish Software design description (continued)	
Sprint 6	Week 6	18/06 – 24/06/2026	• Finish Software testing document • Finish Release package and user guide	
Sprint 7	Week 7	25/06 – 01/07/2026	• Testing • Finish document • Submitting	

3. **Responsibility Assignment**

Do~D; Review~R; Support~S

Responsibility	Thanh Tú	Bảo Trâm	Huyền Trân	Trung Hiếu	Văn Học	Nhật Quang
Project Planning & Tracking	S	D	S	S	S	S
Prepare Project Introduction Document	R	D	R	R	R	R
Prepare Software Requirement Document	S	D	S	S	S	S
Prepare Software Design Document	D	D	D	D	D	D
Prepare Test Document	D	D	D	D	D	D
Prepare Software User Guides	D	D	D	D	D	D
Prepare Final Report	D	D	D	D	D	D

4. **Project Communications**

Communication Item	Who	Purpose	When	Type
Team Weekly Meeting	Group	• Report work project • Assign work for next week • Control project deadline	Every Sunday	Meeting on Google Mee
Weekly Report	Member	• Report work process	1 day / week	Report via Za
Group Meeting	Group	• Report work process	Every Friday	Offline on cla

5. Configuration Management

5.1 Document Management

- Document tool: Confluence
- Task management tool: Jira

5.2 Source Code Management

- Source code management tool: GitHub

5.3 Tools & Infrastructure

Category	Tools / Infrastructure
Programming Languages	PHP, HTML5, CSS3, JavaScript
Database	MySQL
Database Administration	phpMyAdmin
Academic APIs	OpenAlex API, Crossref REST API
API Testing	Postman
Version Control	GitHub
Project Management	Jira, Confluence
Diagramming	Draw.io, PlantUML
IDE / Code Editor	Visual Studio Code
Local Server	WAMP Server

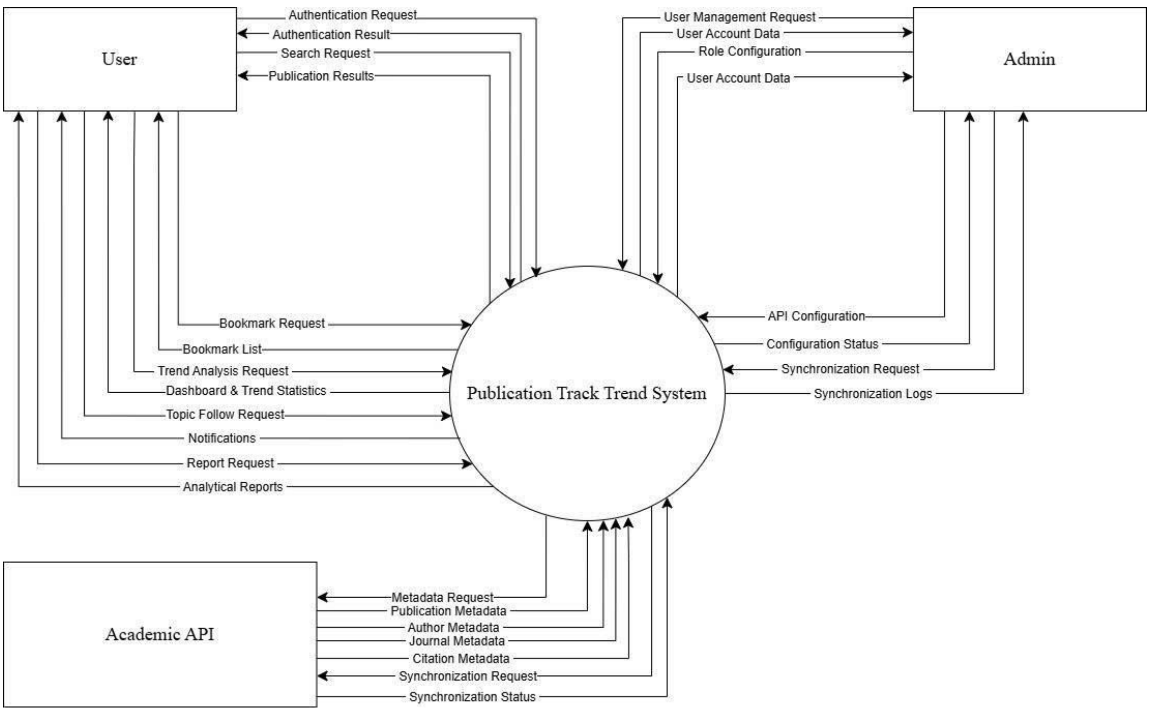
III. SOFTWARE REQUIREMENT SPECIFICATION

1 Product Overview

The Scientific Journal Publication Trend Tracking System (SJPTTS) is a web-based application developed to help researchers, students, lecturers, and academic institutions search, monitor, and analyze scientific publication trends. The system integrates academic data from OpenAlex, and Crossref to provide comprehensive and up-to-date publication information.

The system allows users to search scientific publications using keywords, authors, journals, research fields, or publication years. Users can view detailed publication information, bookmark publications, follow research topics, receive notifications about newly published papers, analyze publication trends through interactive dashboards, and export reports in PDF or CSV format.

Administrators can manage user accounts, configure academic API sources, synchronize publication data, monitor system operations, and maintain the overall performance of the platform. The system aims to provide an efficient solution for discovering scientific literature and supporting research through publication trend analysis.



Hình 1: Publication Trend Tracking System

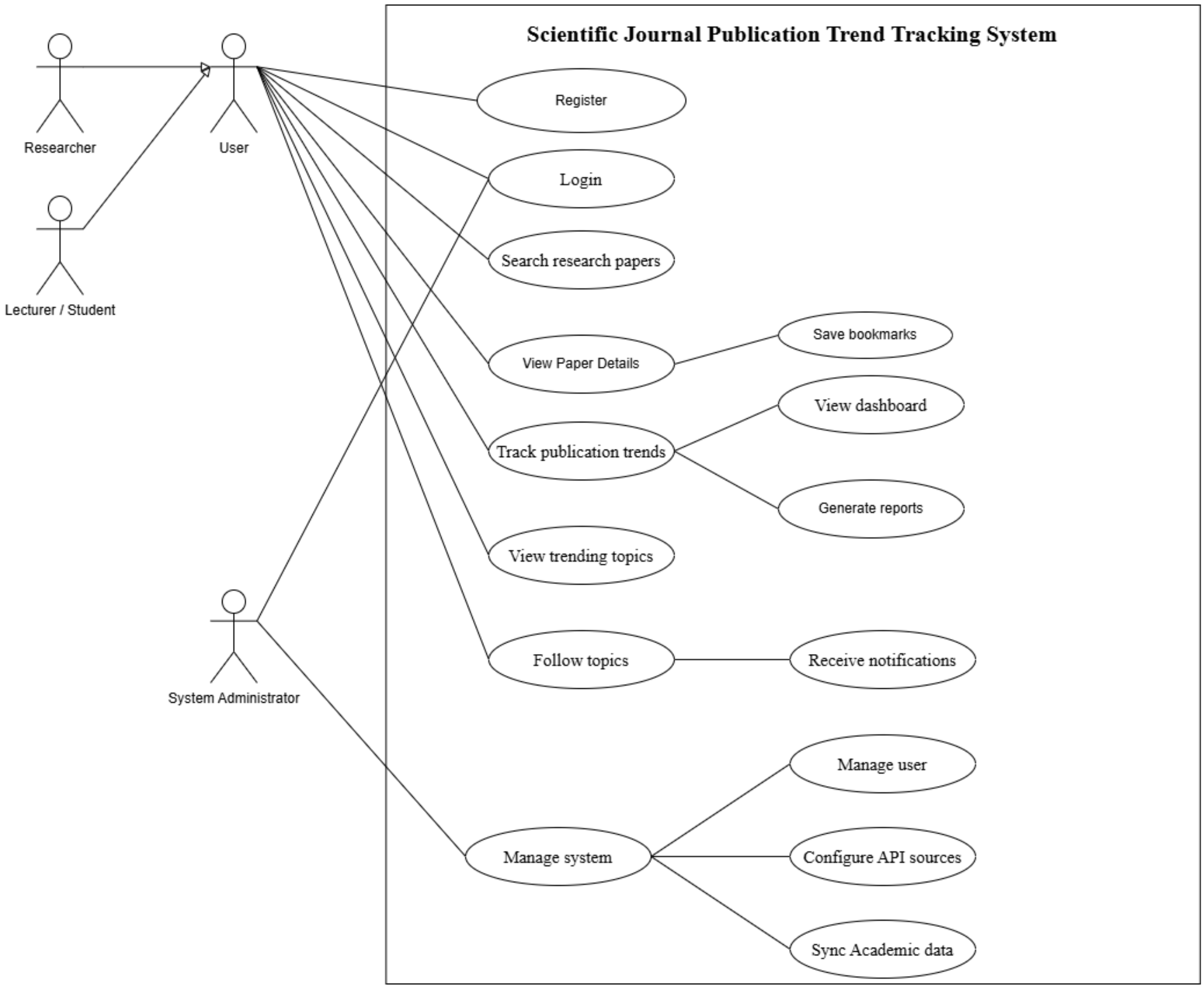
2 User Requirements

2.1 Actors

#	Actor	Description
1	User	System user who can search scientific publications, view publication details, save bookmarks, follow research topics, receive notifications, and analyze publication trends.
2	Administrator	System administrator responsible for managing users, configuring the system, setting up and validating academic data sources (OpenAlex, Crossref), synchronizing data, and monitoring system operations.
3	Publication Trend Tracking System	The core system that handles all functionalities such as paper search, trend analysis, dashboard generation, trending topic detection, bookmark management, notification delivery, and report generation.
4	Academic API	External services such as OpenAlex and Crossref that provide scientific publication metadata (authors, journals, citations) for system synchronization and analysis.

2.2 Use Case Diagrams

2.2.1 Diagram



Hình 2: Use Case Diagram

2.2.2 Use Case Descriptions

Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Manage Account	User	Register, Login	Users create an account and authenticate to access the Scientific Journal Publication Trend Tracking System.	A registered account is created and a secure user session is established.

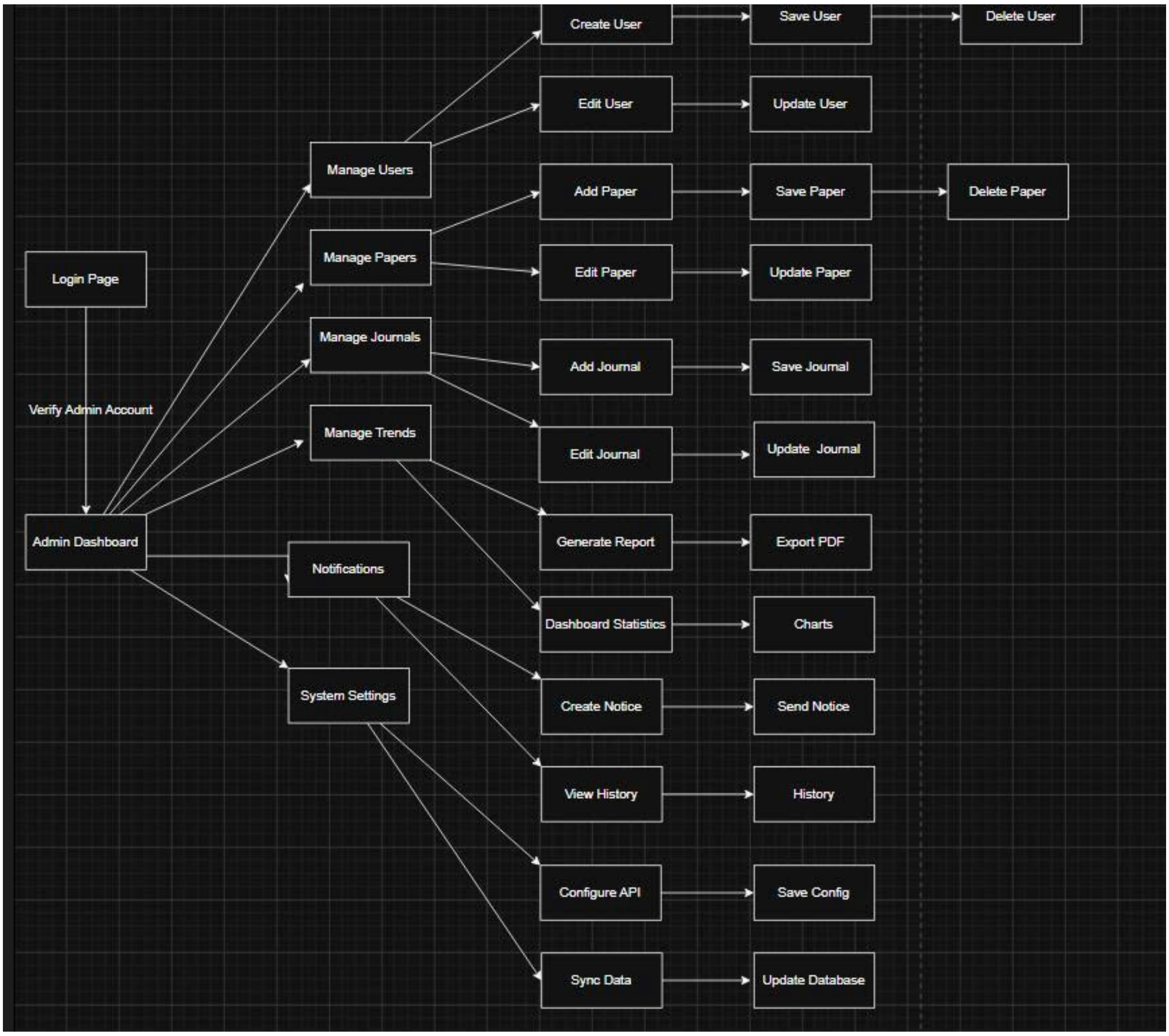
Main UC Group	Main Agent (Actor)	Sub-UCs	Brief Description of the Process	Output
Search Publications	User	Search Research Papers, View Paper Details	Users search scientific publications using keywords, author names, or journal names and view detailed publication metadata.	Matching publications and detailed publication information are displayed.
Manage Bookmarks	User	Save Bookmarks	Users save selected publications into their personal bookmark collection for future reference.	The selected publication is stored in the user's bookmark list.
Track Publication Trends	User	Track Publication Trends, View Dashboard, View Trending Topics	Users analyze publication trends by keyword, topic, author, journal, and time range. The system generates statistical charts and identifies trending research topics.	Trend statistics, dashboards and trending topics are displayed.
Follow Topics	User	Follow Topics, Receive Notifications	Users follow research topics to receive notifications about tracked activities	The selected topic is added to the user's followed-topic list and notifications become available.
Generate Reports	User	Generate Reports	Users generate analytical reports based on publication trend analysis and selected report parameters.	An analytical report summarizing publication trends is generated.
Manage System	Administrator	Manage System, Configure API Sources, Synchronize Academic Data	Administrators manage system settings, configure academic APIs, synchronize publication metadata and maintain system operations.	System settings and publication metadata are successfully updated.
Manage Users	Administrator	Manage Users	Administrators manage user accounts, roles, permissions and account status.	User information and access permissions are updated.

Main UC Group	**Main Agent (Actor)**	**Sub-UCs**	**Brief Description of the Process**	**Output**
Configure API Sources	Administrator	Configure API Sources	Administrators configure, update and test external academic APIs such as OpenAlex, Crossref and Semantic Scholar.	API configuration settings are successfully validated and stored.
Synchronize Academic Data	Administrator	Synchronize Academic Data	Administrators synchronize publication metadata from external academic APIs into the local database for searching and trend analysis.	The publication metadata database is updated with the latest synchronized records.

3 Functional Requirement

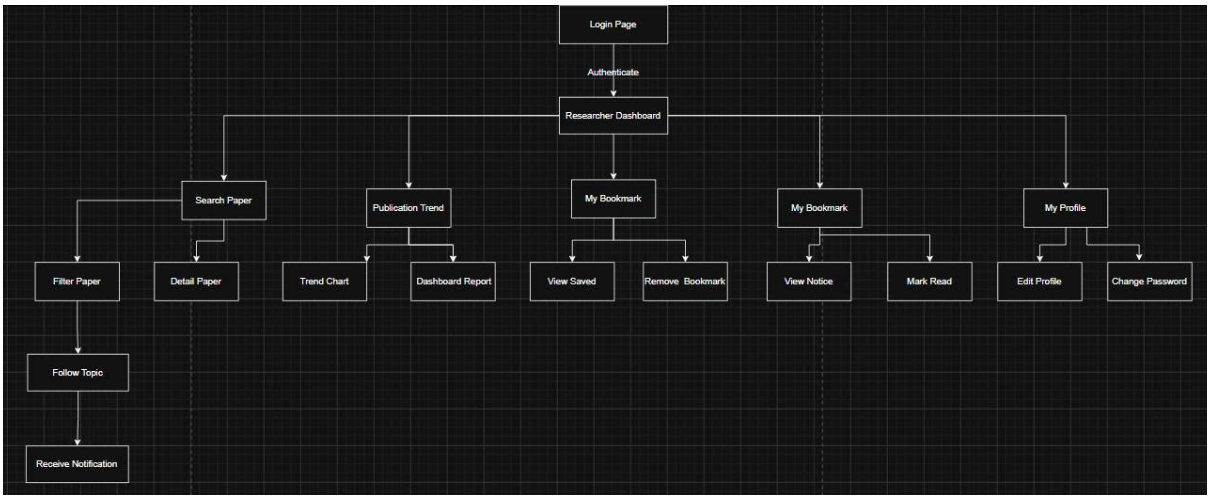
3.1 System Functional Overview

3.1.1 Screens Flow



Hình 3: Screens Flow

3.1.2 Screen Description



Hình 4: Screen Description

Functional Group	**Screen Name**	**Detailed Description**
User Authentication	Login	Purpose: Allows users to log into the system. Components: Username, Password, **Login** button, and **Register** link. Flow: The user enters credentials → The system authenticates the user → The main system interface is displayed. Rule: Username and password are required; invalid credentials will generate an error message.
User Authentication	Register	Purpose: Allows new users to create an account. Components: Full Name, Email, Username, Password, and **Register** button. Flow: The user enters the required information → The system validates the email and username → A new account is created → The user is redirected to the Login page. Rules: Email and username must be unique; passwords are securely hashed before storage.
Paper Management	Search Research Papers	Purpose: Allows users to search scientific publications by keyword, author, or journal. Components: Search box, filters, **Search** button, and result list. Flow: The user enters search criteria → The system searches publication metadata → Matching publications are displayed.

Functional Group	**Screen Name**	**Detailed Description**
Paper Management	View Paper Details	Purpose: Displays detailed metadata of a selected research paper. Components: Title, Authors, Abstract, Keywords, Publication Year, Journal Name, Citation Information, and **Save Bookmark** button. Flow: The user selects a paper → The system retrieves publication metadata → Paper details are displayed.
Paper Management	Bookmarks	Purpose: Manages the user's bookmarked papers. Components: Bookmark list, View Details button, Delete Bookmark button. Flow: The user views or removes bookmarked papers. Rule: Duplicate bookmarks are not allowed.
Publication Trend Analysis	Track Publication Trends	Purpose: Analyzes publication trends based on keywords, research topics, authors, or journals. Components: Search filters, time range selector, and **Analyze** button. Flow: The user enters analysis criteria → The system analyzes publication metadata → Statistical results and charts are displayed.
Publication Trend Analysis	Dashboard	Purpose: Displays an overview of publication statistics. Components: Publications by year chart, Topic distribution chart, Journal distribution chart, and Growth trend chart. Flow: The system generates dashboard visualizations after trend analysis is completed.
Publication Trend Analysis	Trending Topics	Purpose: Displays the most popular research topics. Components: Topic Name, Number of Publications, Trend Direction, and **Follow Topic** button. Flow: The system analyzes publication data → Ranks research topics → Displays trending topics.
Topic Management	Follow Topics	Purpose: Allows users to follow research topics of interest. Components: Topic list and **Follow/Unfollow** button. Flow: The user selects a topic → The system stores it in the followed-topic list.

Functional Group	**Screen Name**	**Detailed Description**
Notifications	Notifications	Purpose: Displays notifications about newly published papers and publication trend updates. Components: Notification list, Timestamp, Notification Content, and links to related papers or trend analysis results. Flow: The system generates notifications → The user views notification details.
Reporting	Generate Reports	Purpose: Generates analytical reports based on publication trend analysis. Components: Topic filter, Keyword filter, Journal filter, Time range selector, and **Generate Report** button. Flow: The user selects report criteria → The system compiles publication statistics → The analytical report is generated and displayed.
System Administration	Manage System	Purpose: Provides administrative functions for system management. Components: User Management, API Configuration, and Academic Data Synchronization. Flow: The administrator selects a management function → The system performs the requested operation.
System Administration	Manage Users	Purpose: Manages user accounts. Components: User list, Edit User Information, Change User Role, Activate Account, and Deactivate Account. Flow: The administrator selects a user → Updates account information → Saves the changes.
System Administration	Configure API Sources	Purpose: Configures external academic data sources. Components: API List, Endpoint URL, API Key, **Test Connection** button, and **Save** button. Flow: The administrator updates API settings → The system tests the connection → The configuration is saved.
System Administration	Synchronize Academic Data	Purpose: Synchronizes publication metadata from external academic APIs into the system database. Components: API List, **Synchronize** button, Synchronization Log, and Synchronization Statistics. Flow: The administrator starts synchronization → The system retrieves metadata from academic APIs → The database is updated → Synchronization results are displayed.

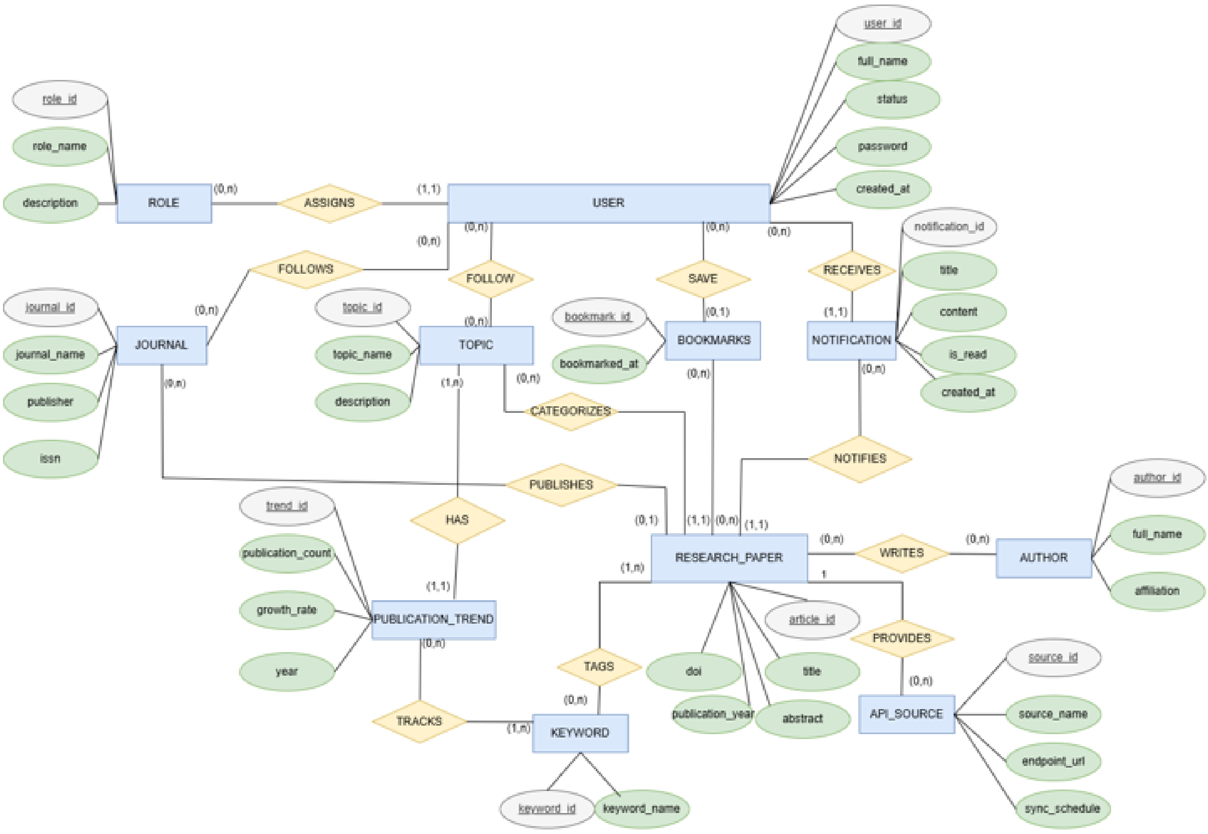
3.2 Screen Authorization

Screen	Researcher	Lecturer/Studen	System Administrator
Login	x	x	x
Register	x	x	
Search Research Papers	x	x	
View Paper Details	x	x	
Save Bookmarks	x	x	
Track Publication Trends	x	x	
View Dashboard	x	x	
Generate Reports	x	x	
View Trending Topics	x	x	
Follow Topics	x	x	
Receive Notifications	x	x	
Manage System			x
Manage Users			x
Configure API Sources			x
Sync Academic Data			x

3.3 Non-Screen Functions

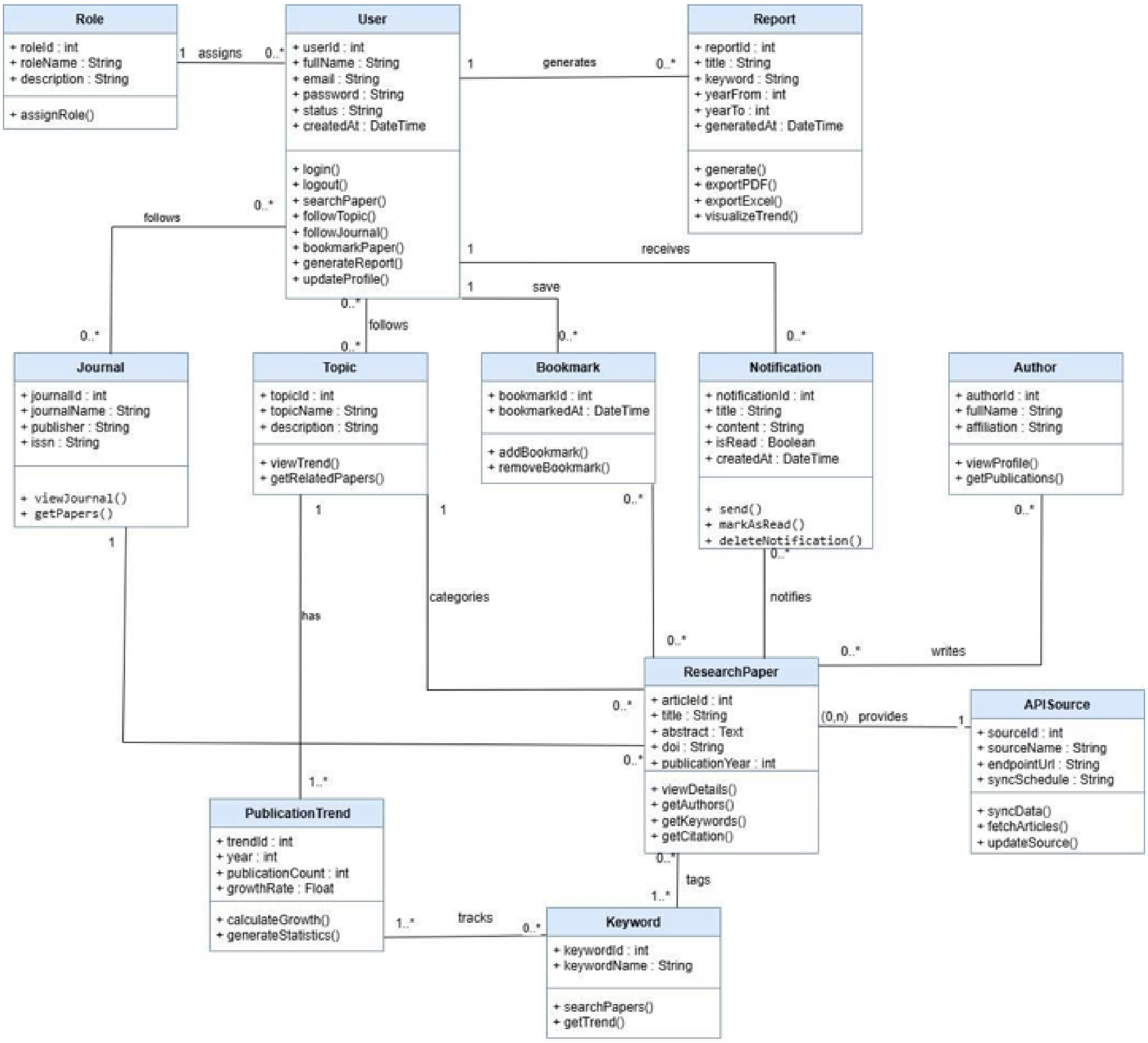
#	Feature	System Function	Description
1	Automated Academic Data Synchronization	Scheduled Job	Scheduled job that automatically retrieves publication metadata directly from external academic APIs (OpenAlex and Crossref). The retrieved data is used for publication search, statistical analysis, and publication trend tracking.
2	Automated Trend Calculation	Scheduled Job	Scheduled job that automatically recalculates publication trend statistics after each data synchronization from external academic APIs. The system analyzes total publications, total citations, Open Access rate, and research topic trends, then updates the data displayed on the dashboard and Trending Topics.

3.4 Entity Relationship Diagram



Hình 5: Entity Relationship Diagram (ERD)

3.5 Class Diagram

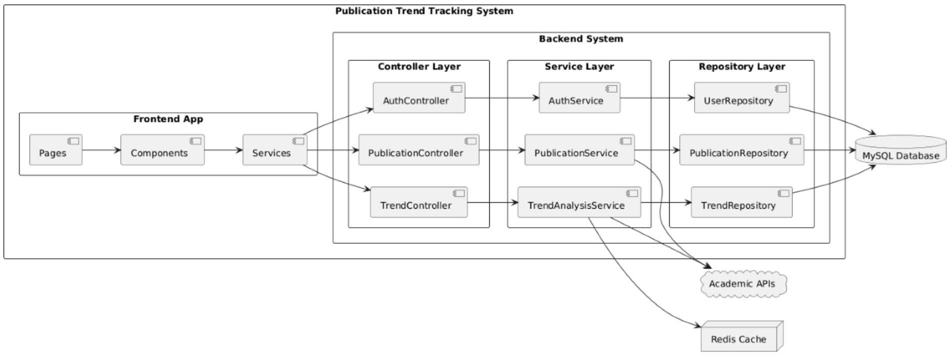


Hình 6: Class Diagram

IV. Software Design Description

1 System Design

1.1 System Architecture



1.1.1 Frontend App

TypeScript (TS): Core technology used for frontend development, enabling the implementation of scalable, maintainable, and type-safe user interfaces.

Pages: Responsible for rendering the main user interfaces, including Login Page, Dashboard Page, Publication Search Page, and Trend Analysis Page.

Components: Reusable user interface elements used to construct application pages, such as search bars, filter panels, tables, charts, and publication cards.

Services: Responsible for handling communication between the frontend and backend through HTTP requests and responses.

1.1.2 Backend System

The backend system follows a layered architecture consisting of Controller Layer, Service Layer, and Repository Layer.

Controller Layer

AuthController: Handles authentication requests such as login, logout, and user authorization.

PublicationController: Processes publication-related requests, including publication searching, filtering, and detailed information retrieval.

TrendController: Handles requests related to publication trend analysis, dashboard visualization, and trend monitoring.

Service Layer

AuthService: Implements authentication logic, including user verification and access control.

PublicationService: Handles publication-related business logic such as processing search queries, filtering publications, and managing publication metadata.

TrendAnalysisService: Responsible for trend analysis, topic identification, dashboard analytics, and report generation.

Repository Layer

UserRepository: Manages data access operations related to user accounts and authentication data.

PublicationRepository: Handles database operations related to publication metadata, authors, and research fields.

TrendRepository: Responsible for storing and retrieving trend-related analytical data.

1.1.3 Database

MySQL Database: Relational database management system used to store system data, including user accounts, publication metadata, trend records, and system-generated reports.

1.1.4 External Services

Academic APIs: External academic data sources used to retrieve scientific publication metadata, including publication titles, authors, research fields, publication years, and citation-related information.

Examples include:

- OpenAlex
- Crossref
- Google Scholar

These APIs support publication synchronization and provide up-to-date research data for the system.

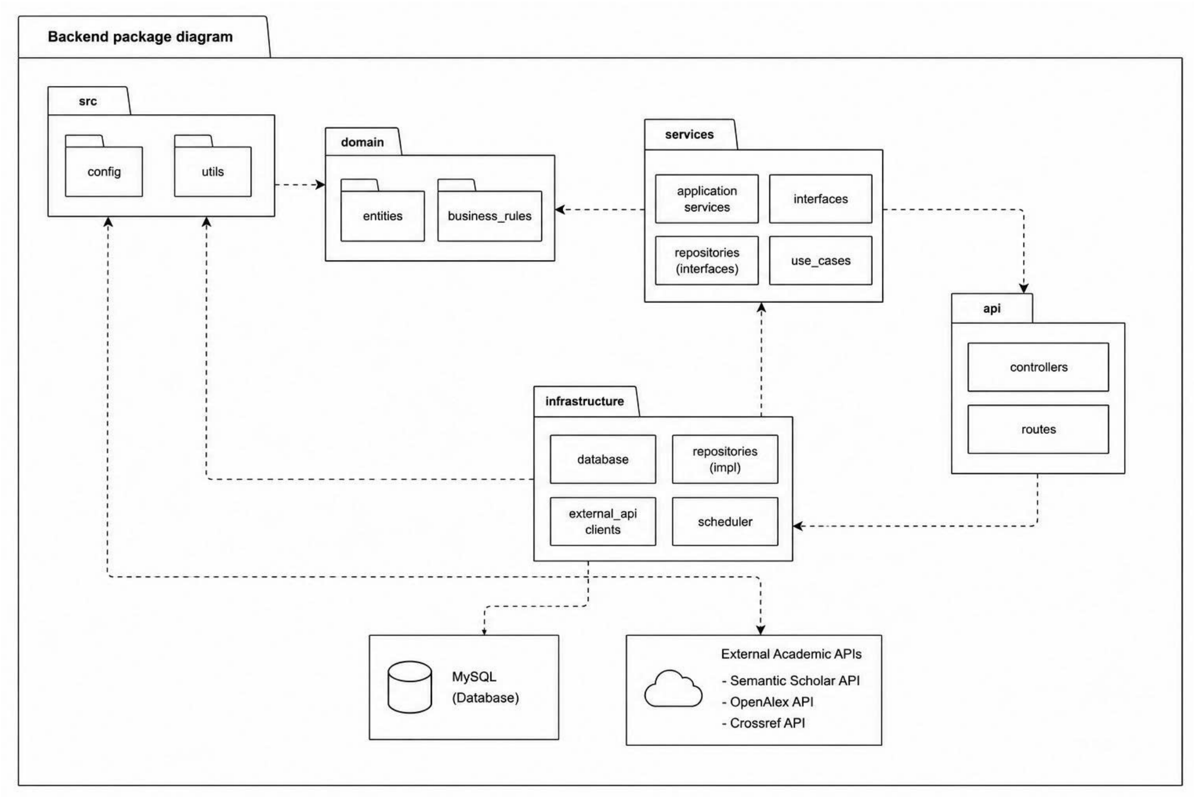
1.1.5 Cache Management

Redis Cache: In-memory caching system used to store frequently accessed data such as dashboard statistics, publication trends, and popular search results to improve system performance and reduce database workload.

1.2 Package Diagram

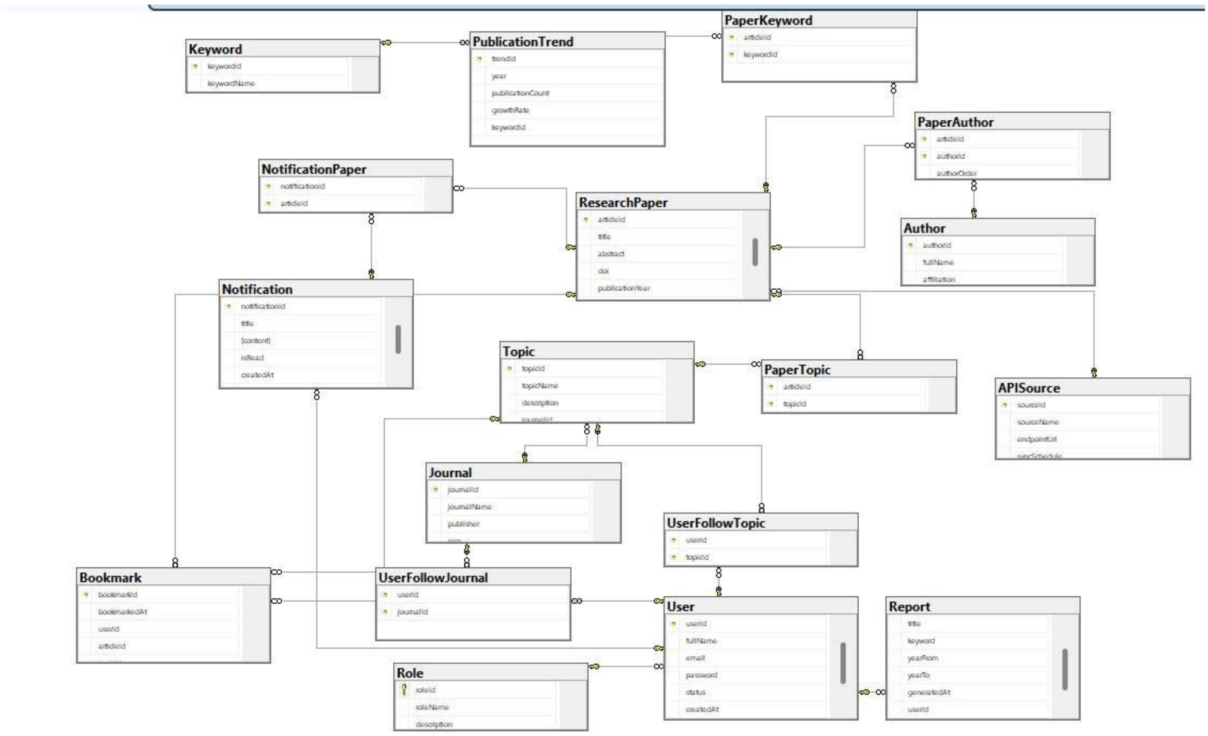
1.2.1 Front End

1.2.2 Back End



Hình 7: **Backend package diagram**

2 Database Design



Hình 8: Database Design

Entities Description

2.1 Role Table

Field	Type	Description	PK	FK
roleId	INT	Role ID	Yes	No

roleName	NVARCHAR(100)	Role name	No	No
description	NVARCHAR(255)	Description	No	No

2.2 User Table

Field	Type	Description	PK	FK
userId	INT	User ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
email	NVARCHAR(150)	Email	No	No
password	NVARCHAR(255)	Password	No	No
status	NVARCHAR(50)	Status	No	No
createdAt	DATETIME2	Created time	No	No
roleId	INT	Role reference	No	Yes → Role

2.3 Journal Table

Field	Type	Description	PK	FK
journalId	INT	Journal ID	Yes	No
journalName	NVARCHAR(200)	Journal name	No	No
publisher	NVARCHAR(150)	Publisher	No	No
issn	NVARCHAR(20)	ISSN	No	No

2.4 UserFollowJournal (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User
journalId	INT	Journal	Yes (Composite)	Yes → Journal

2.5 Topic Table

Field	Type	Description	PK	FK
topicId	INT	Topic ID	Yes	No
topicName	NVARCHAR(150)	Topic name	No	No
description	NVARCHAR(MAX)	Description	No	No
journalId	INT	Journal reference	No	Yes → Journal

2.6 UserFollowTopic (M:N)

Field	Type	Description	PK	FK
userId	INT	User	Yes (Composite)	Yes → User
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.7 APISource Table

Field	Type	Description	PK	FK
sourceId	INT	Source ID	Yes	No
sourceName	NVARCHAR(150)	Source name	No	No
endpointUrl	NVARCHAR(500)	API URL	No	No
syncSchedule	NVARCHAR(100)	Sync schedule	No	No

2.8 Author Table

Field	Type	Description	PK	FK
authorId	INT	Author ID	Yes	No
fullName	NVARCHAR(150)	Full name	No	No
affiliation	NVARCHAR(255)	Affiliation	No	No

2.9 Keyword Table

Field	Type	Description	PK	FK
keywordId	INT	Keyword ID	Yes	No
keywordName	NVARCHAR(100)	Keyword	No	No

2.10 Research Paper Table

Field	Type	Description	PK	FK
articleId	INT	Paper ID	Yes	No
title	NVARCHAR(500)	Title	No	No
abstract	NVARCHAR(MAX)	Abstract	No	No
doi	NVARCHAR(200)	DOI	No	No
publicationYear	INT	Year	No	No
sourceId	INT	API Source	No	Yes → APISource

2.11 PaperTopic (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper
topicId	INT	Topic	Yes (Composite)	Yes → Topic

2.12 PaperKeyword (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper

keywordId	INT	Keyword	Yes (Composite)	Yes → Keyword

2.13 PaperAuthor (M:N)

Field	Type	Description	PK	FK
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper
authorId	INT	Author	Yes (Composite)	Yes → Author
authorOrder	INT	Author order	No	No

2.14 PublicationTrend Table

Field	Type	Description	PK	FK
trendId	INT	Trend ID	Yes	No
year	INT	Year	No	No
publicationCount	INT	Count	No	No
growthRate	FLOAT	Growth rate	No	No
keywordId	INT	Keyword	No	Yes → Keyword

2.15 Report Table

Field	Type	Description	PK	FK
reportId	INT	Report ID	Yes	No
title	NVARCHAR(300)	Title	No	No
keyword	NVARCHAR(200)	Keyword	No	No
yearFrom	INT	From year	No	No
yearTo	INT	To year	No	No
generatedAt	DATETIME2	Created time	No	No

userId	INT	User	No	Yes → User

2.16 Notification Table

Field	Type	Description	PK	FK
notificationId	INT	Notification ID	Yes	No
title	NVARCHAR(300)	Title	No	No
content	NVARCHAR(MAX)	Content	No	No
isRead	BIT	Read status	No	No
createdAt	DATETIME2	Created time	No	No
userId	INT	User	No	Yes → User

2.17 NotificationPaper (M:N)

Field	Type	Description	PK	FK
notificationId	INT	Notification	Yes (Composite)	Yes → Notification
articleId	INT	Paper	Yes (Composite)	Yes → ResearchPaper

2.18 Bookmark Table

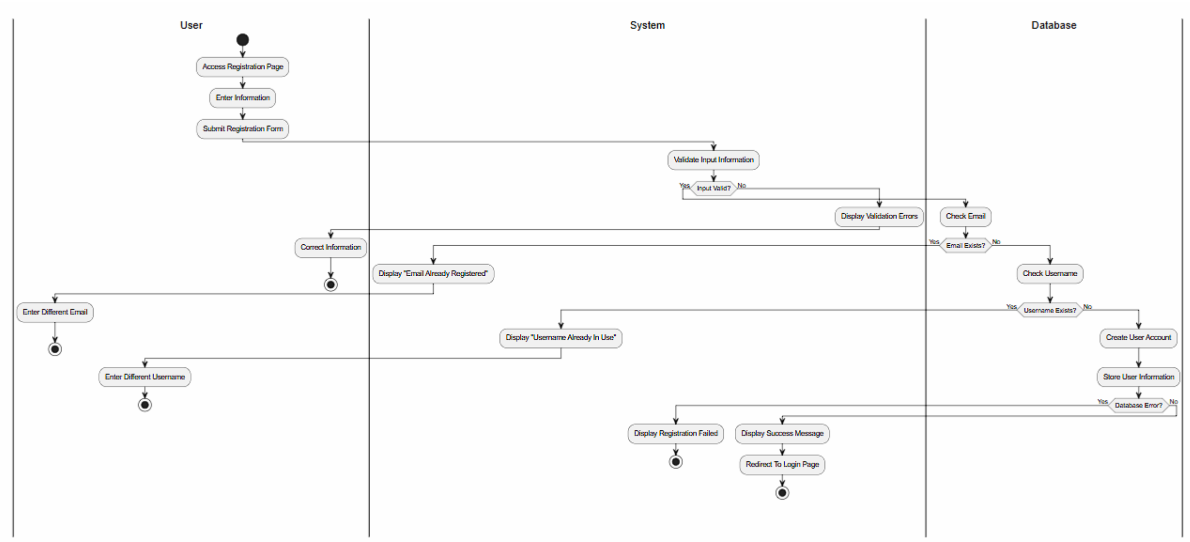
Field	Type	Description	PK	FK
bookmarkId	INT	Bookmark ID	Yes	No
bookmarkedAt	DATETIME2	Time saved	No	No
userId	INT	User	No	Yes → User

articleId	INT	Paper	No	Yes → Re-search-Paper
topicId	INT	Topic	No	Yes → Topic

3 Detail Design

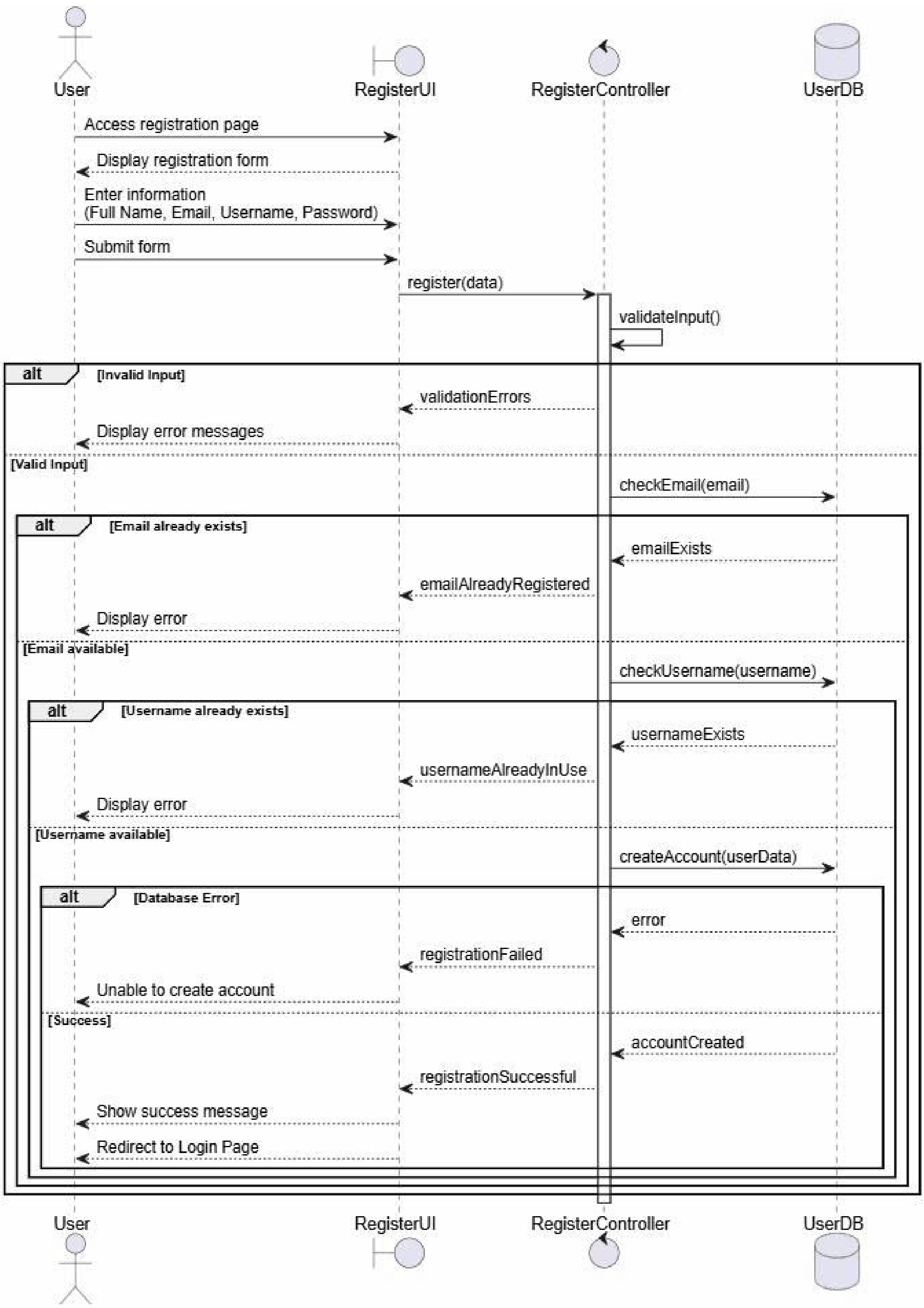
3.1 Register

3.1.1 Activity diagram



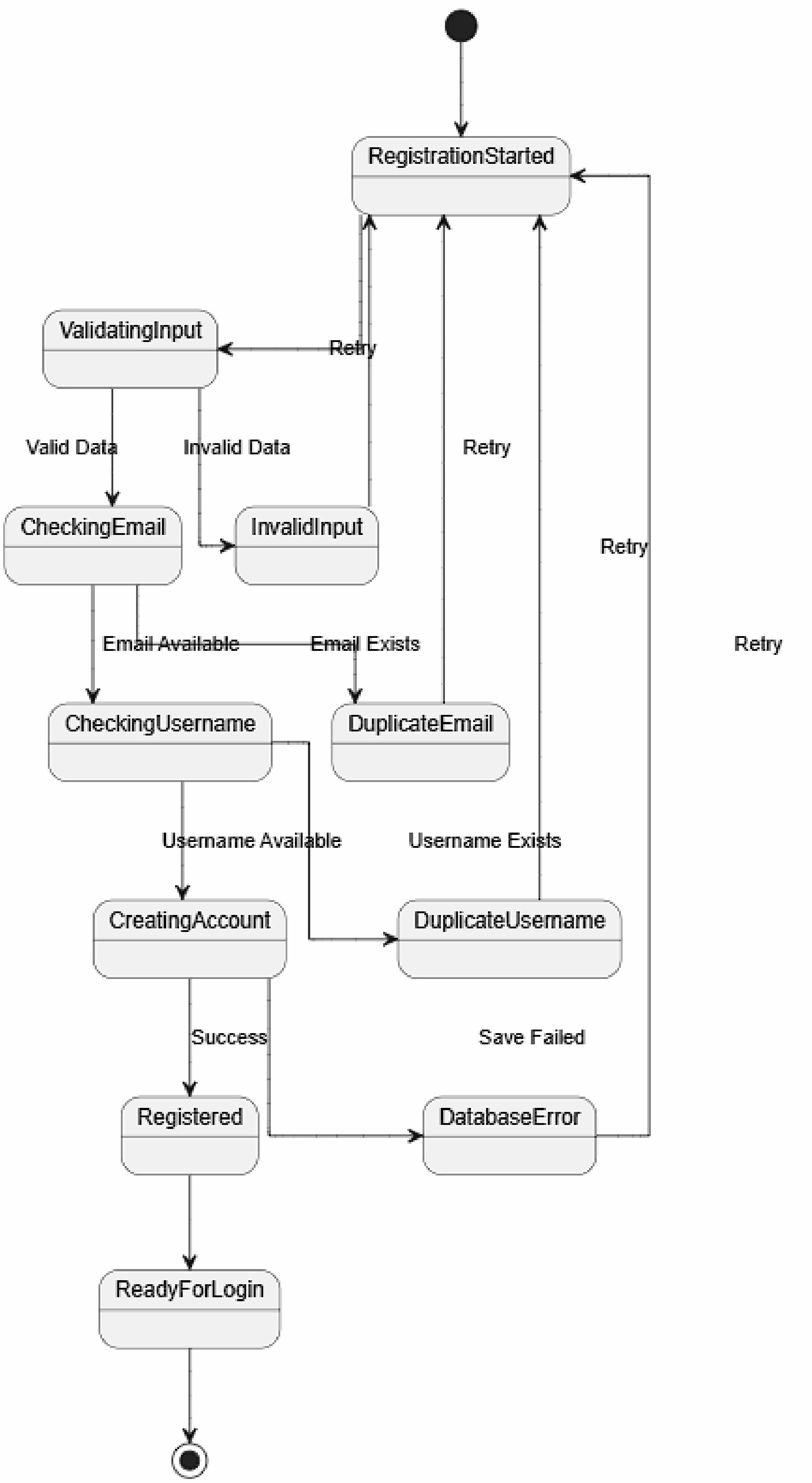
Hình 9: Activity diagram

3.1.2 Sequence Diagram



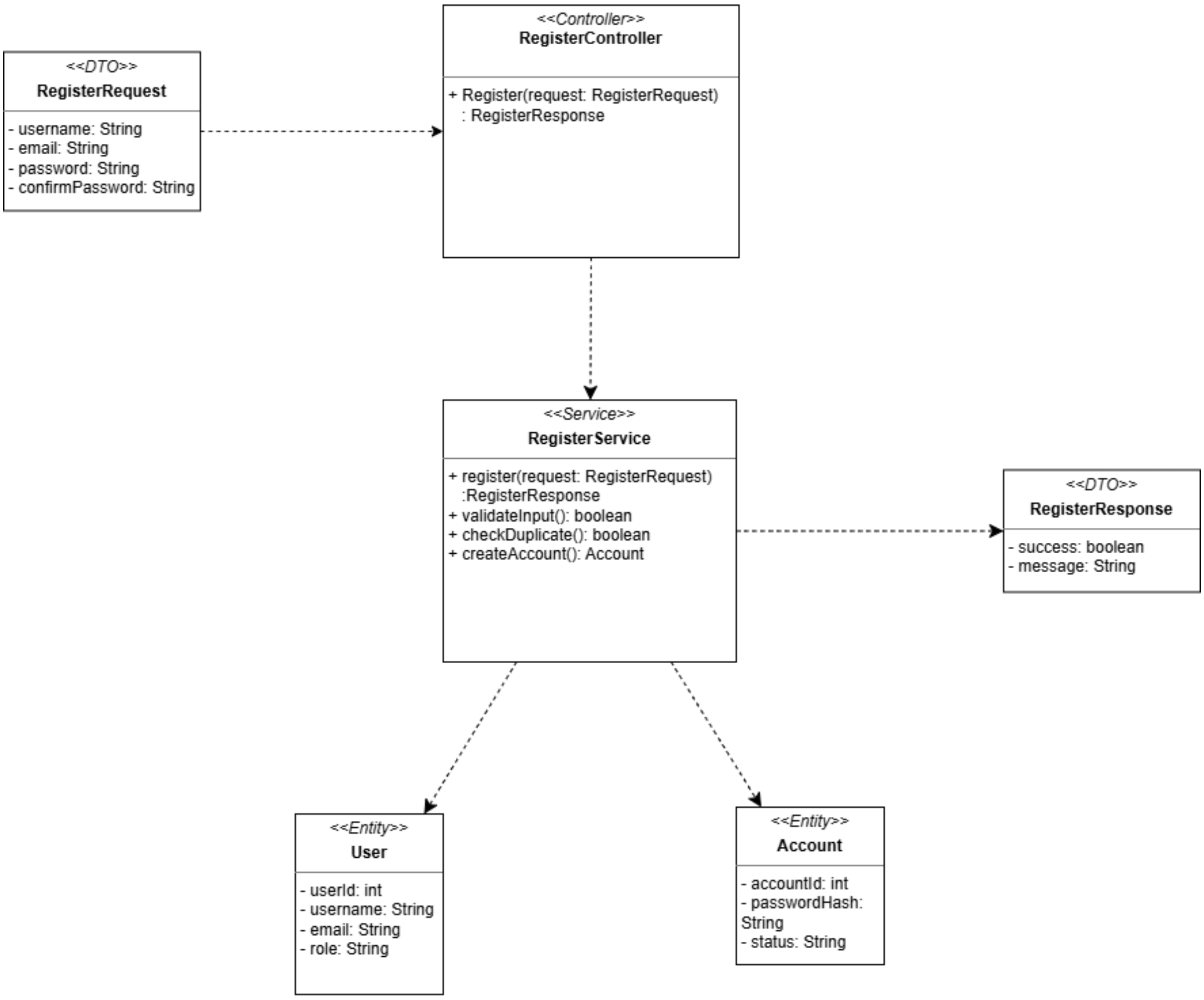
Hình 10: Sequence Diagram

3.1.3 State Diagram



Hình 11: State diagram

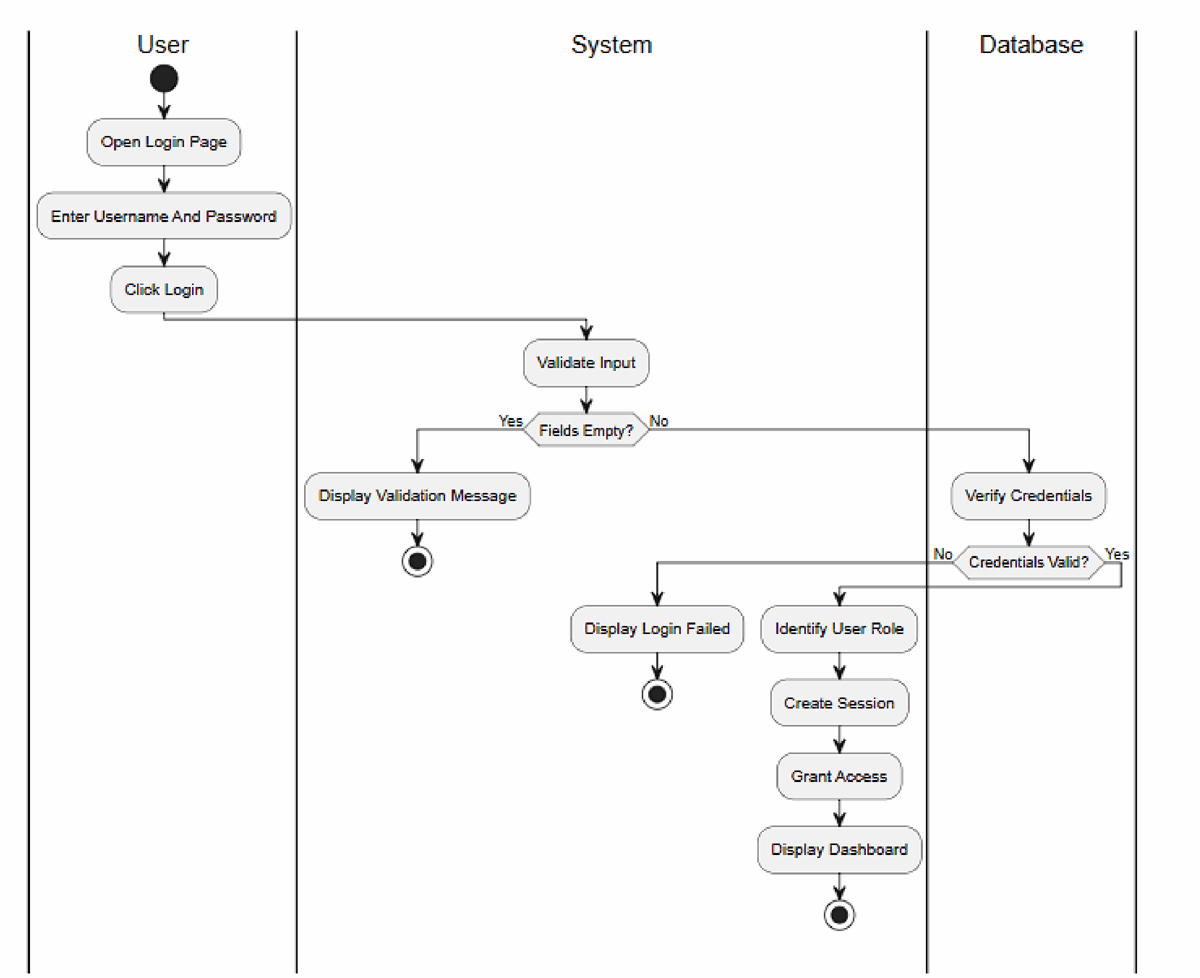
3.1.4 Class Diagram



Hình 12: Class Diagram: Register

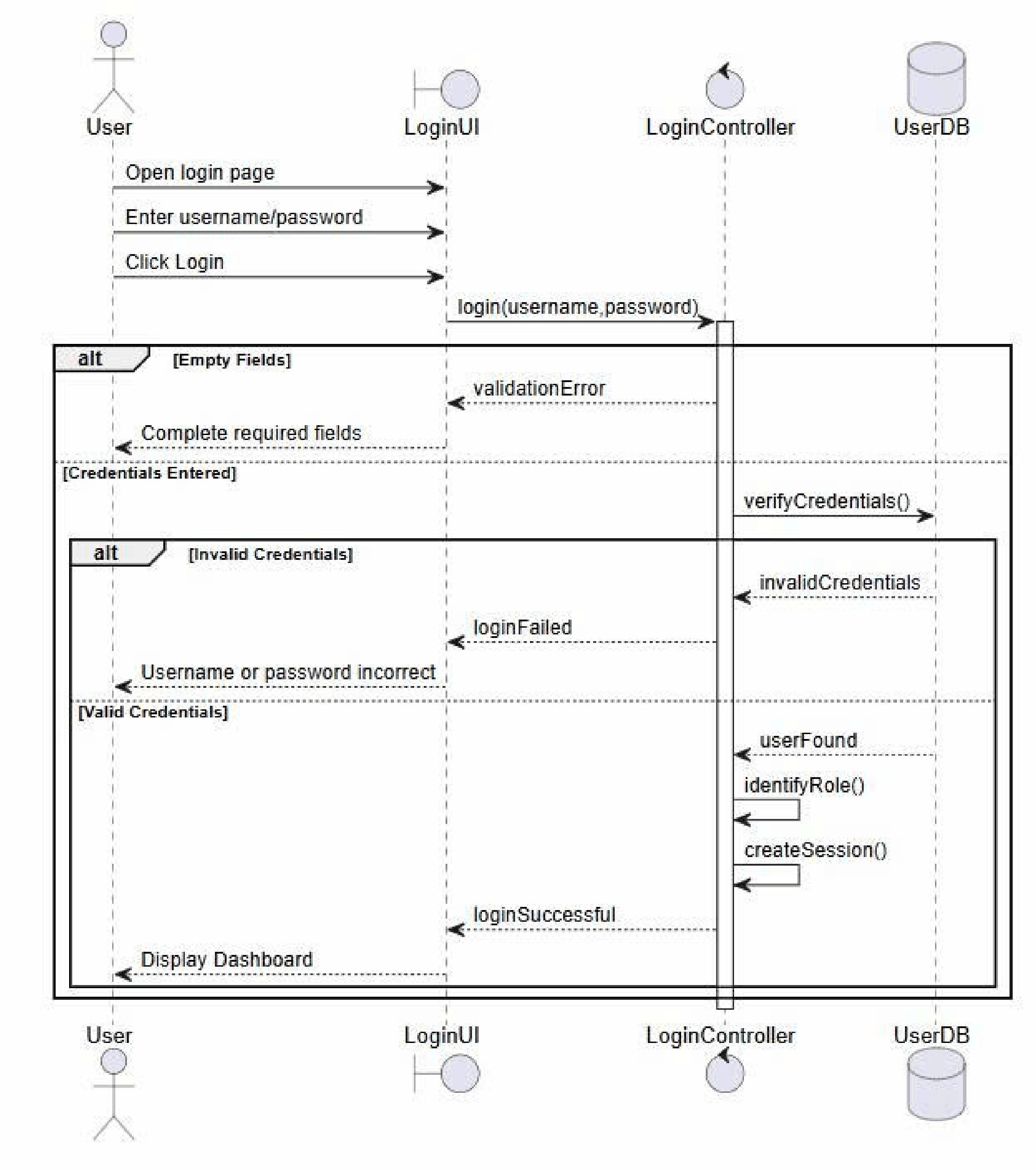
3.2 Login

3.2.1 Activity Diagram



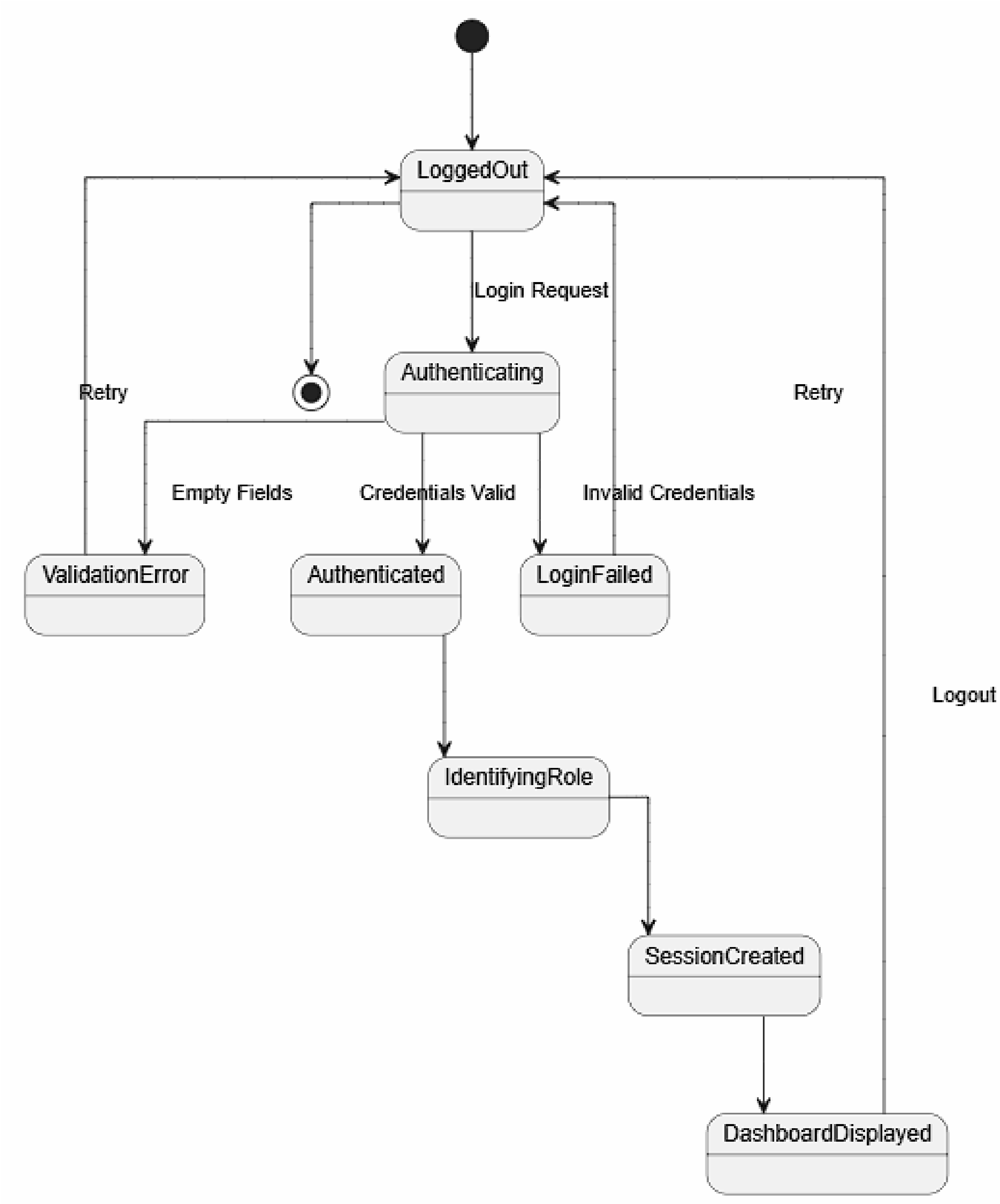
Hình 13: Activity diagram

3.2.2 Sequence Diagram



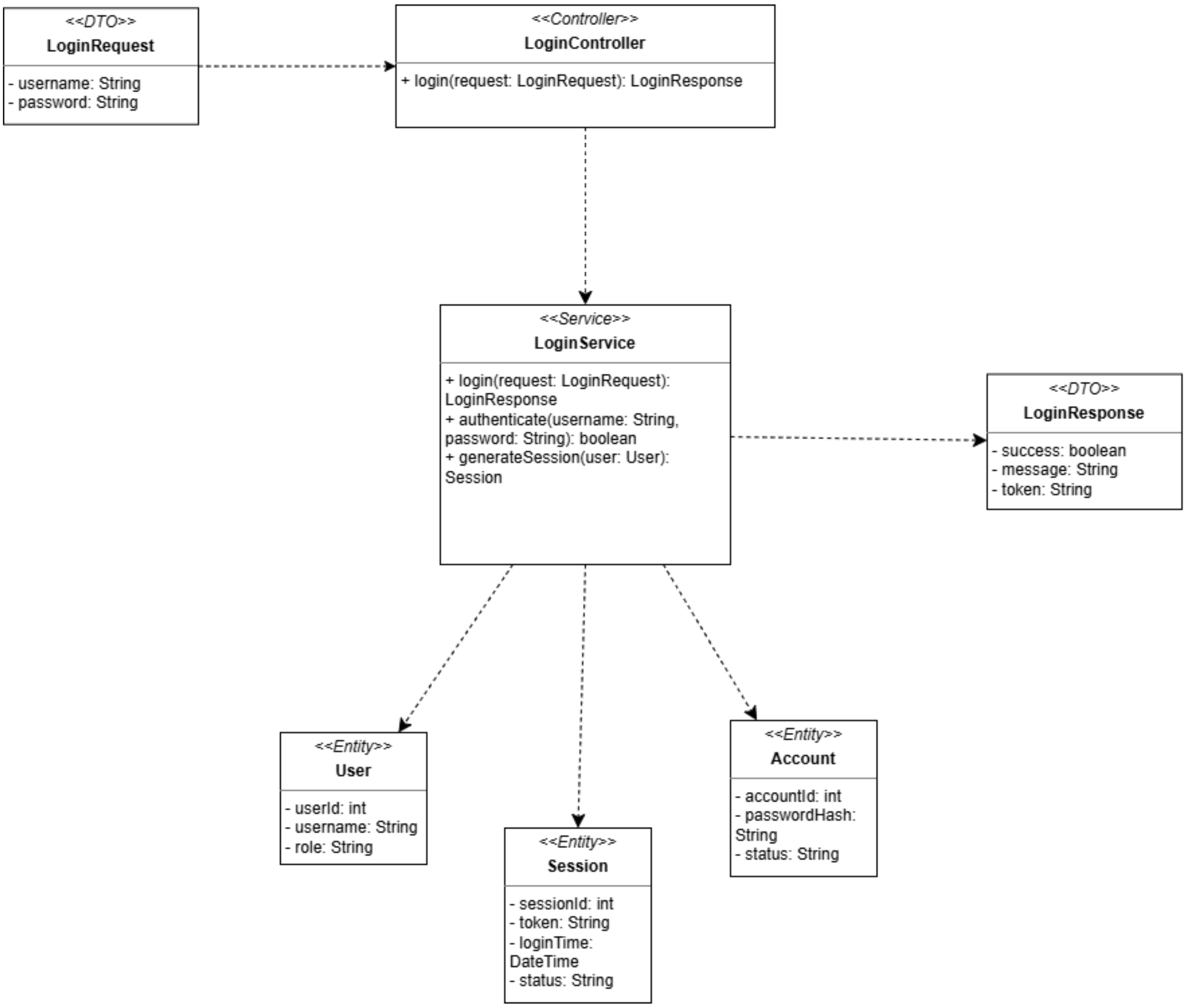
Hình 14: Sequence Diagram

3.2.3 State Diagram



Hình 15: State diagram

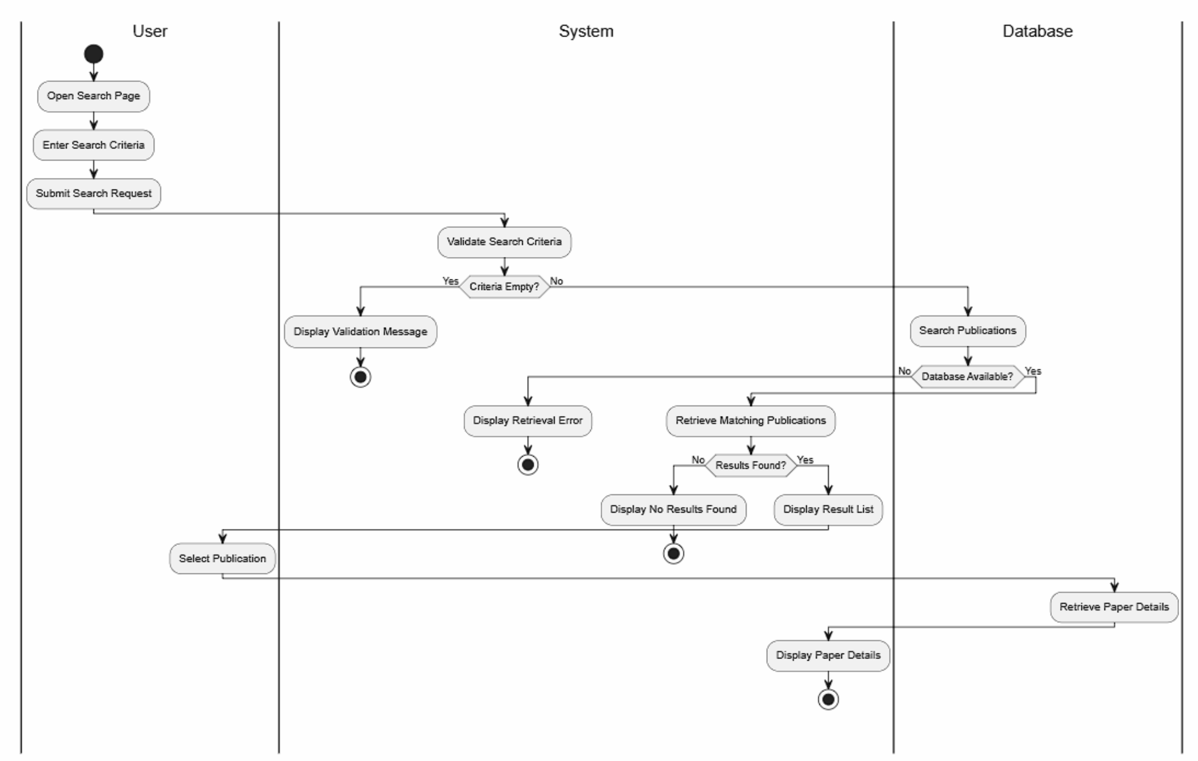
3.2.4 Class Diagram



Hình 16: Class Diagram: Login

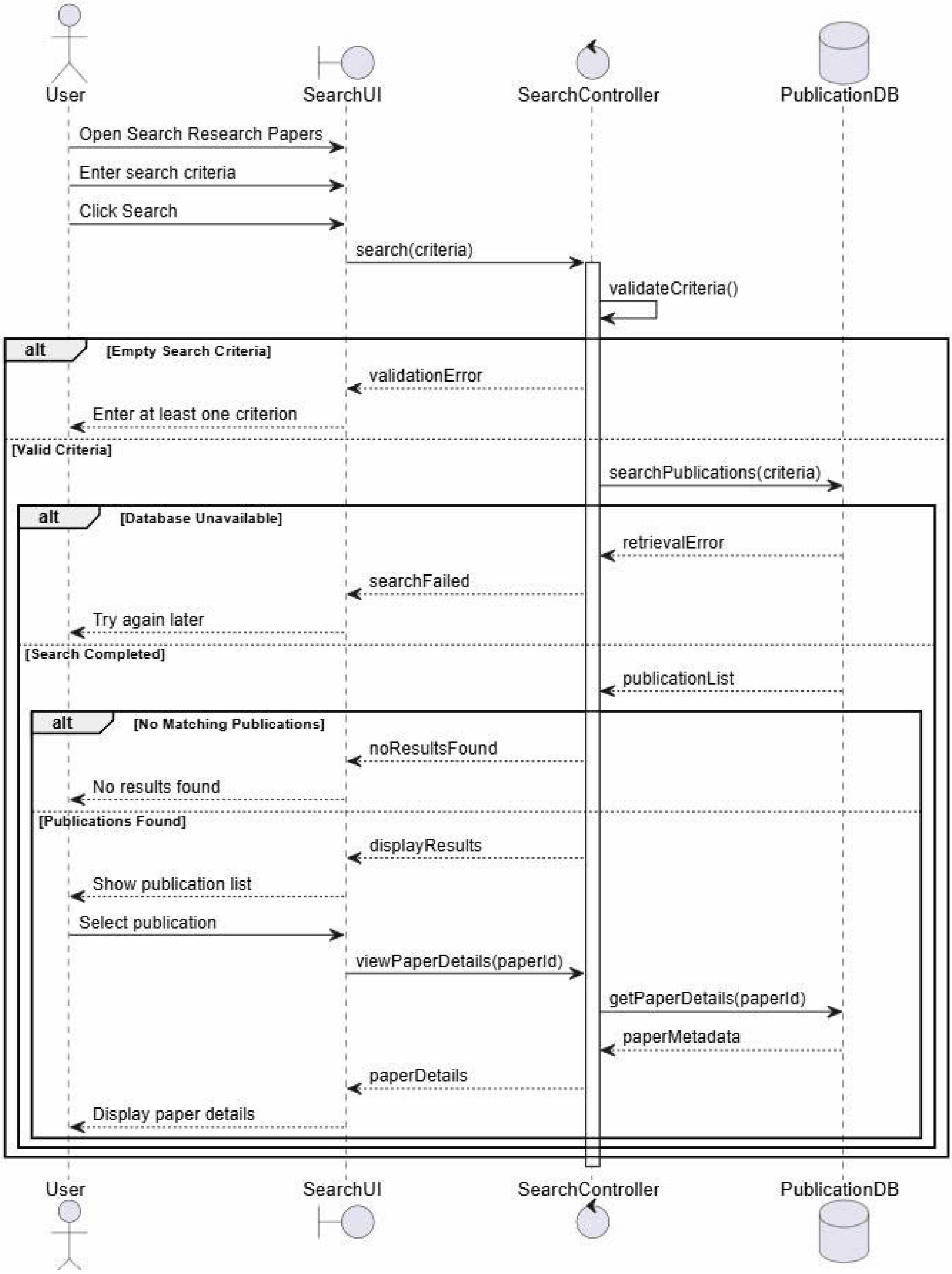
3.3 Search Research Papers

3.3.1 Activity Diagram



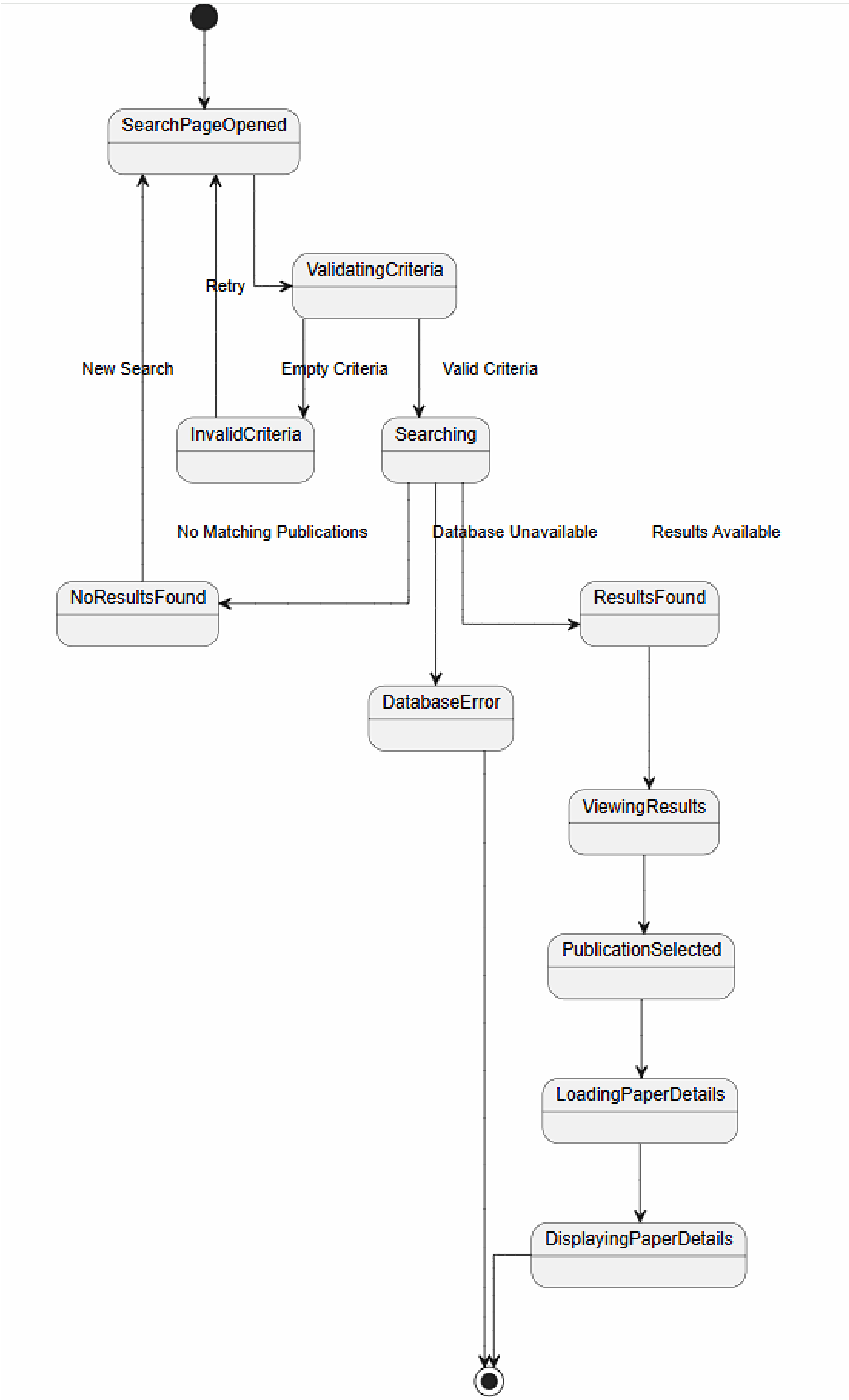
Hình 17: Activity diagram

3.3.2 Sequence Diagram



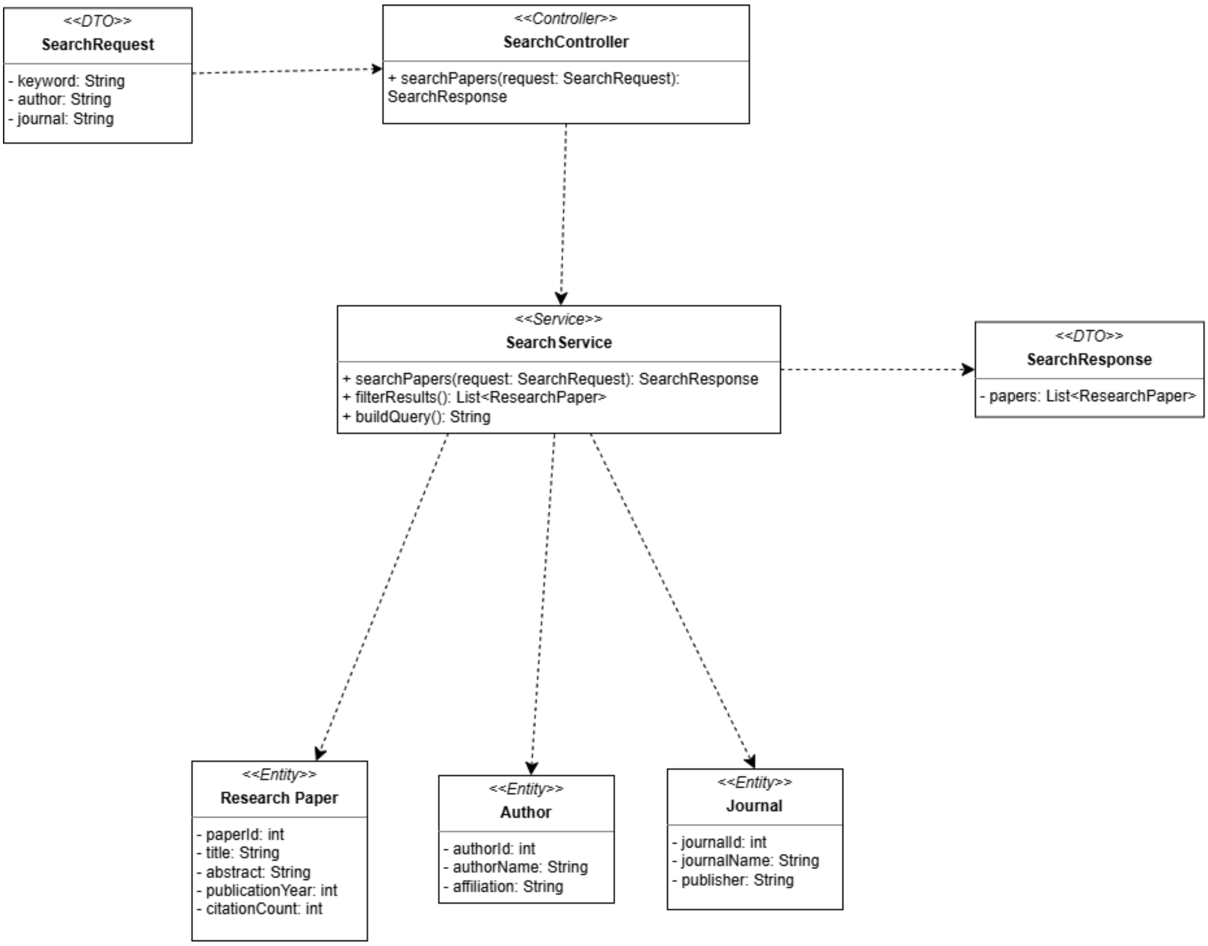
Hình 18: Sequence Diagram

3.3.3 State Diagram



Hình 19: State diagram

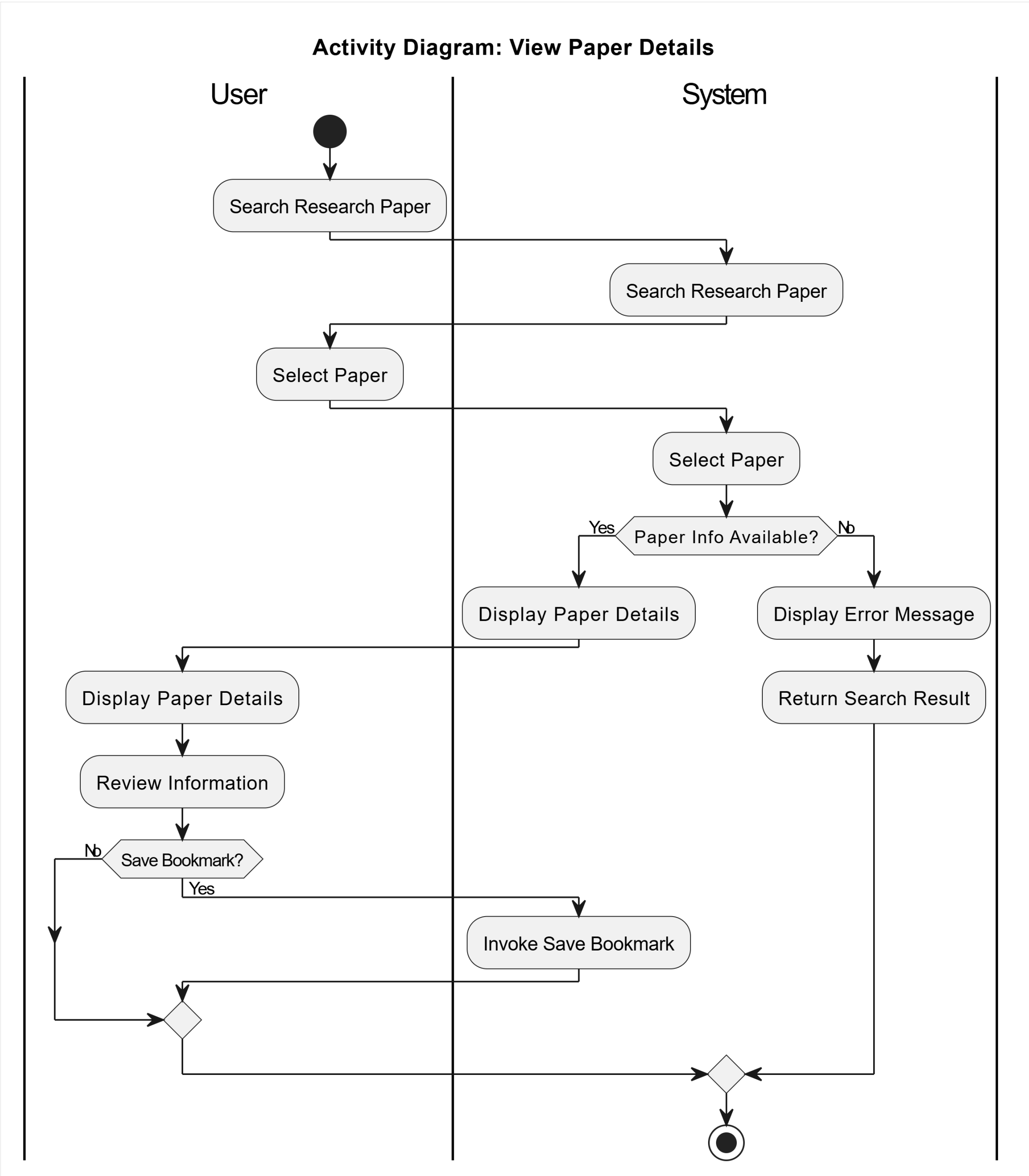
3.3.4 Class Diagram



Hình 20: Class Diagram: Search Research Papers

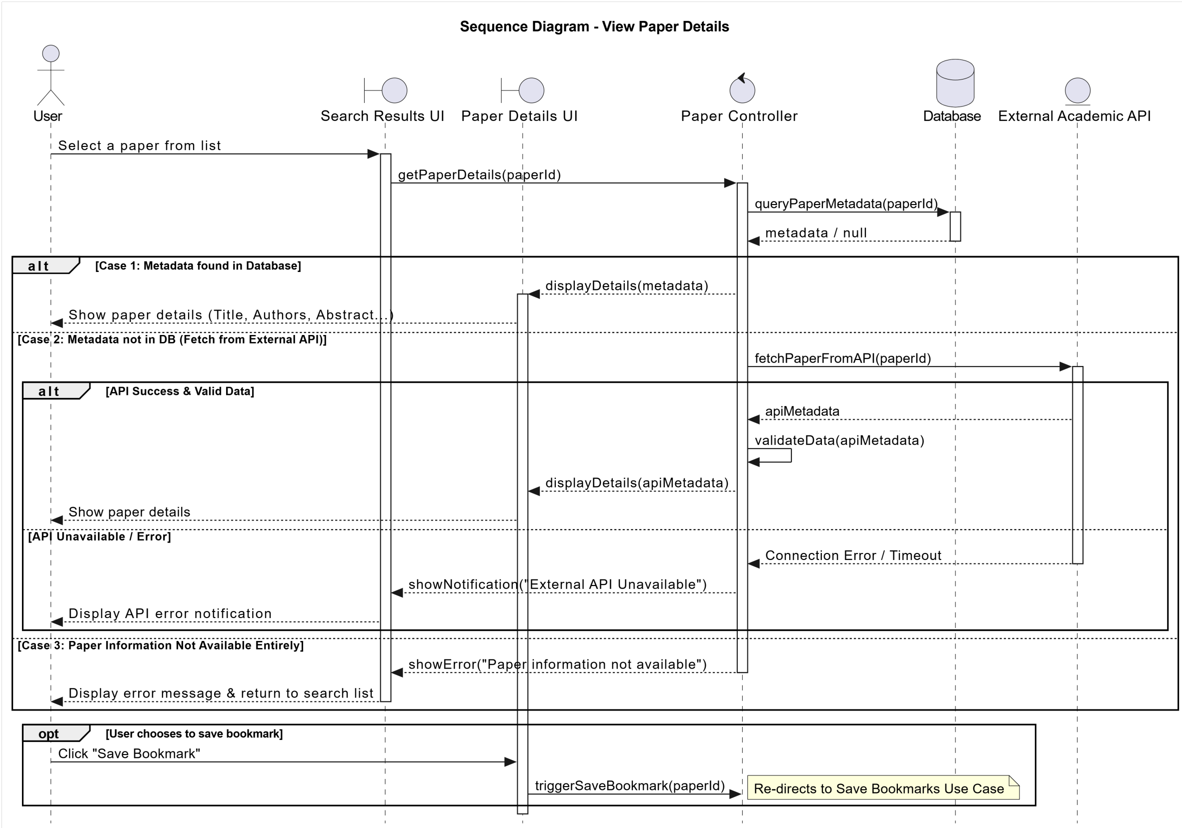
3.4 View Paper Details

3.4.1 Activity Diagram



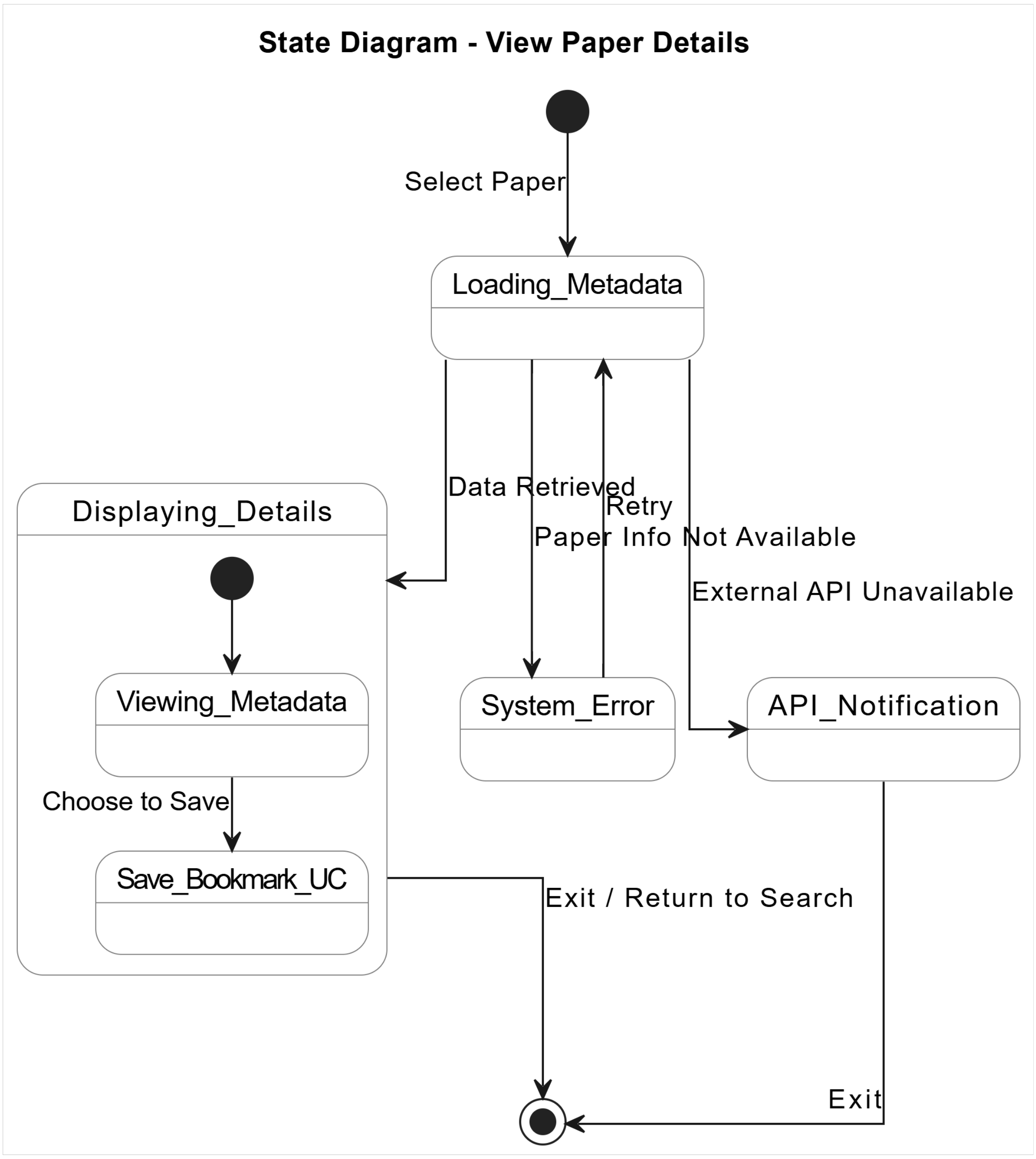
Hình 21: Activity Diagram: View Paper Details

3.4.2 Sequence Diagram



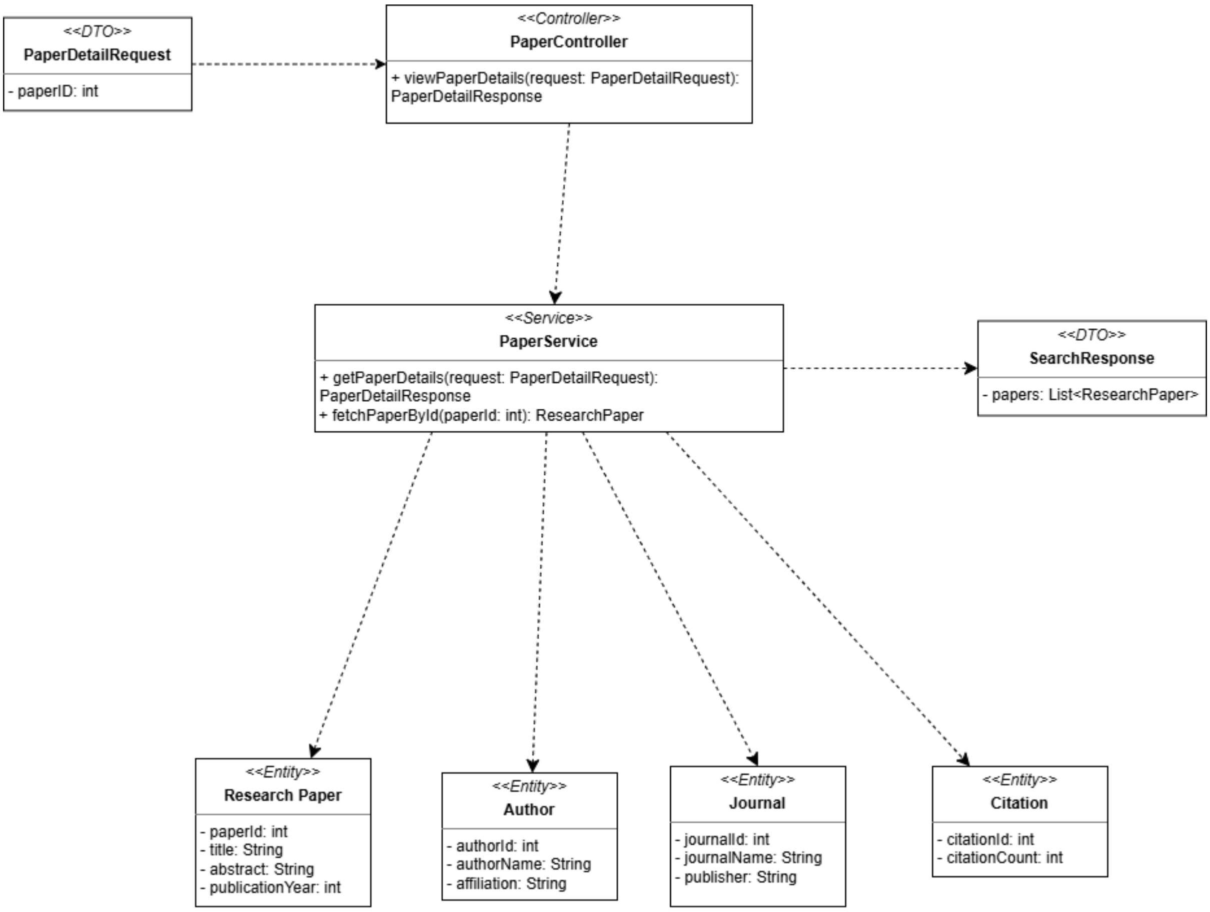
Hình 22: Sequence Diagram: View Paper Details

3.4.3 State Diagram



Hình 23: State Diagram: View Paper Details

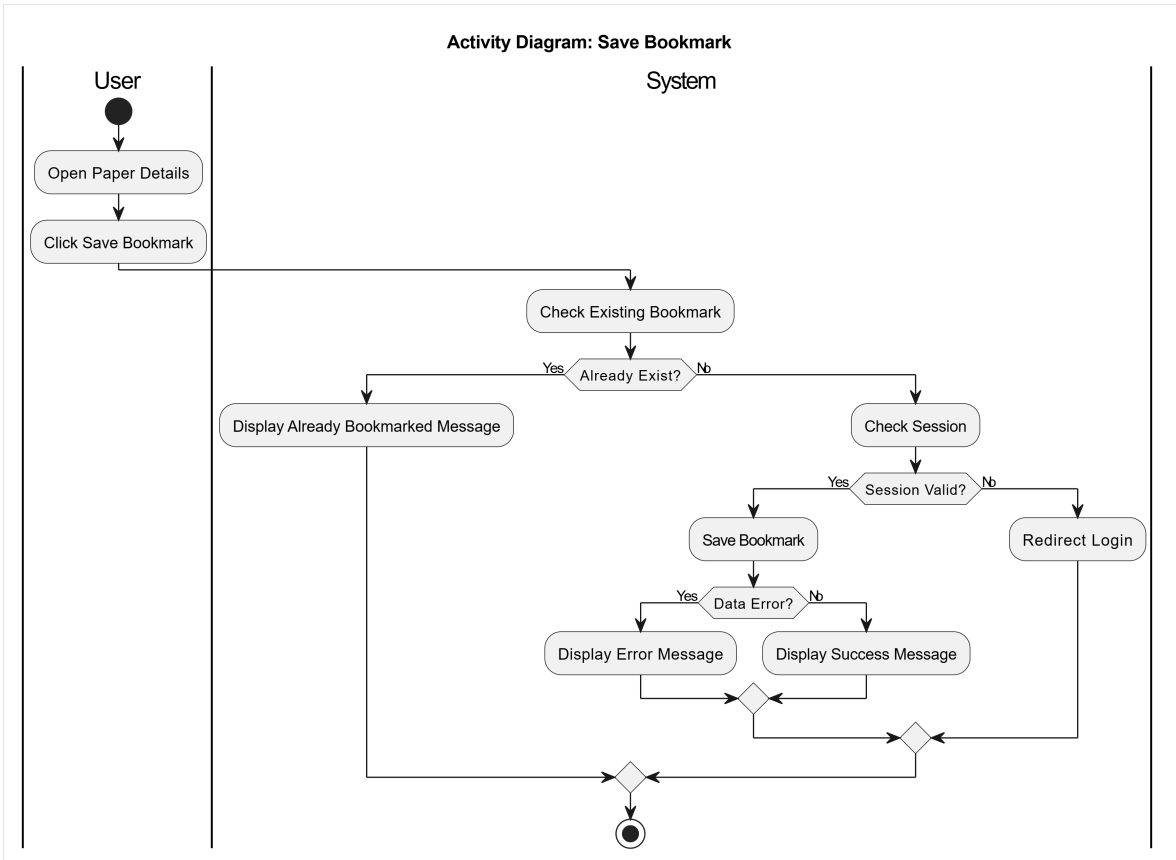
3.4.4 Class Diagram



Hình 24: Class Diagram: View Paper Detail

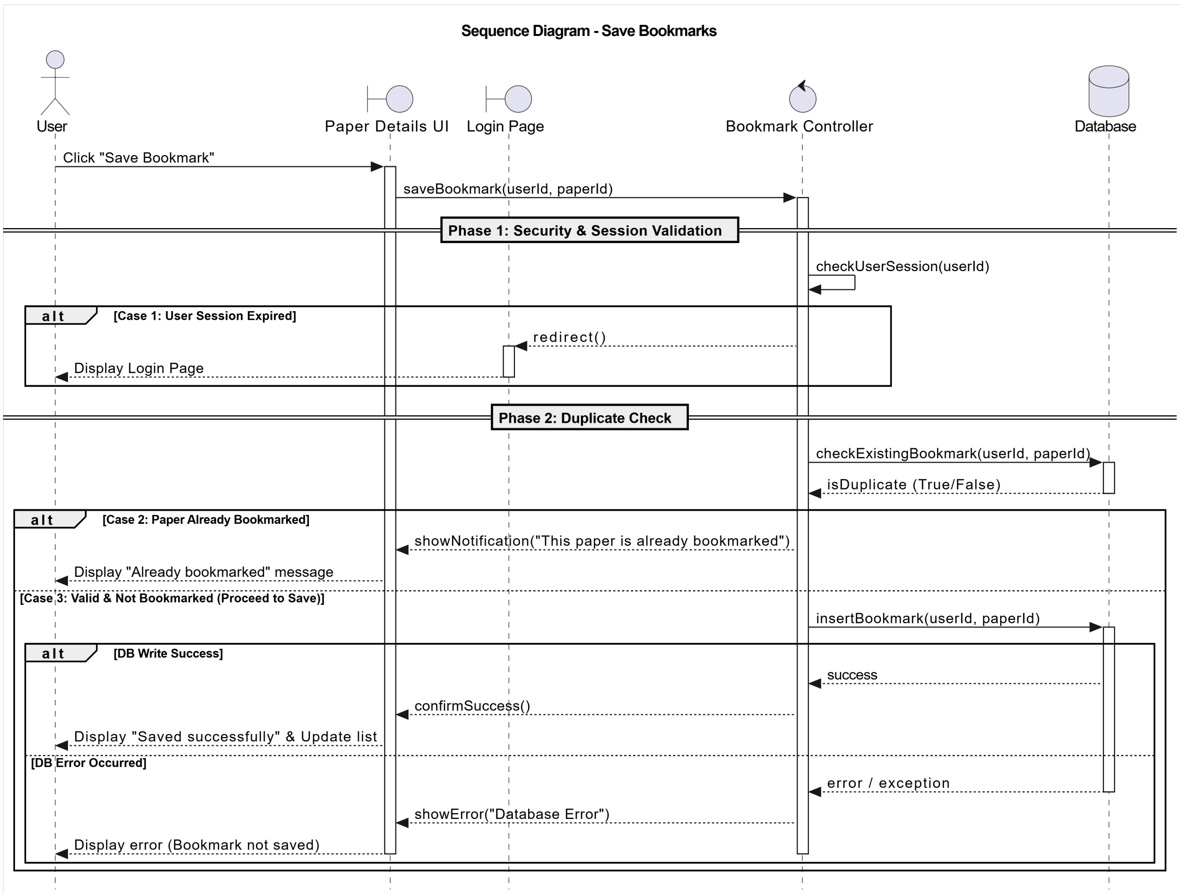
3.5 Save Bookmarks

3.5.1 Activity Diagram



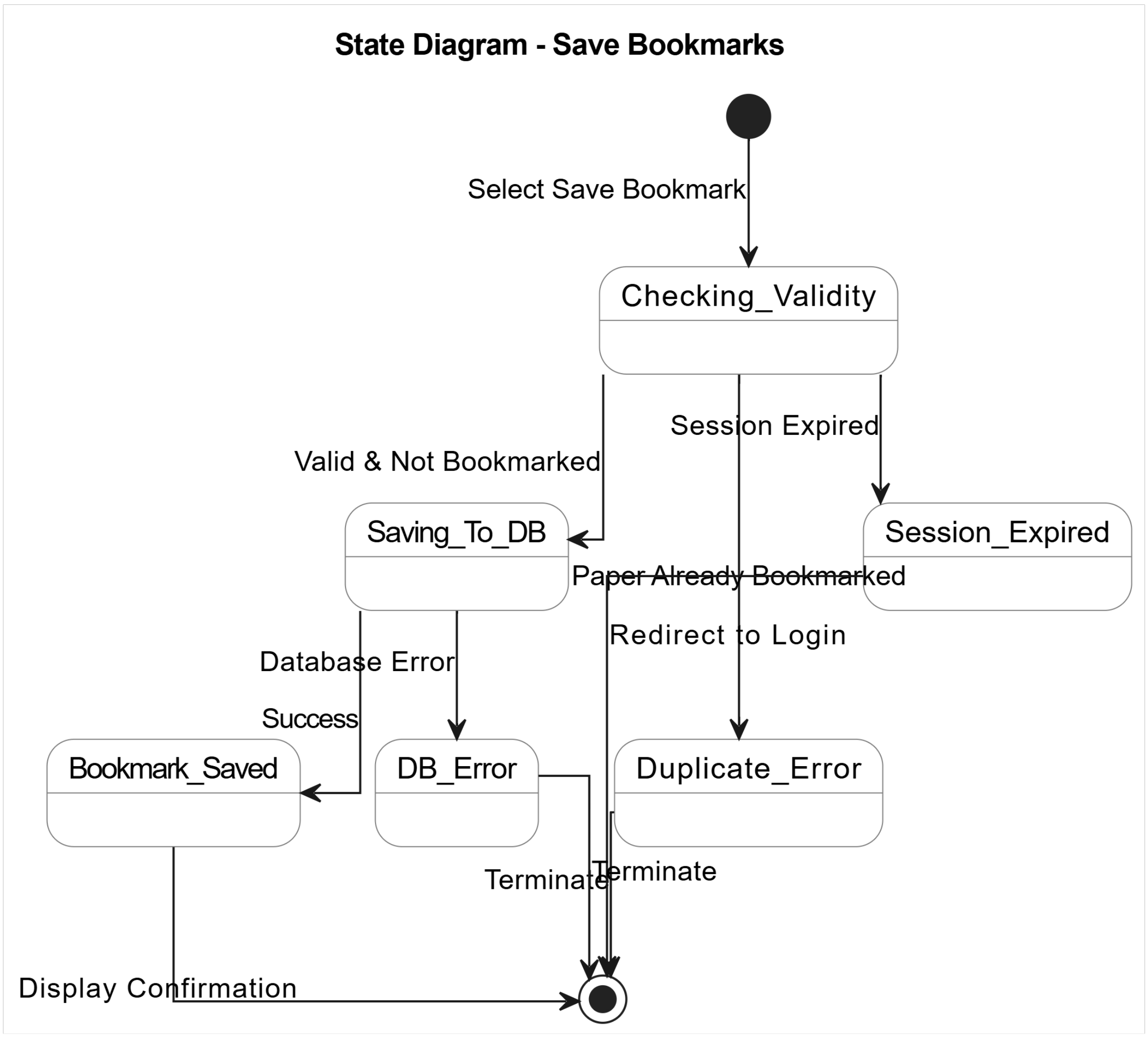
Hình 25: Activity Diagram: Save Bookmarks

3.5.2 Sequence Diagram



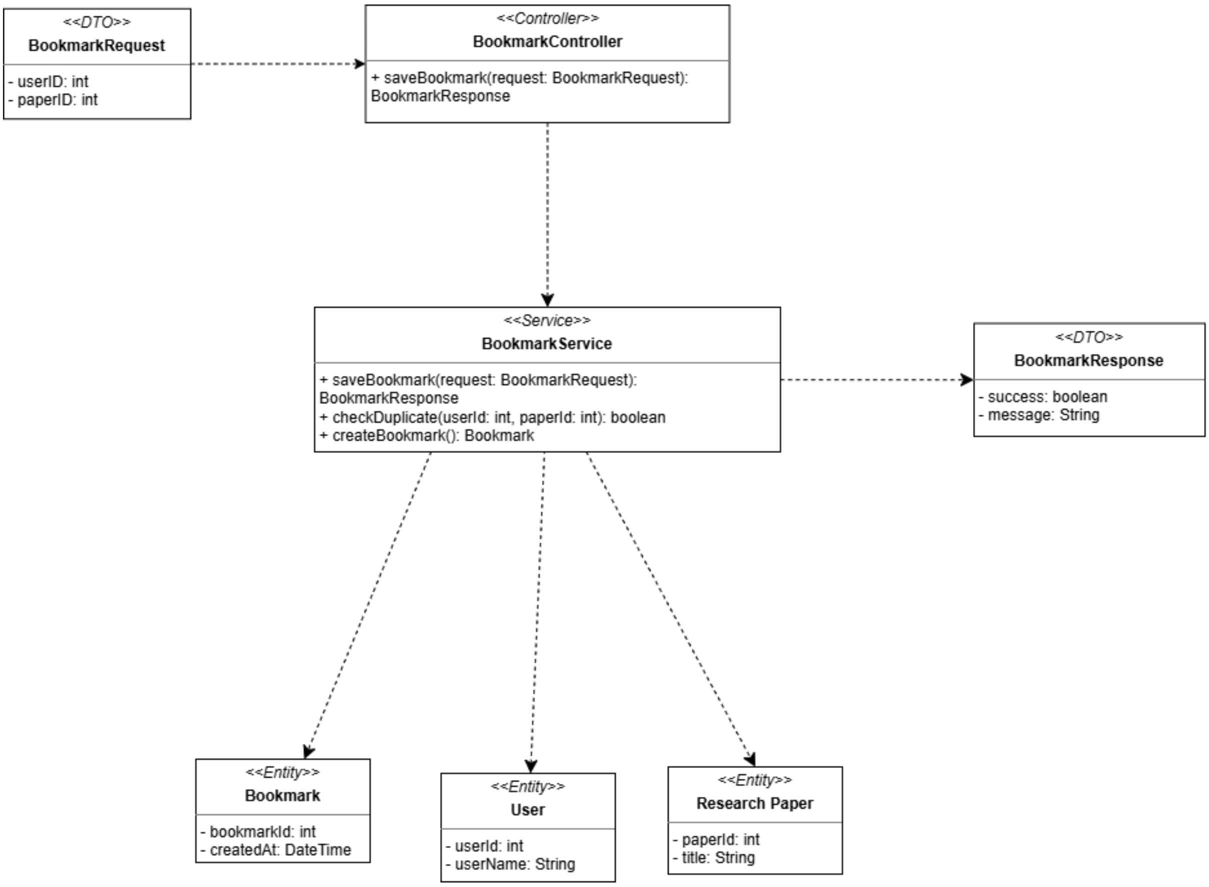
Hình 26: Sequence Diagram: Save Bookmarks

3.5.3 State Diagram



Hình 27: State Diagram: Save Bookmarks

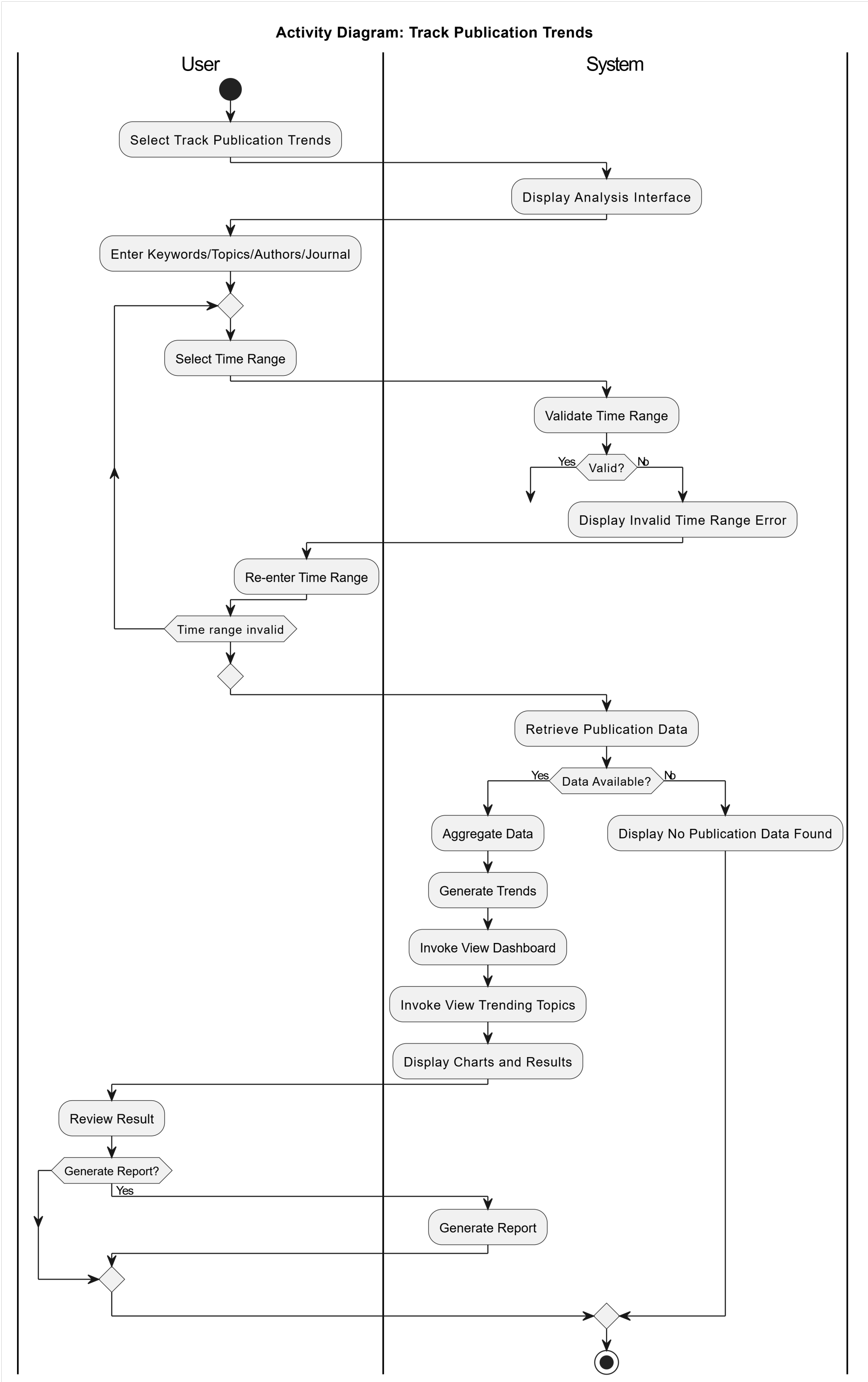
3.5.4 Class Diagram



Hình 28: Class Diagram: Save Bookmarks

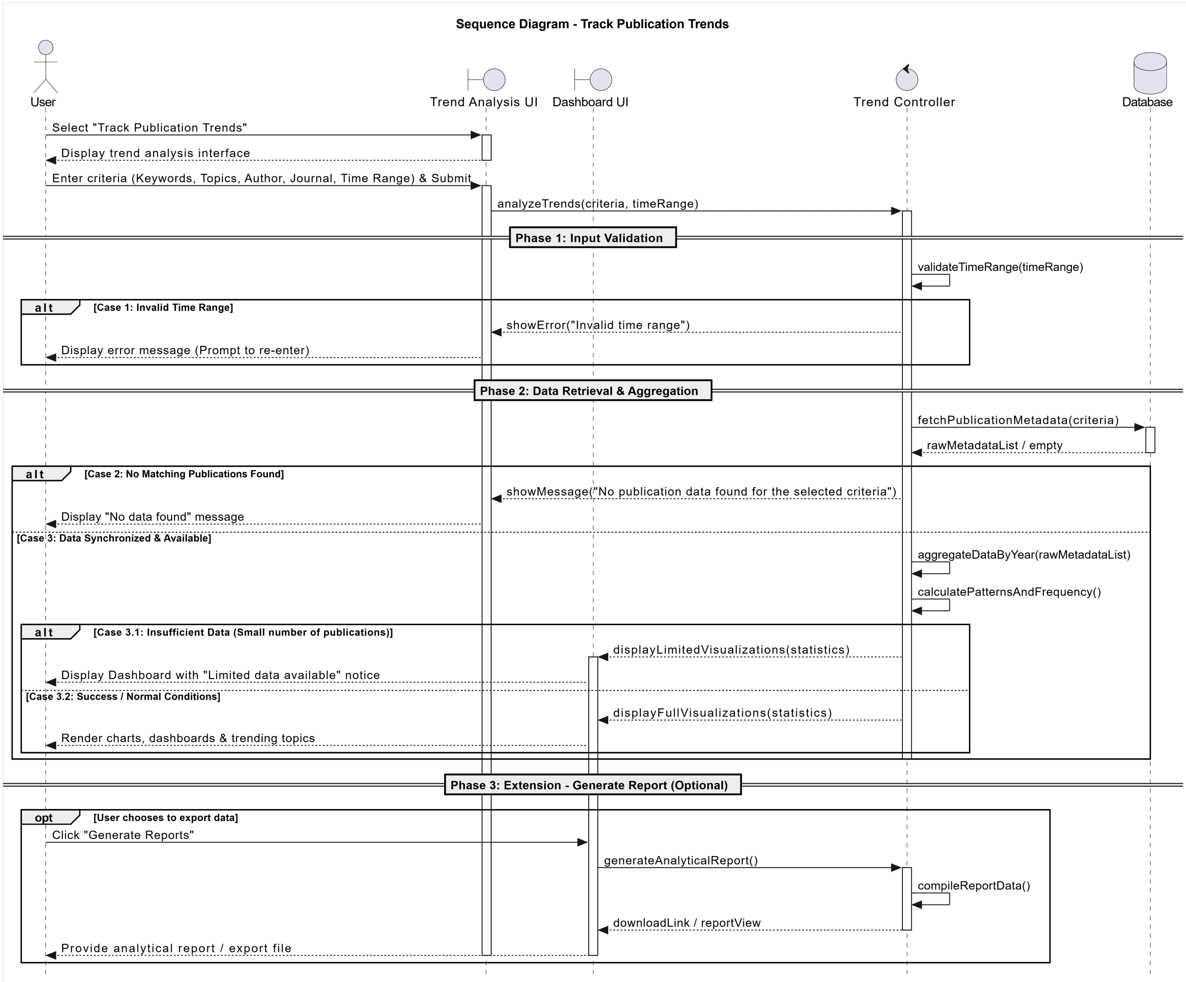
3.6 Track Publication Trends

3.6.1 Activity Diagram



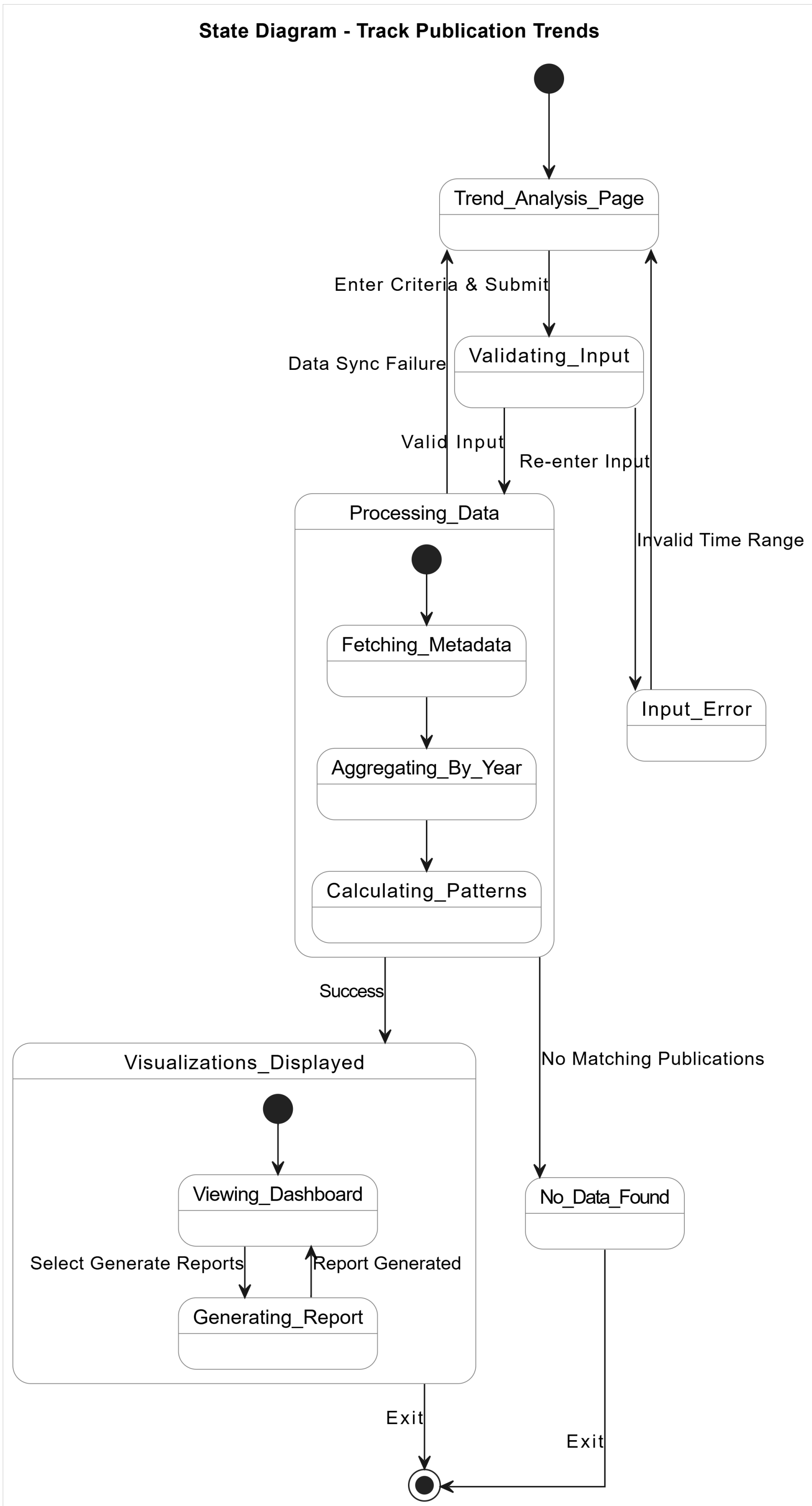
Hình 29: Activity Diagram: Track Publication Trends

3.6.2 Sequence Diagram



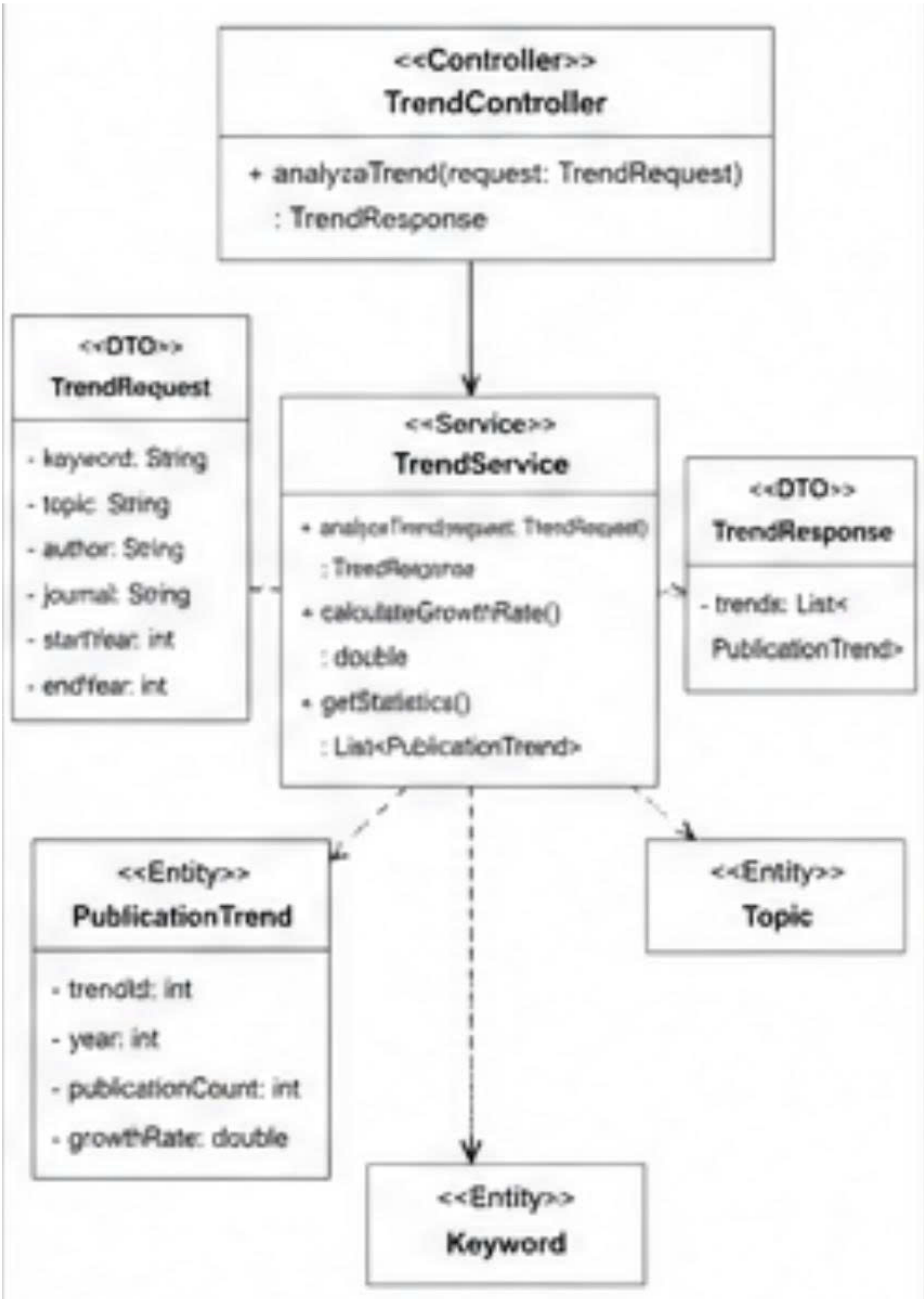
Hình 30: Sequence Diagram: Track Publication Trends

3.6.3 State Diagram



Hình 31: State Diagram: Track Publication Trends

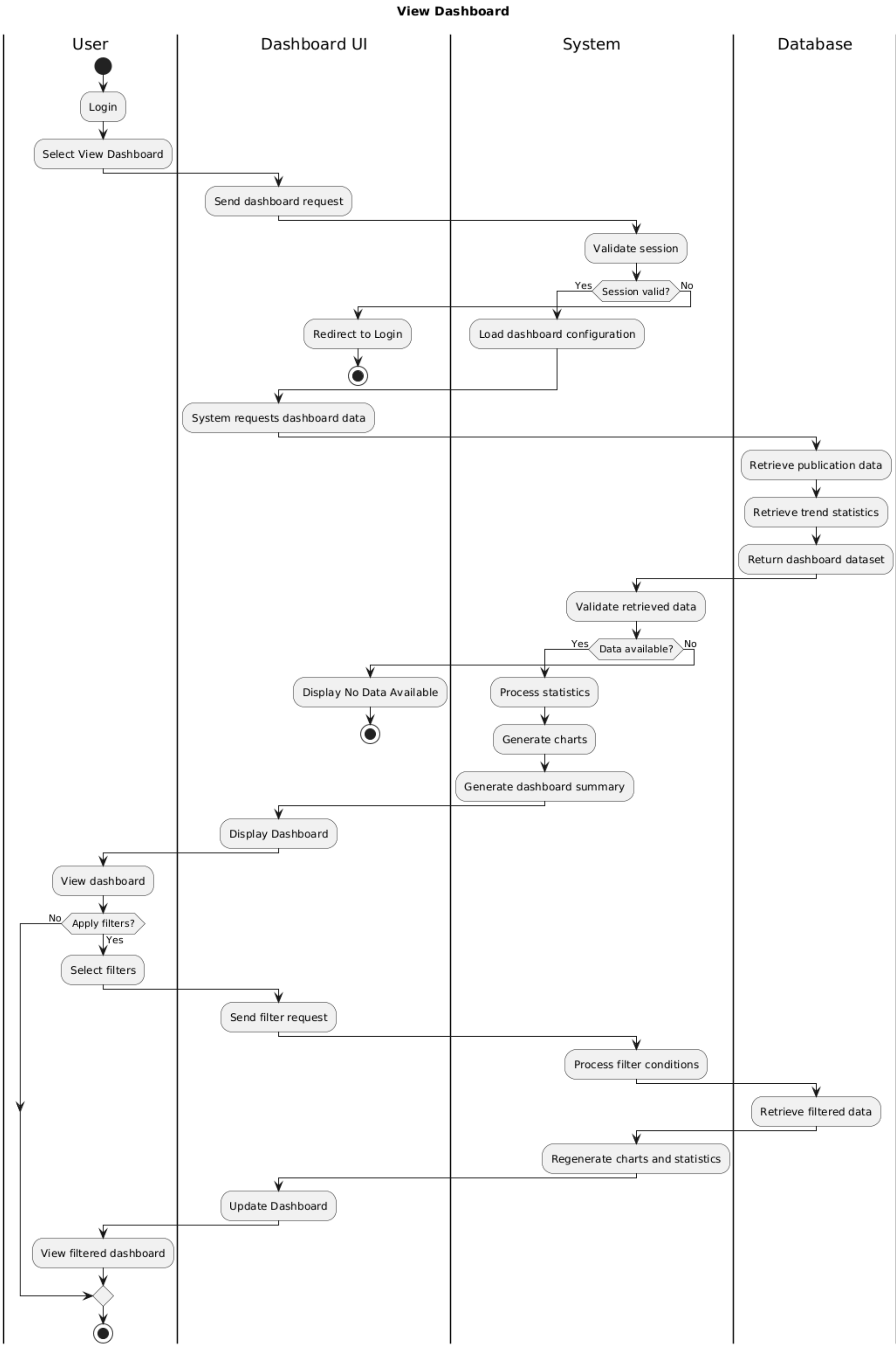
3.6.4 Class Diagram



Hình 32: Class Diagram: Track Publication Trends

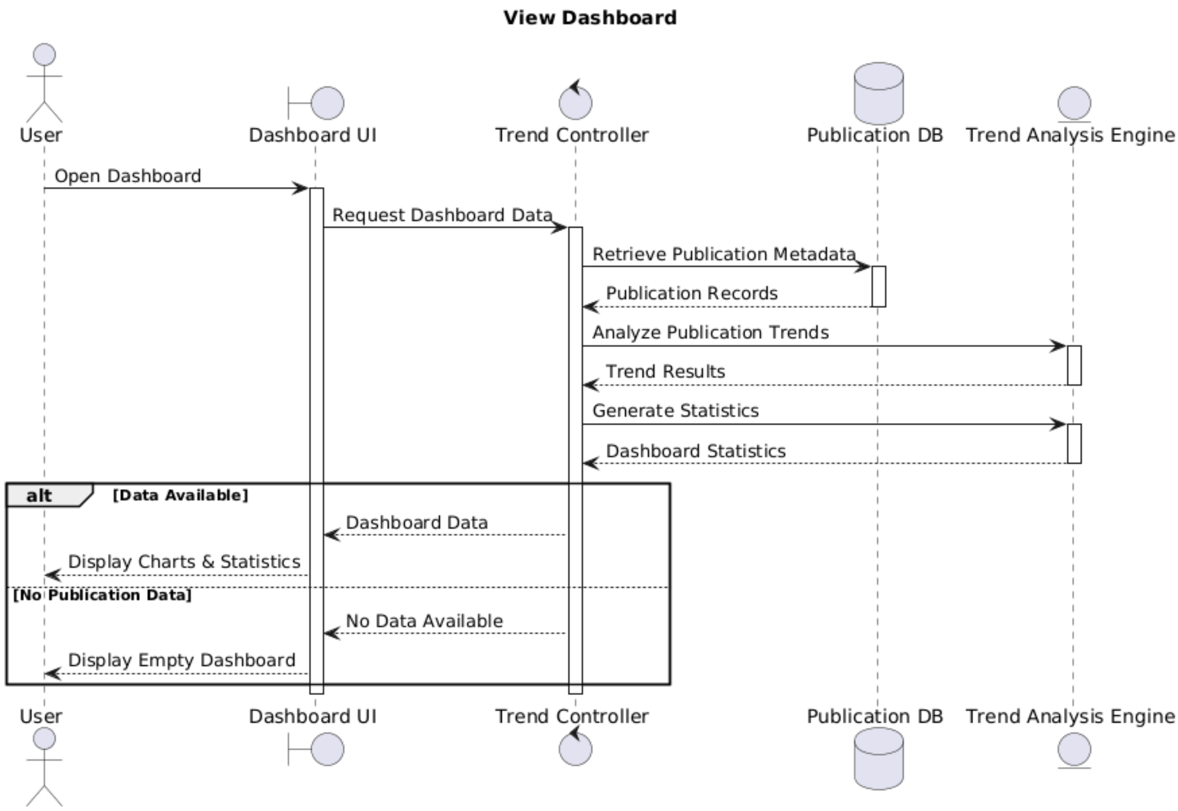
3.7 View Dashboard

3.7.1 Activity Diagram



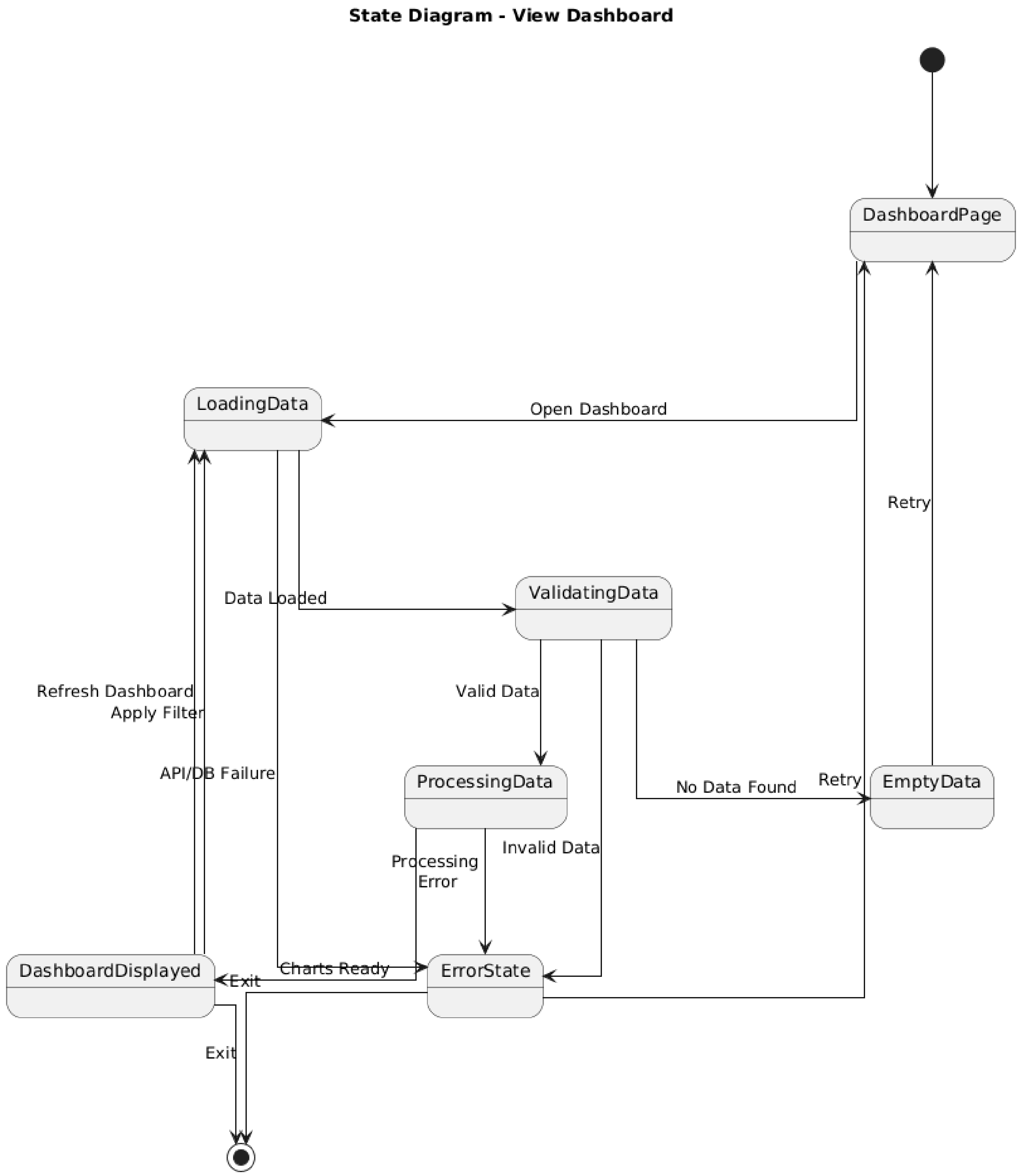
Hình 33: Activity Diagram: View Dashboard

3.7.2 Sequence Diagram



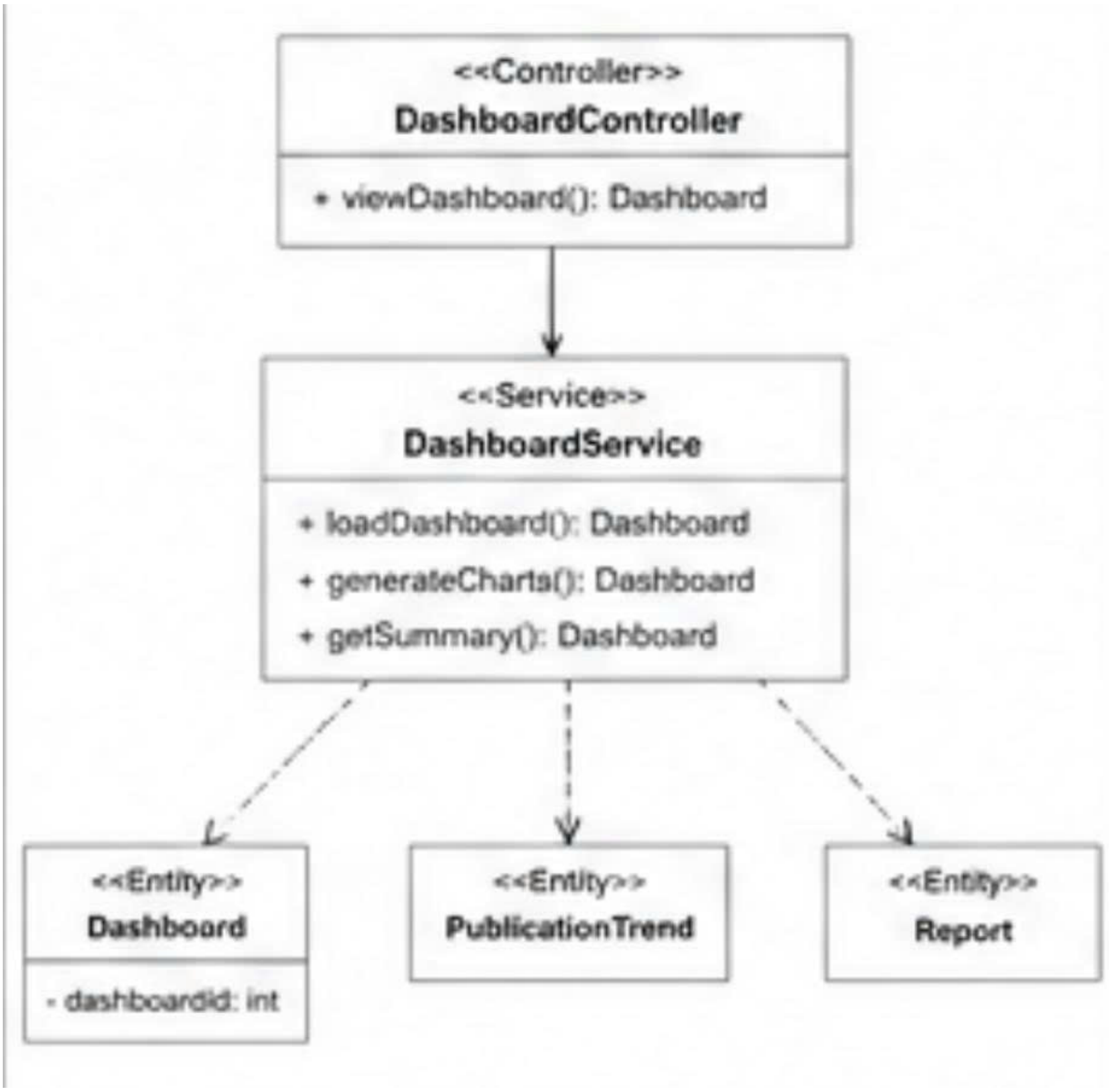
Hình 34: Sequence Diagram: View Dashboard

3.7.3 State Diagram



Hình 35: State Diagram: View Dashboard

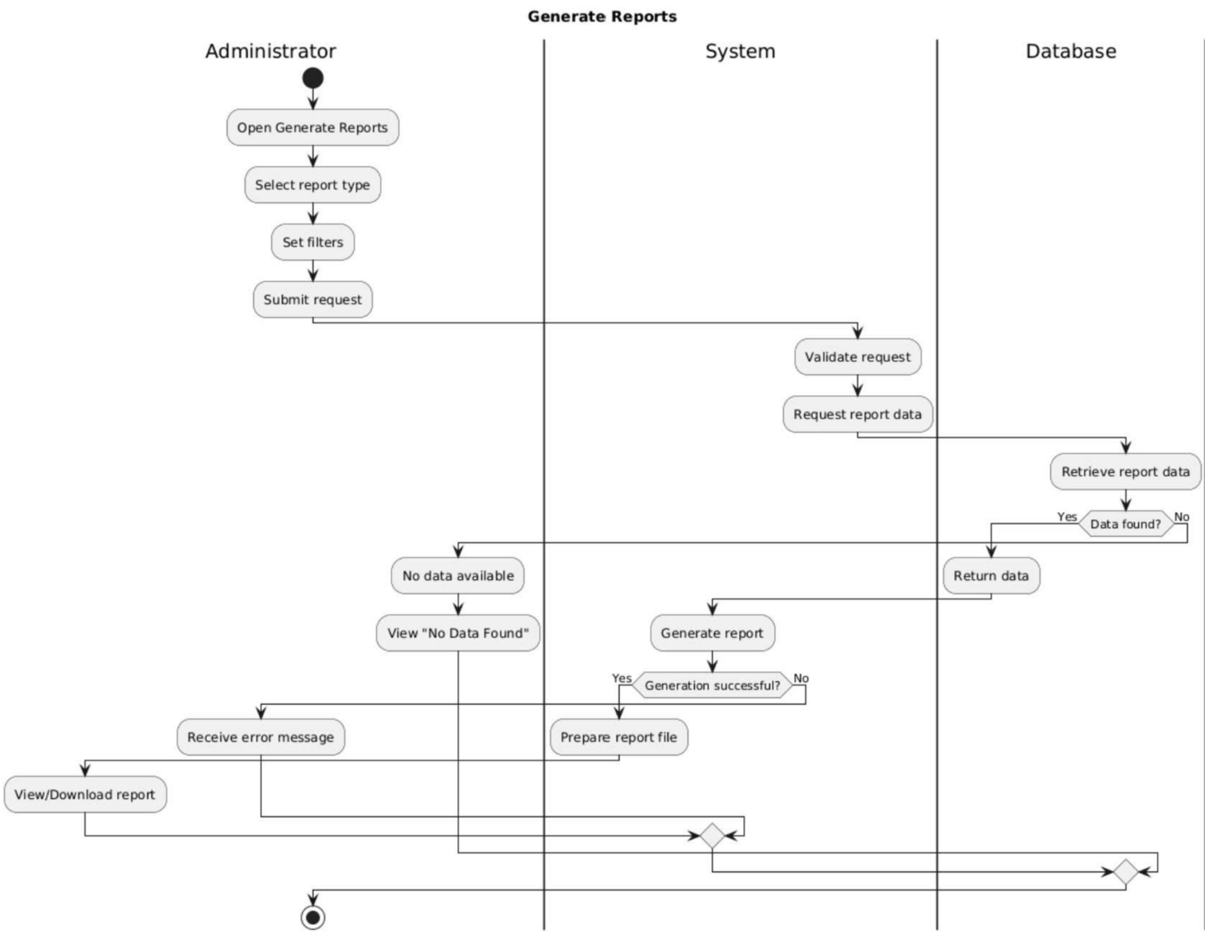
3.7.4 Class Diagram



Hình 36: Class Diagram: View Dashboard

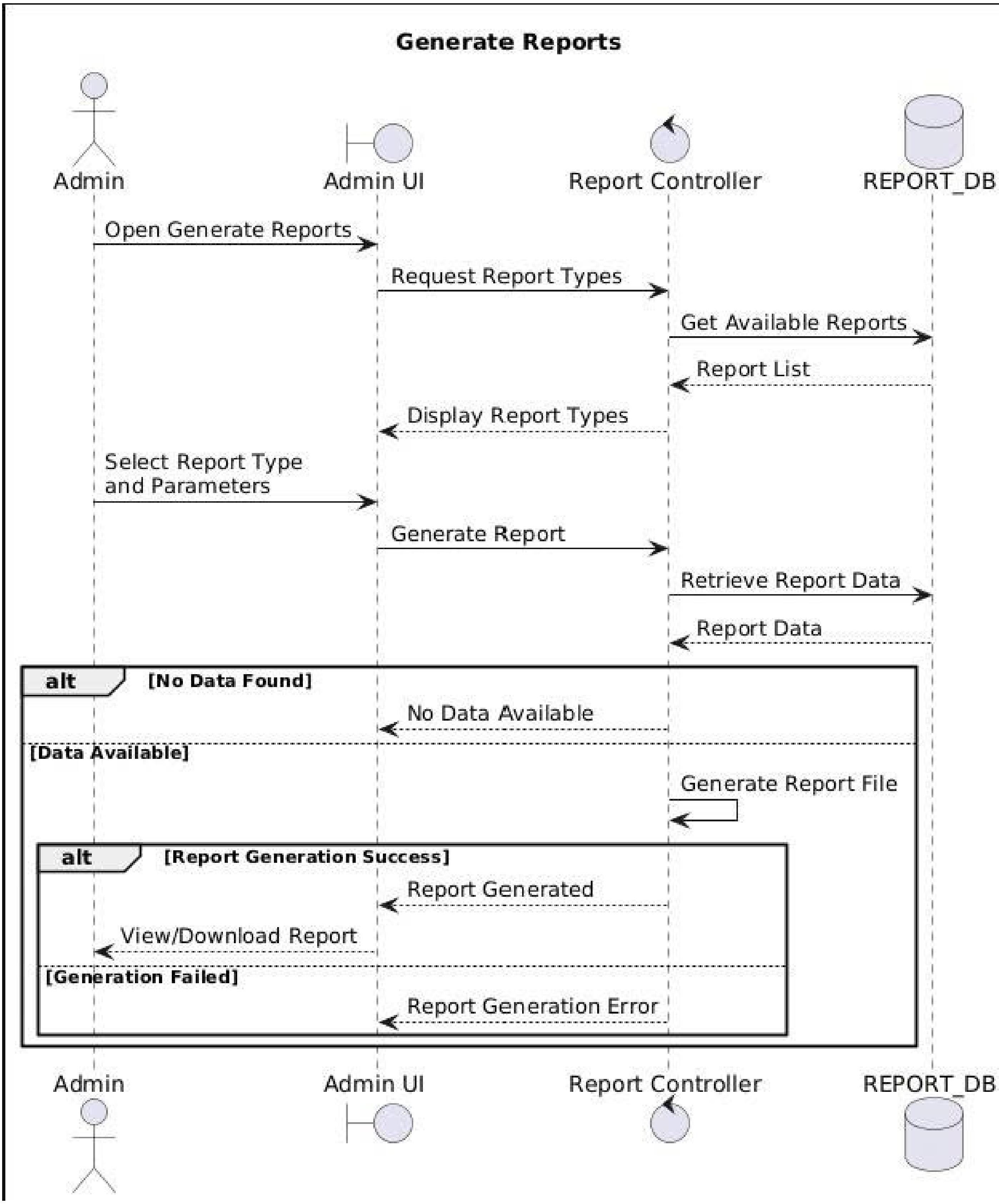
3.8 Generate Reports

3.8.1 Activity Diagram



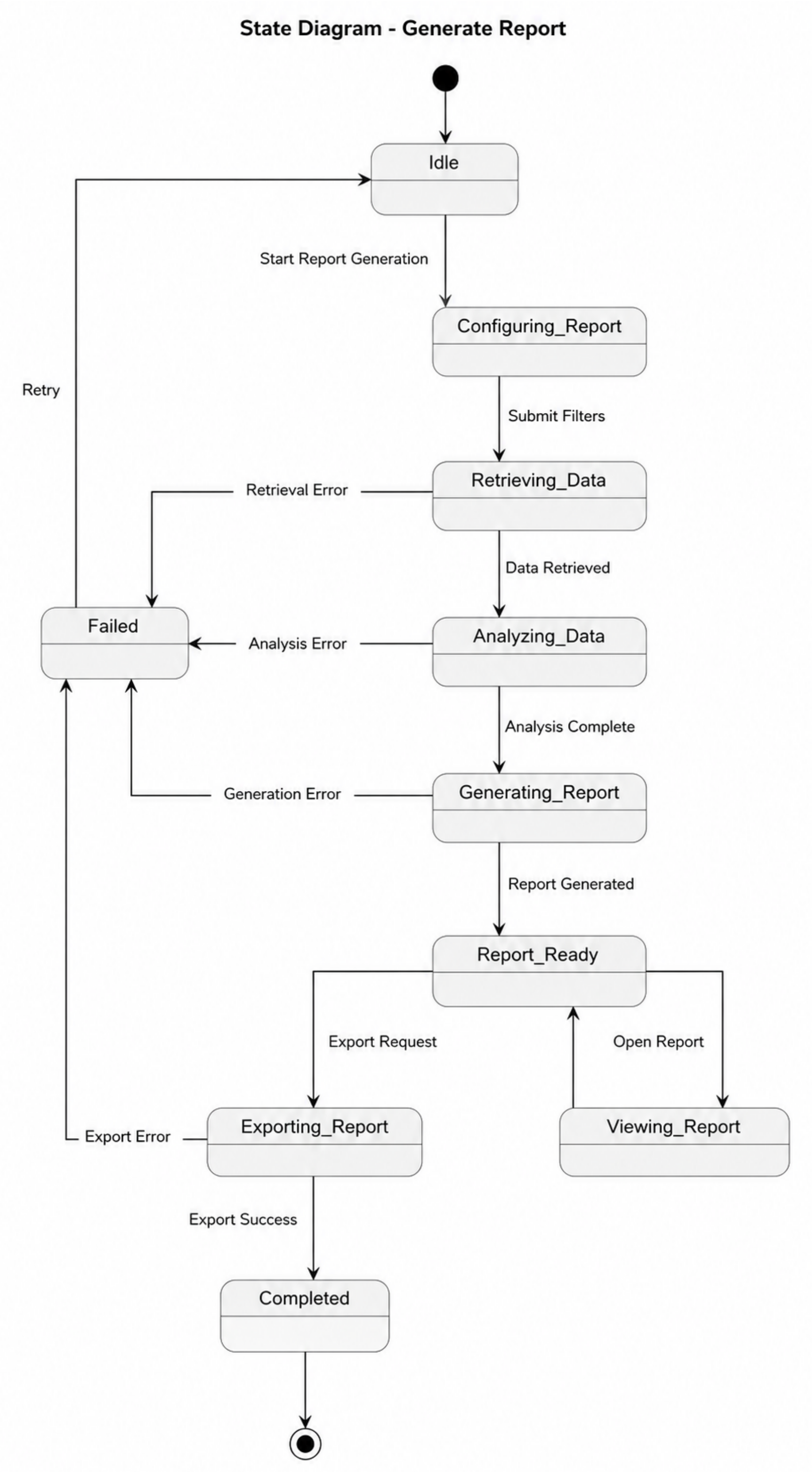
Hình 37: Activity Diagram - Generate reports

3.8.2 Sequence Diagram



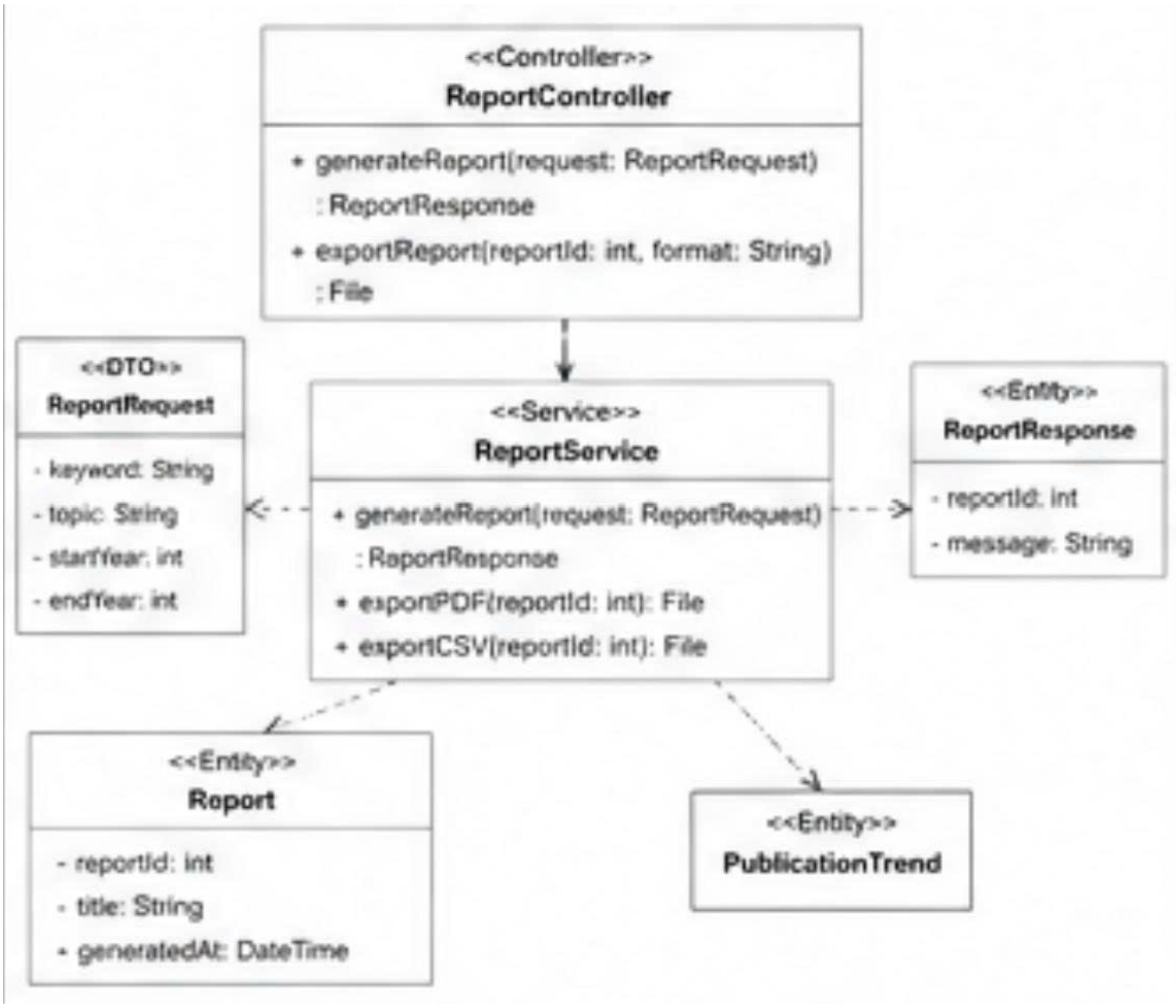
Hình 38: Sequence diagram - Generate reports

3.8.3 State Diagram



Hình 39: State diagram - Generate reports

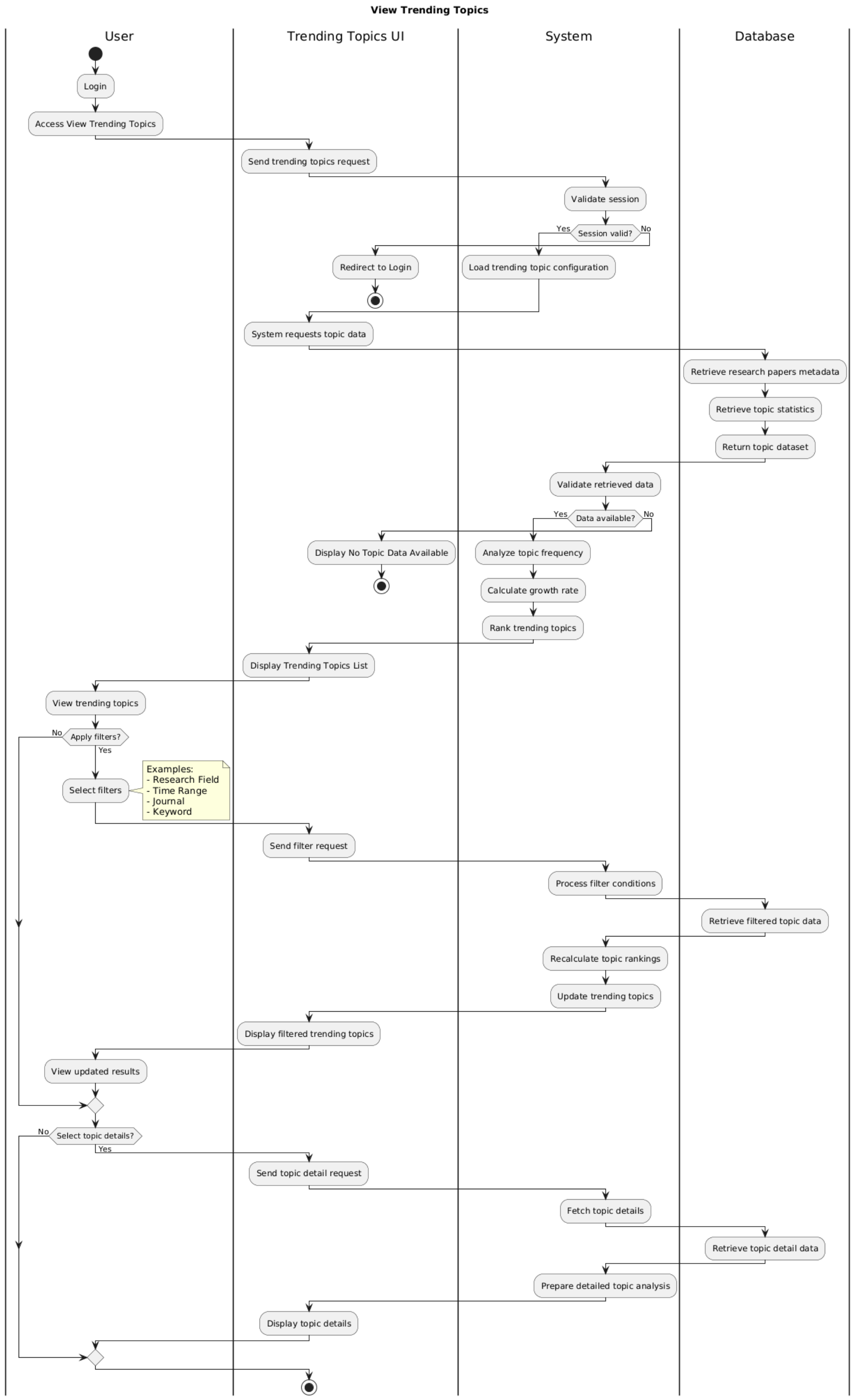
3.8.4 Class Diagram



Hình 40: Class Diagram - Generate Reports

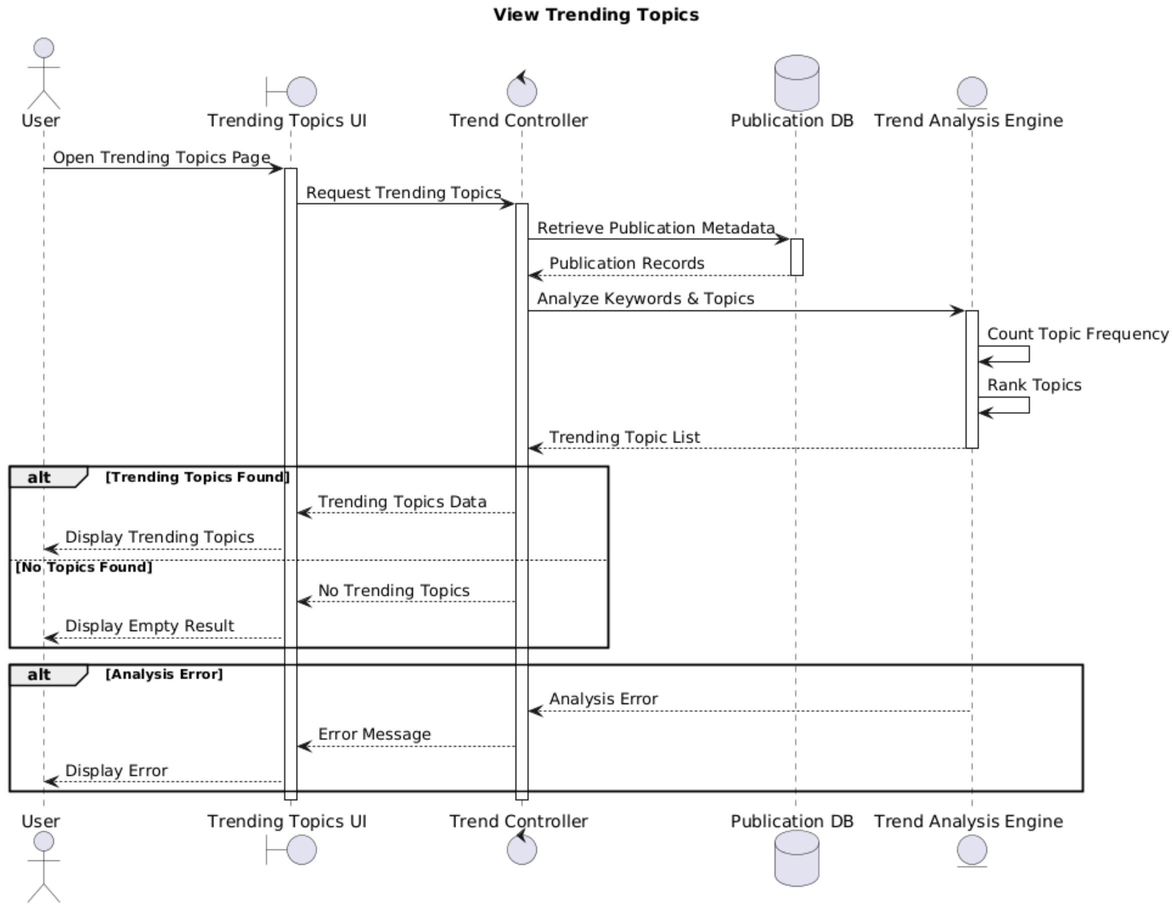
3.9 View Trending Topics

3.9.1 Activity Diagram



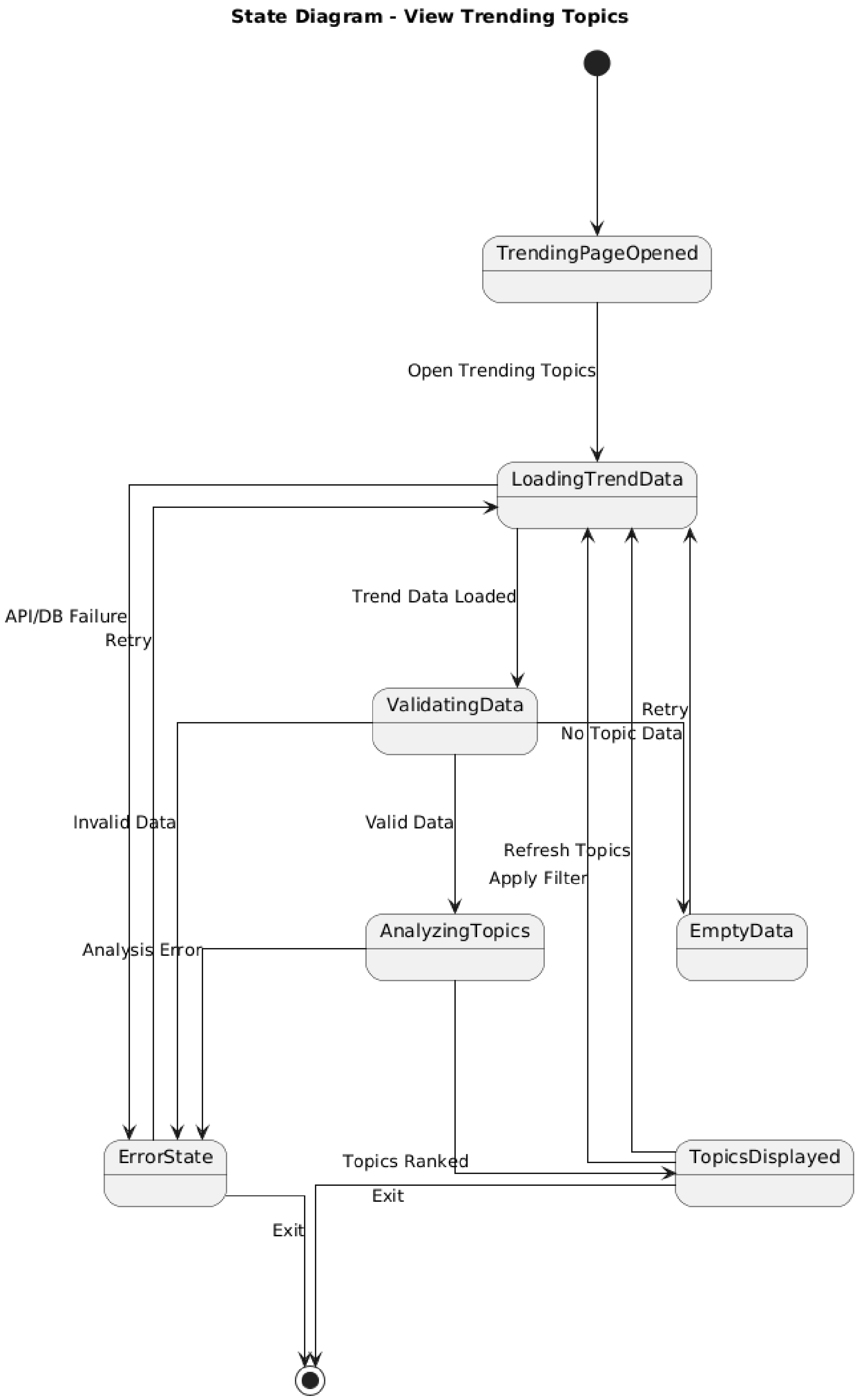
Hình 41: Activity Diagram - View Trending Topics

3.9.2 Sequence Diagram



Hình 42: Sequence diagram

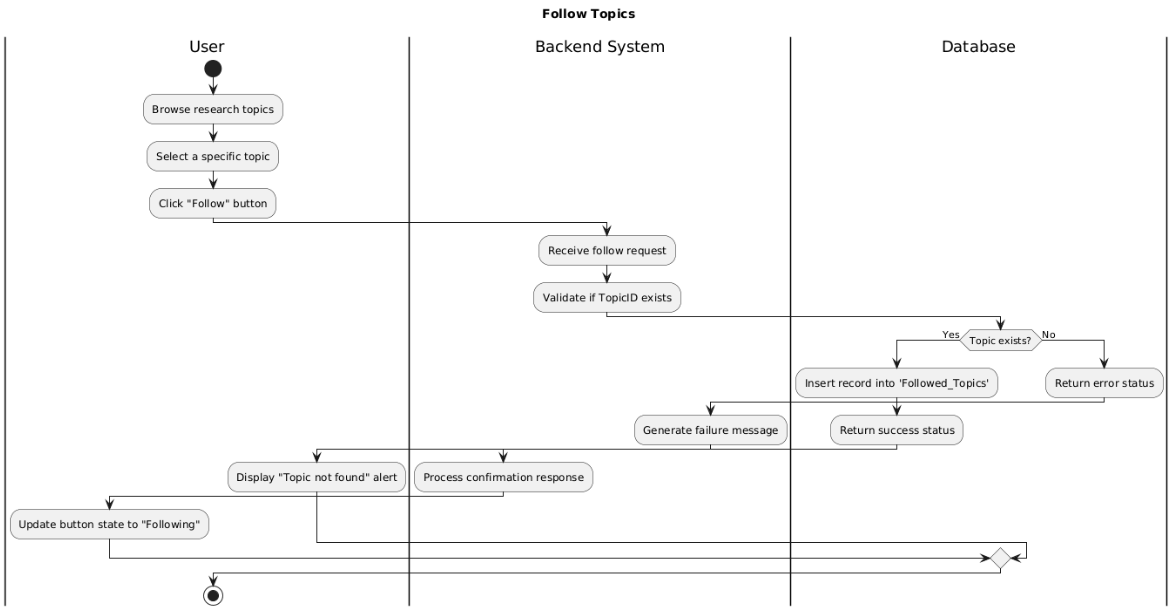
3.9.3 State Diagram



Hình 43: State diagram

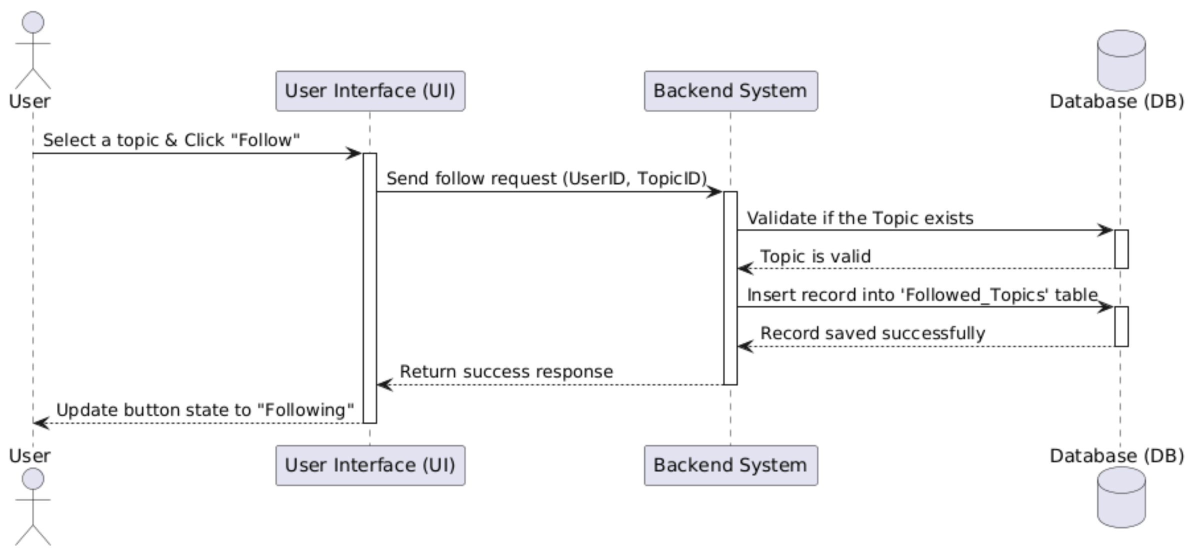
3.10 Follow Topics

3.10.1 Activity Diagram



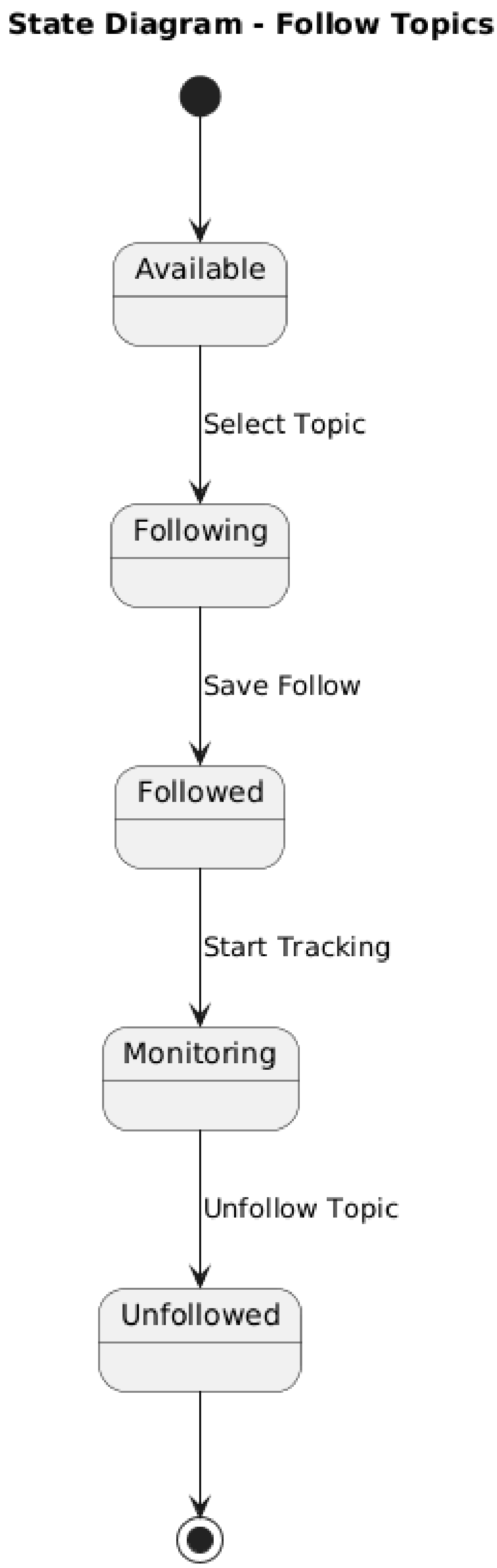
Hình 44: Activity diagram

3.10.2 Sequence Diagram



Hình 45: Sequence diagram

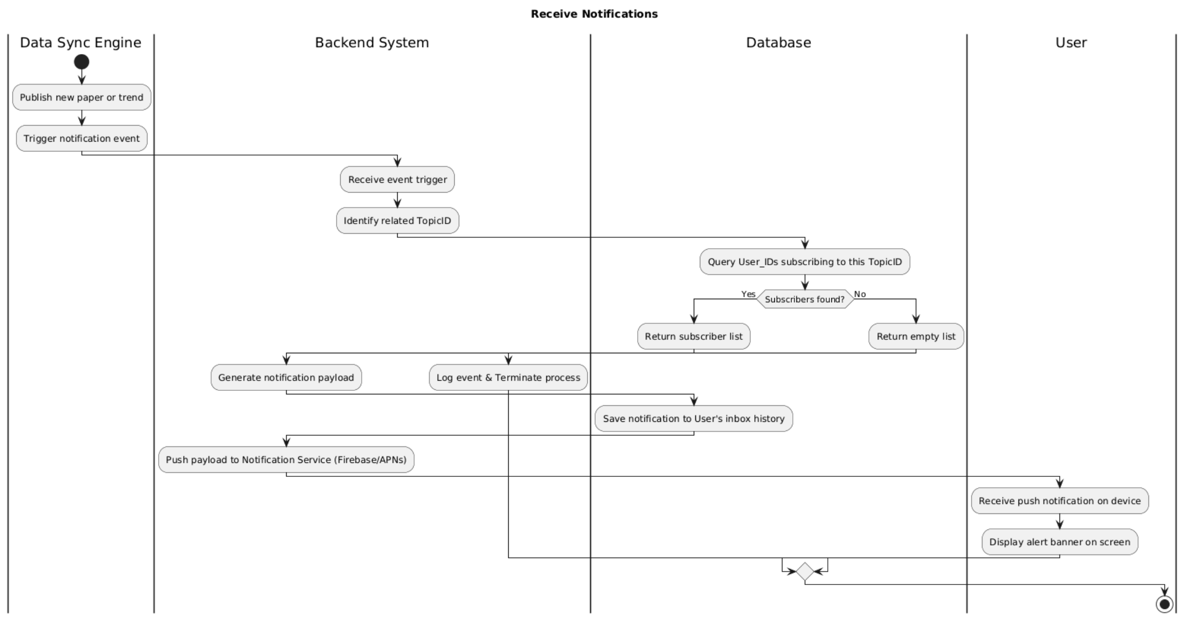
3.10.3 State Diagram



Hình 46: State diagram

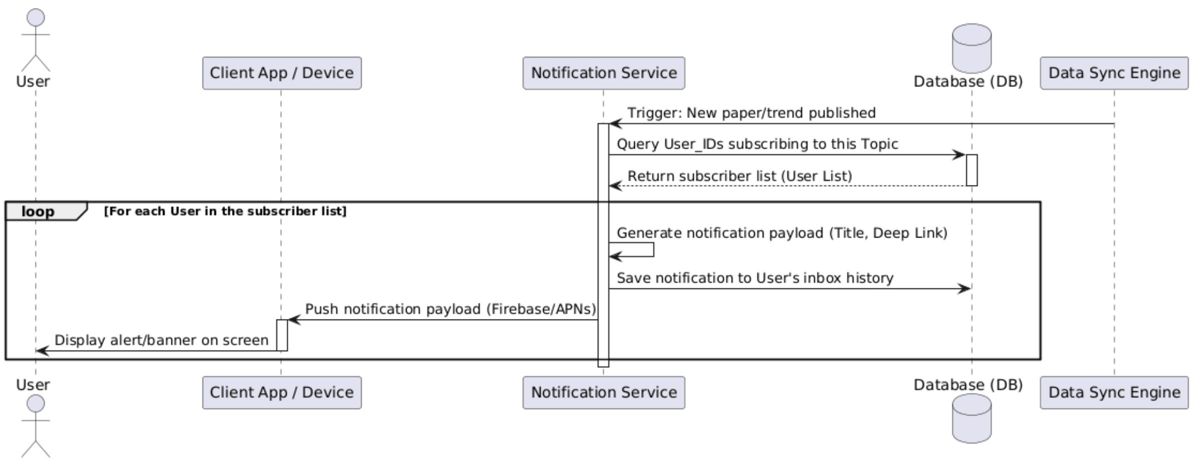
3.11 Receive Notifications

3.11.1 Activity Diagram



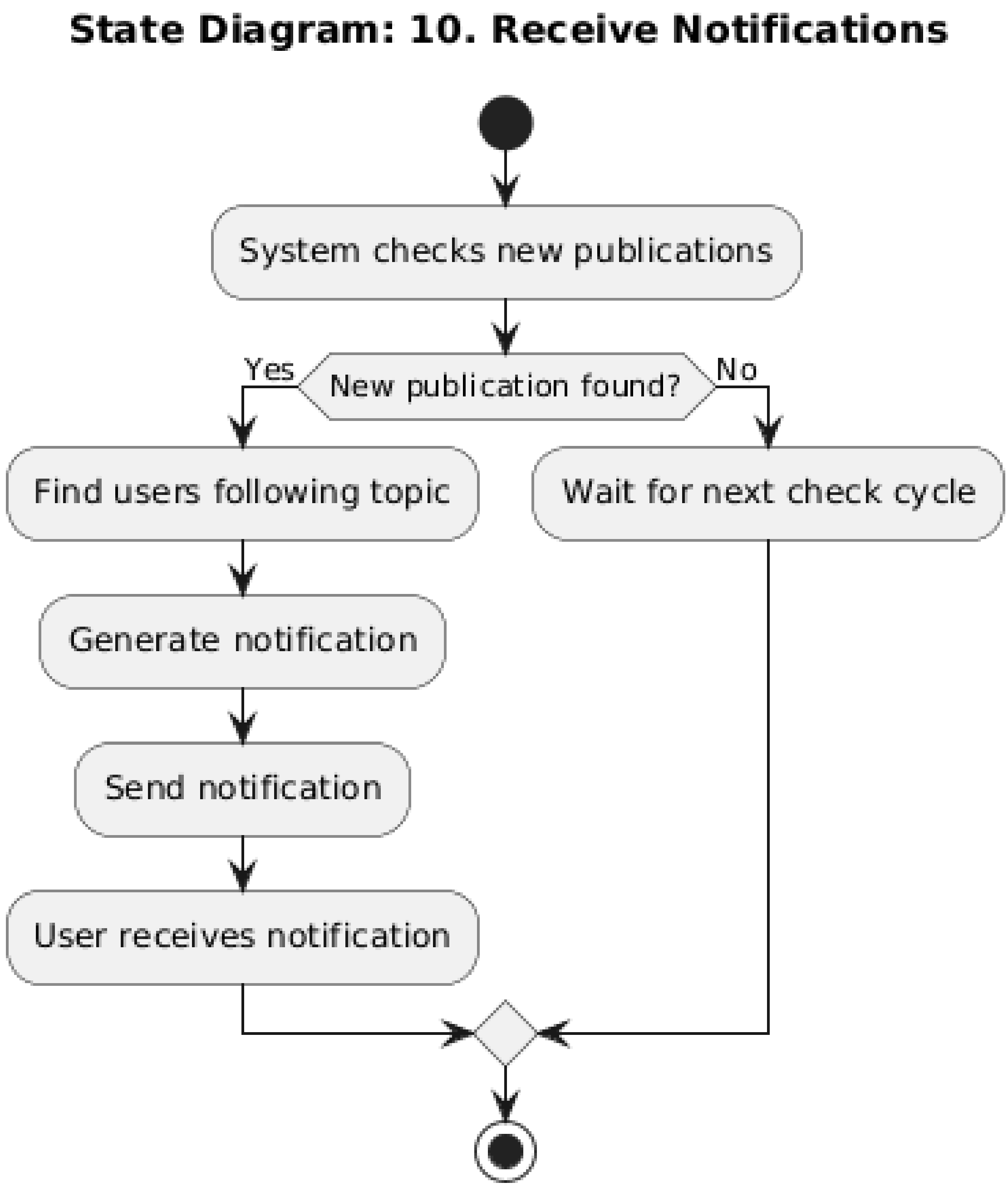
Hình 47: Activity diagram

3.11.2 Sequence Diagram



Hình 48: Sequence diagram

3.11.3 State Diagram



Hình 49: State diagram

3.12 Manage System

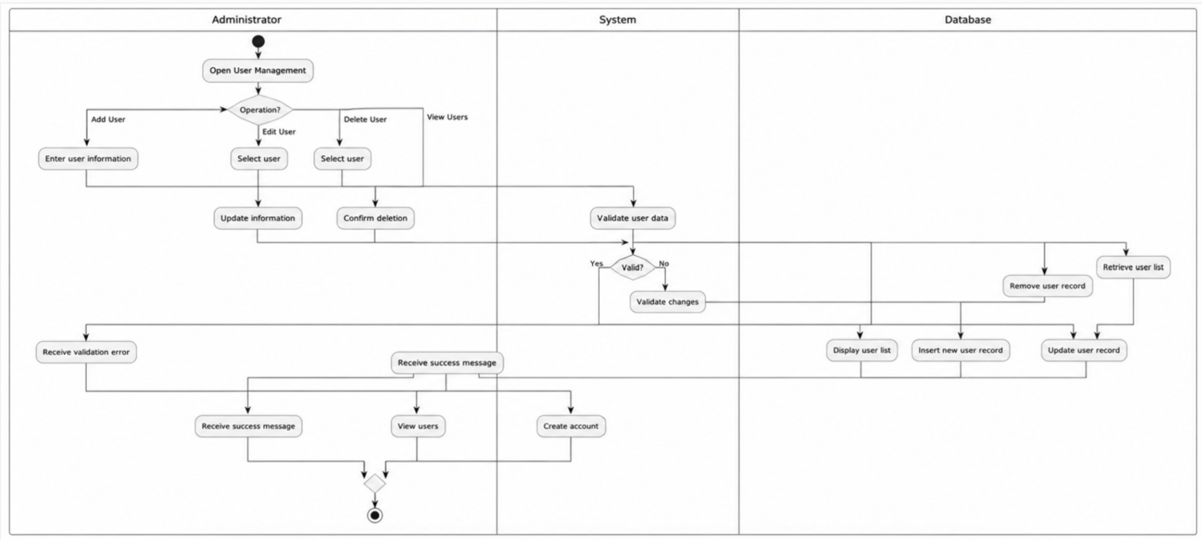
3.12.1 Activity Diagram

3.12.2 Sequence Diagram

3.12.3 State Diagram

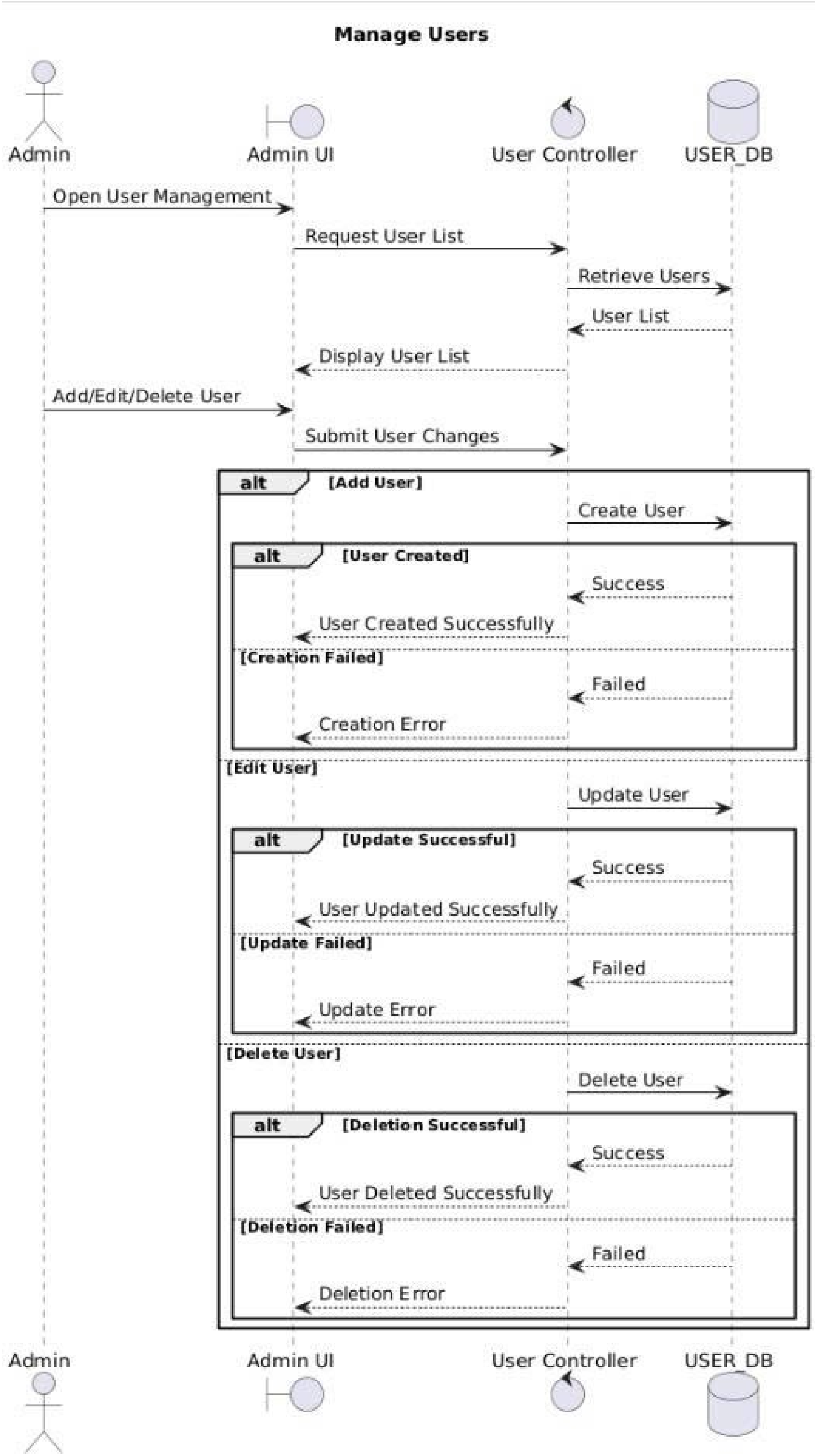
3.13 Manage User

3.13.1 Activity Diagram



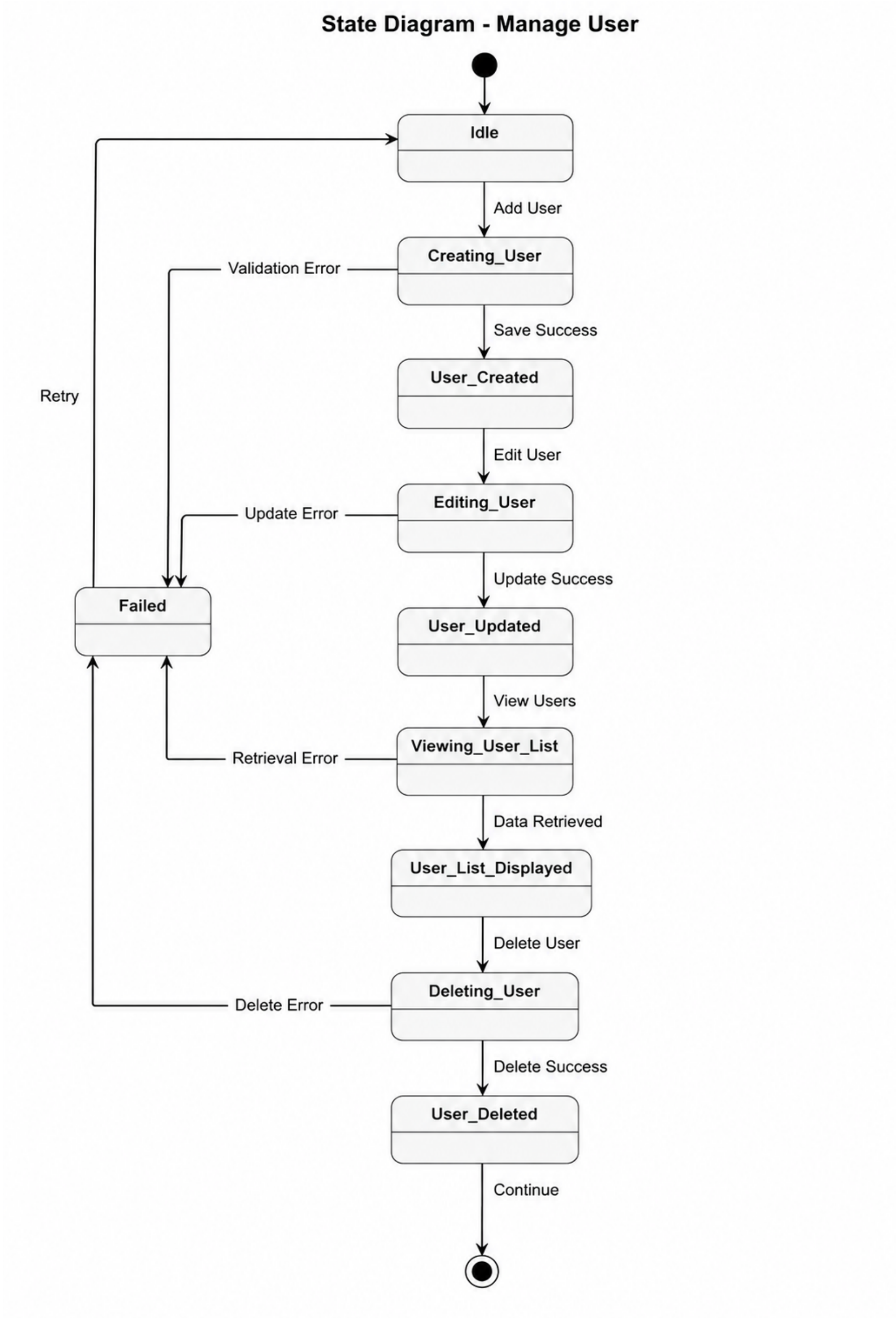
Hình 50: Activity Diagram - Manage Users

3.13.2 Sequence Diagram



Hình 51: Sequence diagram - Manage Users

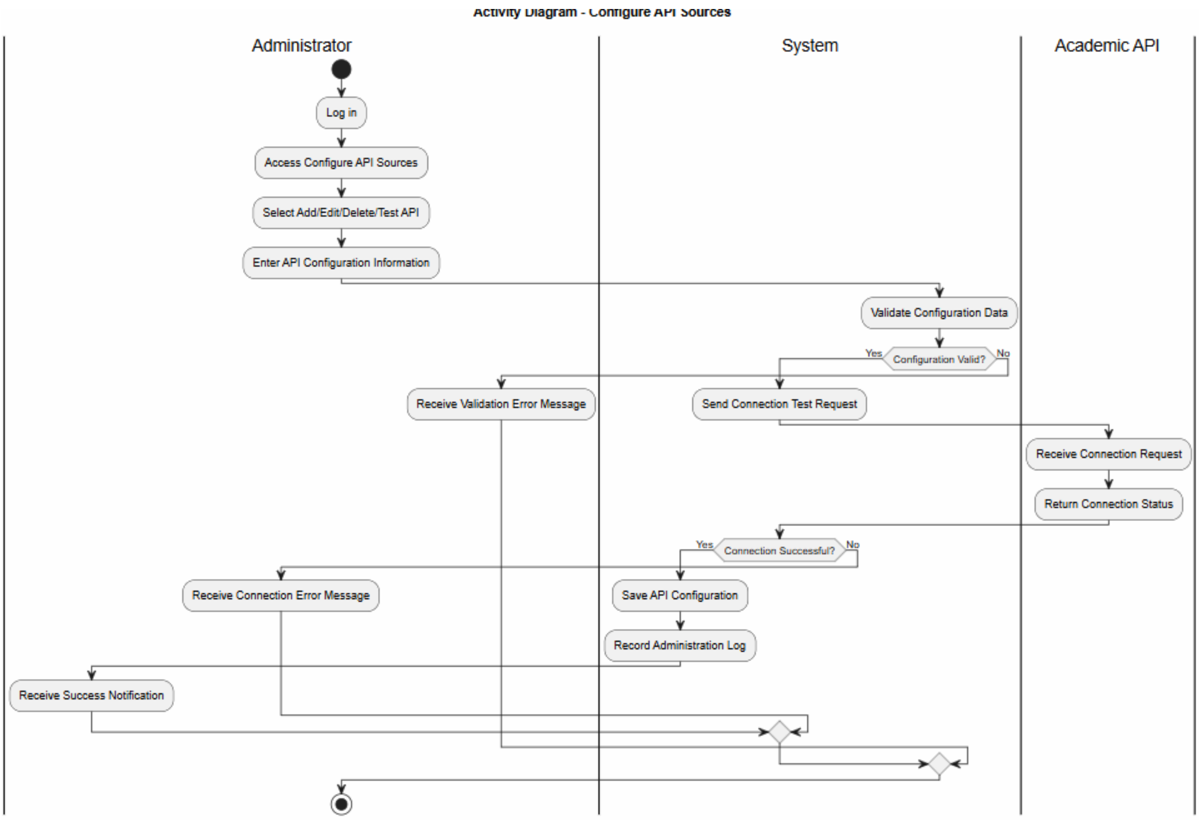
3.13.3 State Diagram



Hình 52: State diagram - Manage Users

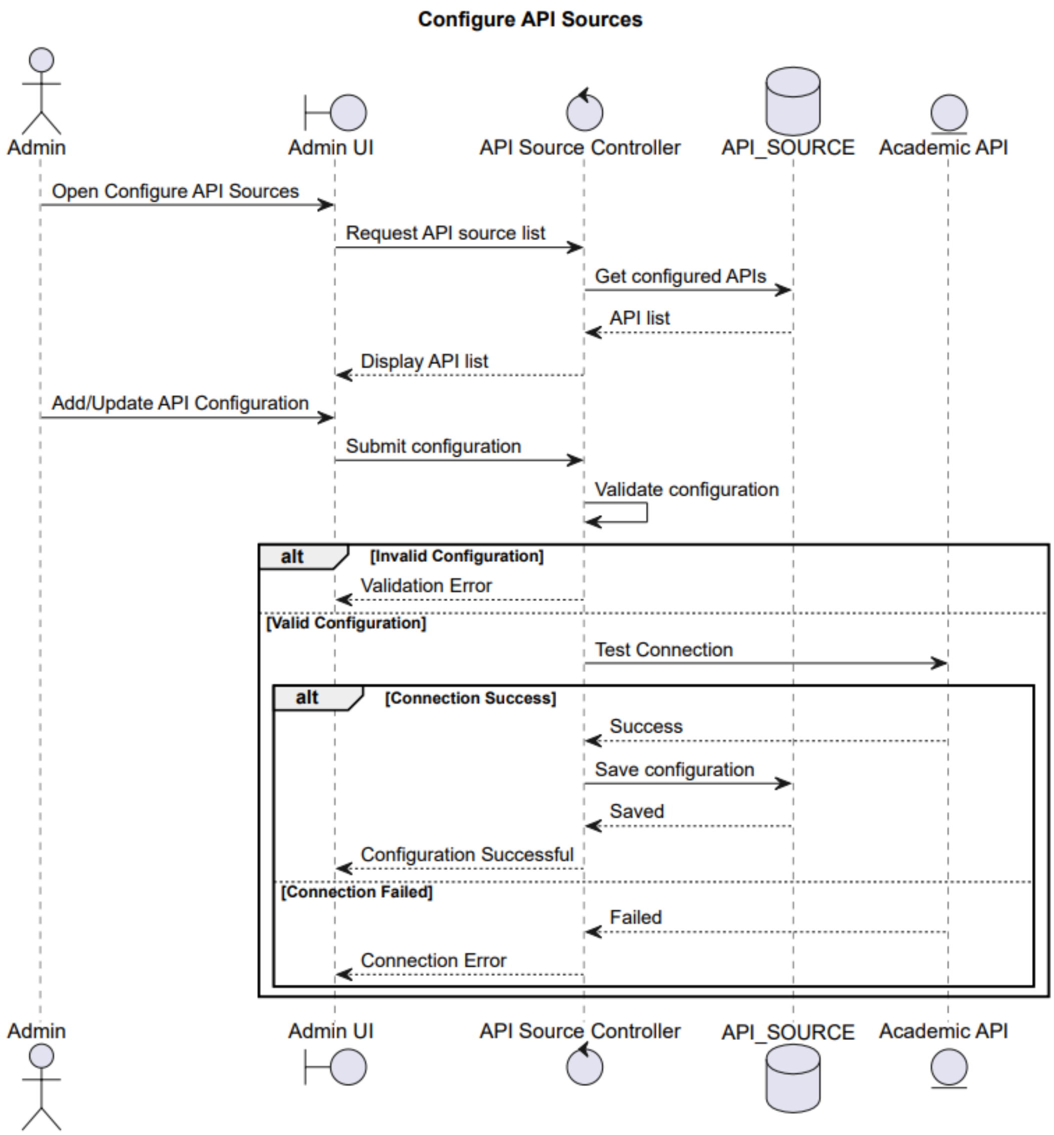
3.14 Configure API Sources

3.14.1 Activity Diagram



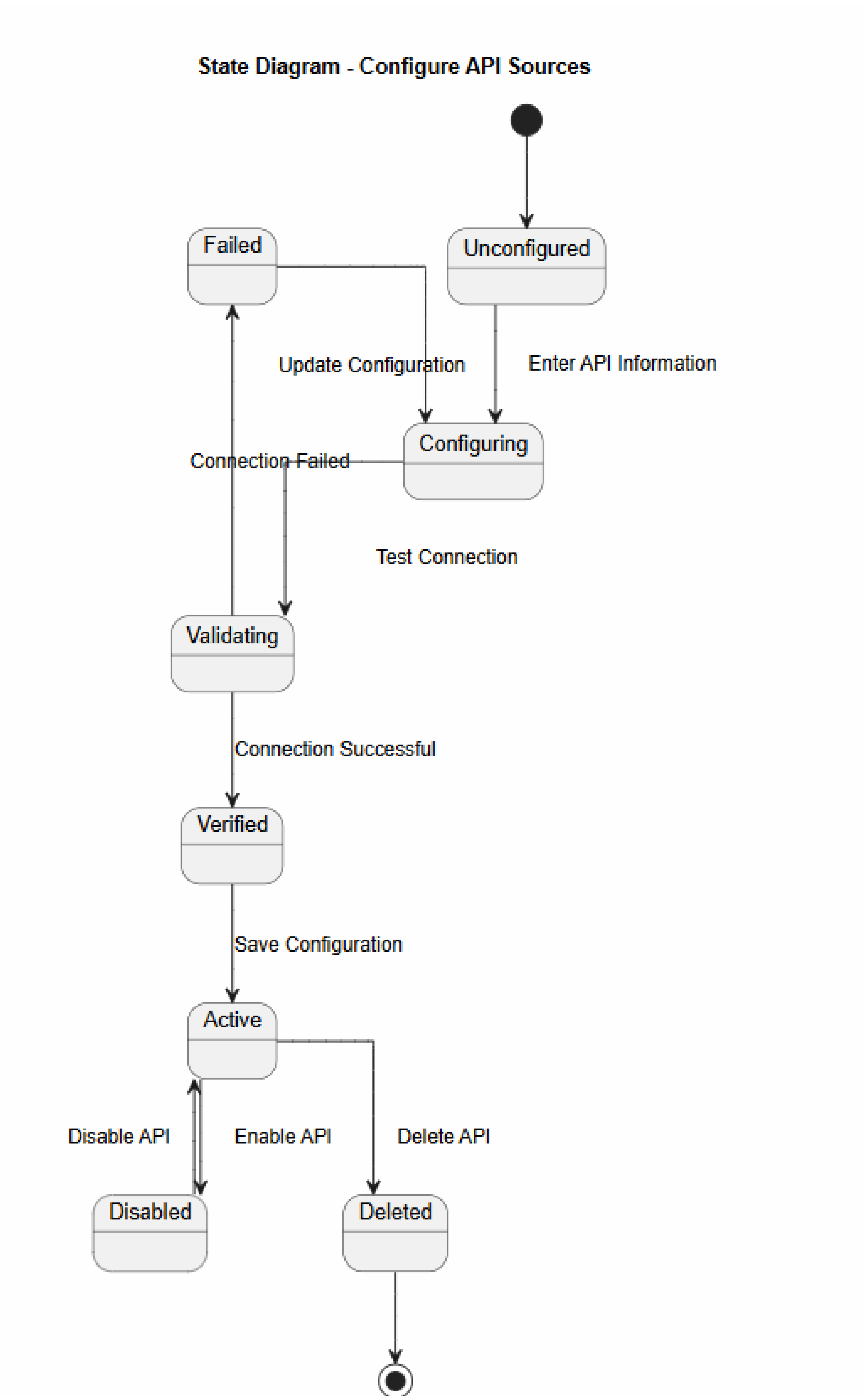
Hình 53: Configure API Sources

3.14.2 Sequence Diagram



Hình 54: Configure API Sources

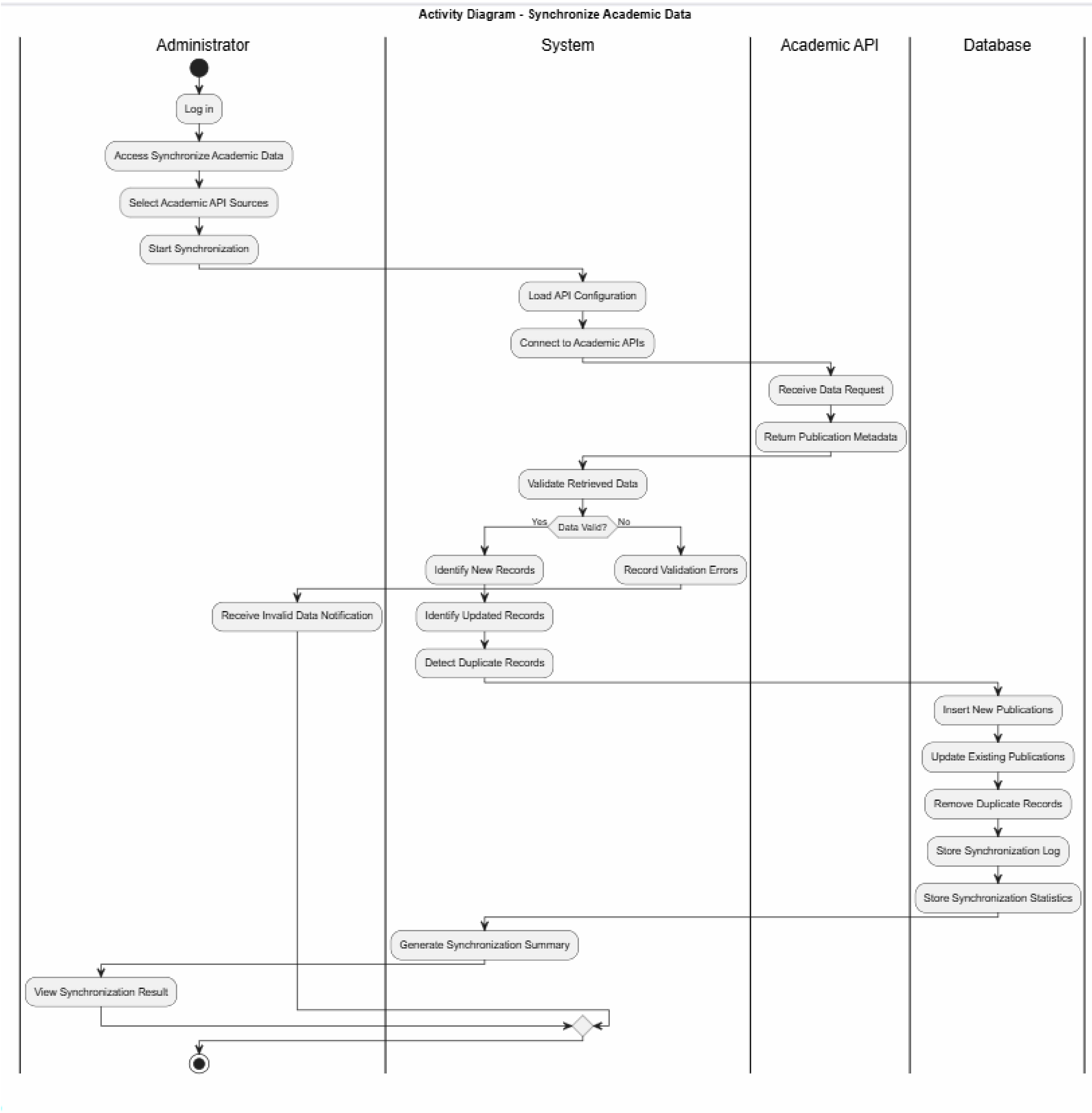
3.14.3 State Diagram



Hình 55: State diagram

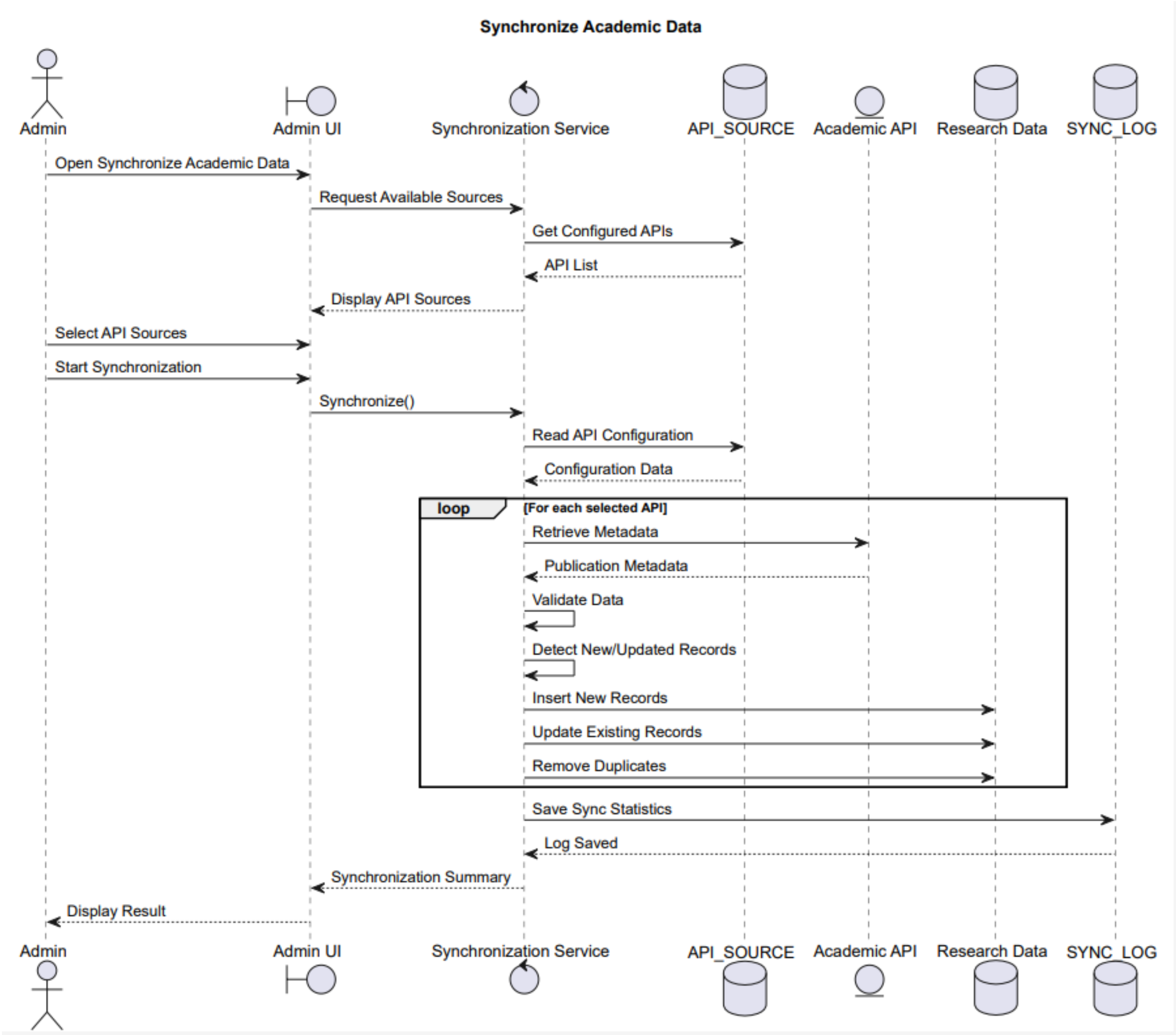
3.15 Sync Academic Data

3.15.1 Activity Diagram



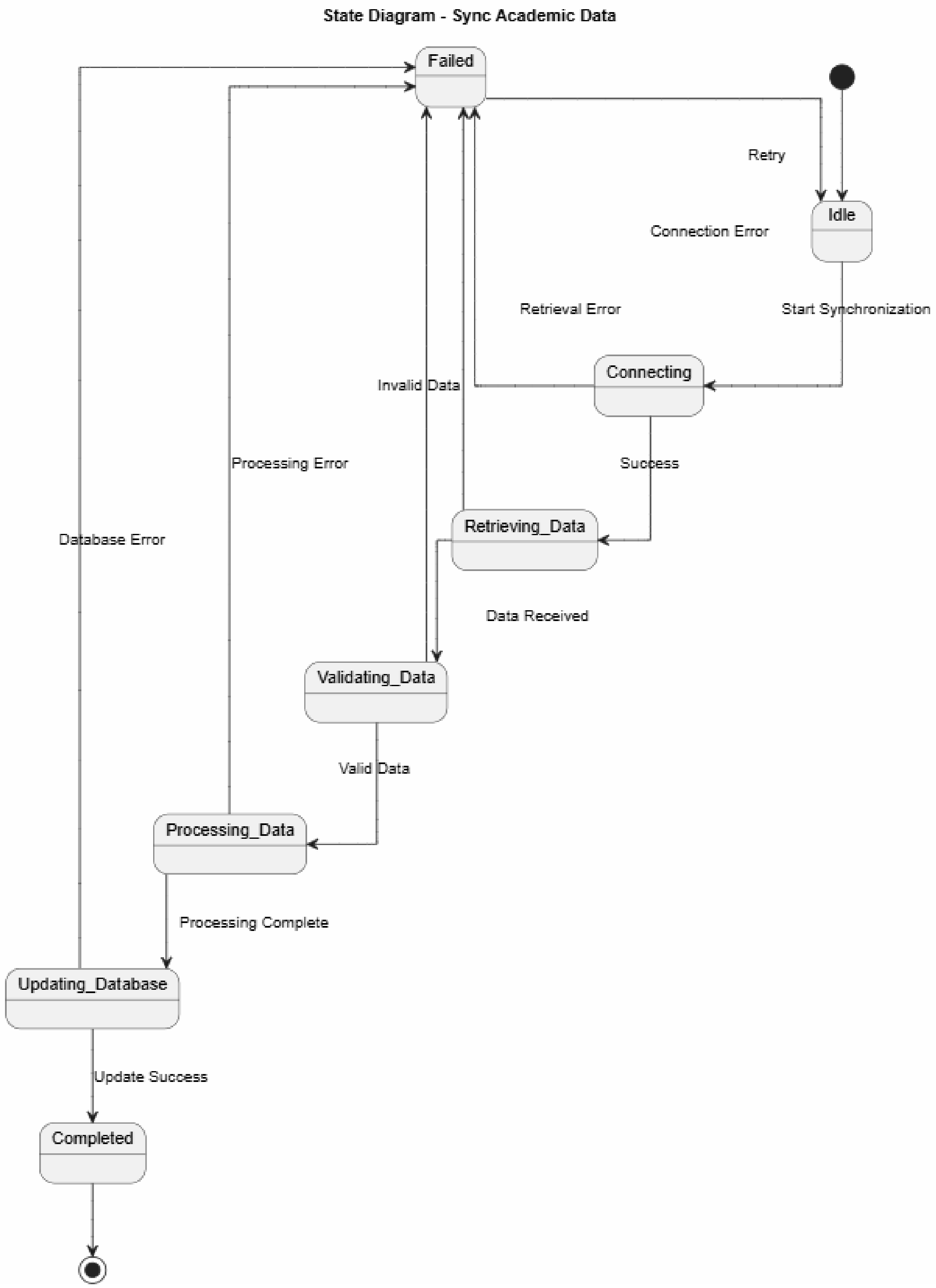
Hình 56: Sync Academic Data

3.15.2 Sequence Diagram



Hình 57: Sync Academic Data

3.15.3 State Diagram



Hình 58: Sync Academic Data